REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION REACTIONS.

Reduction:- • Addition of hydrogen

• Removal of oxygen

✓⇒ Gain of $e^{\ominus}$(s)

• decrease in oxidation no.

REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION REACTIONS.

* Reduction steps for important functional group

① $R{-}X \xrightarrow{2H} R{-}H + HX$ (Alkane)

② $R{-}OH \xrightarrow{2H} R{-}H + H_2O$

③ $R{-}CH{=}O \xrightarrow{2H} R{-}CH_2{-}OH \xrightarrow[-H_2O]{2H} R{-}CH_3$ (1° Alcohol; Alkane)

④ $R{-}\overset{O}{\overset{\|}{C}}{-}R \xrightarrow{2H} R{-}\underset{OH}{CH}{-}R \xrightarrow[-H_2O]{2H} R{-}CH_2{-}R$ (2° alcohol)

⑤ $R{-}\overset{O}{\overset{\|}{C}}{-}Cl \xrightarrow{2H} R{-}\overset{O}{\overset{\|}{C}}{-}H \longrightarrow R{-}CH_2{-}OH \longrightarrow R{-}CH_3$

⑥ $R{-}\overset{O}{\overset{\|}{C}}{-}OH \xrightarrow[-H_2O]{2H} R{-}\overset{O}{\overset{\|}{C}}{-}H \xrightarrow{2H} R{-}CH_2{-}OH \longrightarrow R{-}CH_3$

⑦ $R{-}\overset{O}{\overset{\|}{C}}{-}O{-}\overset{O}{\overset{\|}{C}}{-}R \longrightarrow R{-}\overset{O}{\overset{\|}{C}}{-}H$ (+ $R{-}COOH$) $\longrightarrow R{-}CH_2{-}OH$ (+ $R{-}CHO$) $\longrightarrow R{-}CH_3$

REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION REACTIONS.

* Reduction steps for important functional group

(8) $R-\overset{\overset{O}{\|}}{C}-OR' \xrightarrow{2H} R-\overset{\overset{O}{\|}}{C}-H \longrightarrow R-CH_2-OH \longrightarrow R-CH_3$

$+$

$R'-OH \longrightarrow R'-H$

(9) $R-N^{+}(=O)-O^{-} \xrightarrow[-H_2O]{2H} R-N=O$ (nitoo) $\xrightarrow{2H} R-NH-OH$ (alkyl hydroxyl amine) $\xrightarrow[-H_2O]{2H} R-NH_2$

(10) $R-CH=NH \xrightarrow{Red} R-CH_2-NH_2$

imine → amine

(11) $R-C\equiv N \xrightarrow{2H} R-\overset{H}{\underset{|}{C}}=NH \xrightarrow{Hydrolysis} R-\overset{H}{\underset{|}{C}}=O + NH_3$

similar reduction of isocyanide forming secondary amine

Reduction, Oxidation And Hydrolysis

(I) Reduction Reactions.

* Reduction steps for important functional group



unstable (just like gem-diols)

(12) $R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow{2H} \left[R-\overset{OH}{\overset{|}{C}}H-NH_2\right] \xrightarrow[-H_2O]{} R-CH=NH \longrightarrow R-CH_2-NH_2$



(13) $R-N=C=O \xrightarrow{4H} \left[R-NH-CH_2-OH\right] \xrightarrow[-H_2O]{} R-N=CH \xrightarrow{2H} R-NH-CH_3$

2° amine

(14) $R-N_3$

$R-\overset{\ominus}{N}-\overset{\oplus}{N}\equiv N \xrightarrow{2H} R-NH_2 + N_2\uparrow$

(15) $R-N=N-R \xrightarrow{2H} R-NH-NH-R \longrightarrow 2R-NH_2$

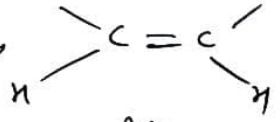
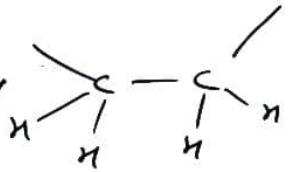
Azoalkane — Hydrazoalkane

(16) $>C=C< \xrightarrow{2H} >\overset{H}{C}-\overset{H}{C}<$ Alkane

Reduction, Oxidation And Hydrolysis

(I) Reduction Reactions.

* Reduction steps for important functional group

→ $-C\equiv C-$ $\xrightarrow{2H}$  Alkene $\longrightarrow$  Alkane

oxygen containing compounds on reduction generally form alcohols
nitrogen containing compounds on reduction generally form amine

REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION REACTIONS.

	H_2 + Metal (Hydrogenation)	$LiAlH_4$ in ether	$NaBH_4$ in ether	Na + EtOH Bouveault Blac Red. (B.B.R)	B_2H_6 in THF (Hydroboration)
① $R-CH=O \xrightarrow{\text{Reduction}} R-CH_2-OH$ (1° alcohol)	✓	✓	✓	✓	✓
② $R-\underset{\underset{O}{\|}}{C}-R \xrightarrow{"} R-\underset{\underset{OH}{\|}}{CH}-R$ (2° ")	✓	✓	✓	✓	✓
③ $R-\overset{\overset{O}{\|}}{C}-Cl \xrightarrow{"} R-CH_2-OH$	✓	✓	✓	✓	NO
④ $R-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R \xrightarrow{"} 2R-CH_2-OH$	X	✓	X	✓	✓ possible in special cases
⑤ $R-\overset{\overset{O}{\|}}{C}+OR' \xrightarrow{"} R-CH_2-OH + R'OH$	X	✓	X	✓	✓
⑥ $R-\overset{\overset{O}{\|}}{C}-OH \xrightarrow{"} R-CH_2-OH$	X	✓	X	X imp.	✓ imp.
⑦ $R-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow{"} R-CH_2-NH_2$	X	✓	X	X	✓
⑧ $R-C\equiv N \xrightarrow{"} R-CH_2-NH_2$	✓	✓	X	✓	✓
⑨ $R-NC \xrightarrow{"} R-NH-CH_3$	✓	✓	X	✓	✓
⑩ $R-NO_2 \xrightarrow{"} R-NH_2$	✓	✓	X	X	X
⑪ $R-X \xrightarrow{"} R-H$	✓	✓	✓	✓	NO

REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION REACTIONS

		H_2 + metal Hydrogenation	$LiAlH_4$ in ether	$NaBH_4$ in ether	Na + EtOH B.B.R	B_2H_6 Hydroboration
(12)	$>C=C<$ —Red→ Alkane	✓	X	X	X	✓
(13)	$-C\equiv C-$ —∴→ Alkane	✓	X	X	X	✓

REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION REACTIONS.

(1) Reduction with HI + Red. P

↑ very-very strong reducing agent sends almost all to alkane

except fluorine

① $R\text{-}X + 2HI \xrightarrow{\text{Red. P}} R\text{-}H + HX + I_2$

② $R\text{-}OH + 2HI \longrightarrow R\text{-}H + H_2O + I_2$

③ $R\text{-}CH{=}O + \text{''} \longrightarrow R\text{-}CH_3 + H_2O + I_2$

④ $R\text{-}\overset{O}{\overset{\|}{C}}\text{-}R + \text{''} \longrightarrow R\text{-}CH_2\text{-}R + \text{''} + I_2$

⑤ $R\text{-}\overset{O}{\overset{\|}{C}}\text{-}OH + \text{''} \longrightarrow R\text{-}CH_3 + \text{''} + I_2$

⑥ $R\text{-}\overset{O}{\overset{\|}{C}}\text{-}Cl + \text{''} \longrightarrow R\text{-}CH_3 + \text{''} + \text{''} + HCl$

⑦ $R\text{-}\overset{O}{\overset{\|}{C}}\text{-}O\text{-}\overset{O}{\overset{\|}{C}}\text{-}R + \text{''} \longrightarrow 2R\text{-}CH_3 + \text{''} + \text{''}$

⑧ $R\text{-}\overset{O}{\overset{\|}{C}}\text{-}O\text{-}R' + \text{''} \longrightarrow R\text{-}CH_3 + \text{''} + \text{''} + R'\text{-}H$

REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION REACTIONS.

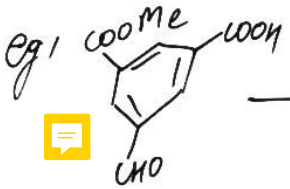
(1) Reduction with HI + Red. P

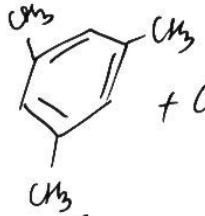
↑ very-very strong reducing agent

→ $R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow{HI + Red\ P} R-CH_2-NH_2$

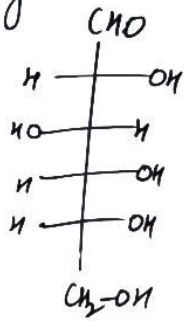
→ $R-O-R \xrightarrow{"} 2\ R-H$

→  epoxide $\xrightarrow{"} CH_3-CH_3$

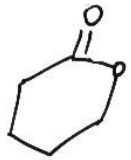
eg 1 

$\xrightarrow{HI + Red\ P}$  + CH_4

mesitylene

eg 2 

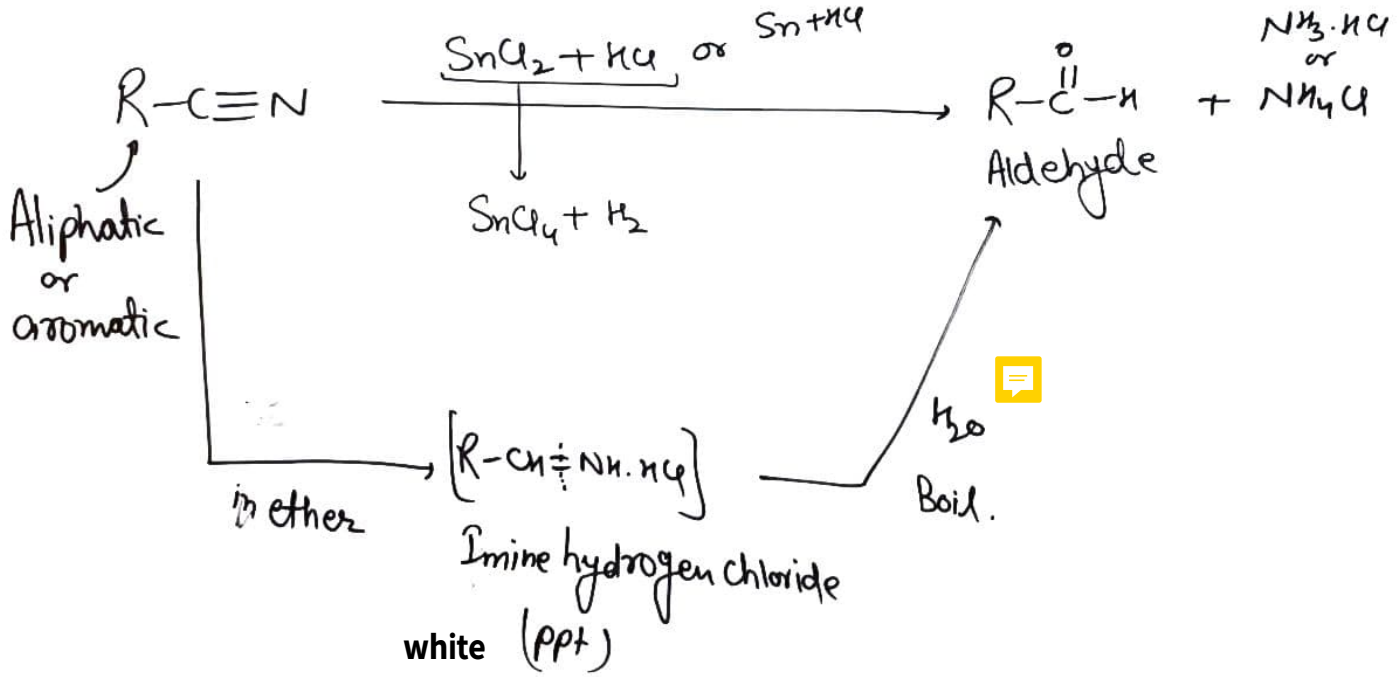
D. Glucose
$\xrightarrow{HI + Red\ P}$ n. Hexane

eg 3  $\xrightarrow{" \ "}$  n-Pentane

REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION REACTIONS.

(II) Stephen's Reduction of cyanide:



REDUCTION, OXIDATION AND HYDROLYSIS

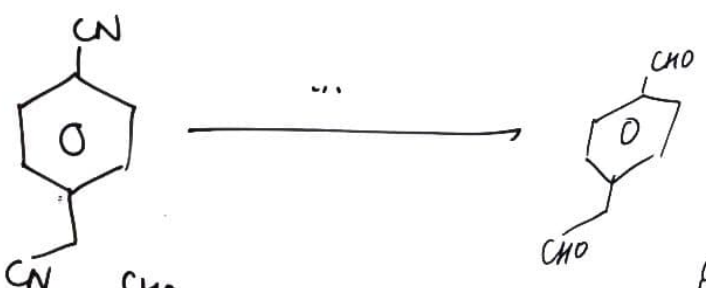
REDUCTION REACTIONS.

(II) Stephen's Reduction of cyanide:

eg1 $CH_3-CN \xrightarrow{SnCl_2 + HCl} CH_3-CHO$

eg2 $Ph-CN \longrightarrow Ph-CHO$

eg3



eg4

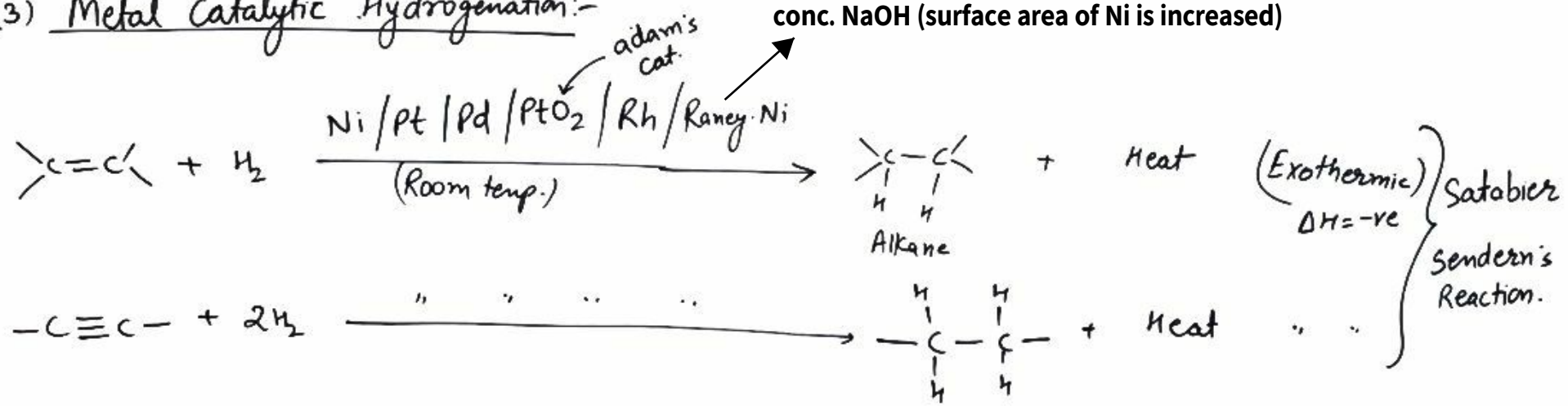


Reduction, Oxidation And Hydrolysis

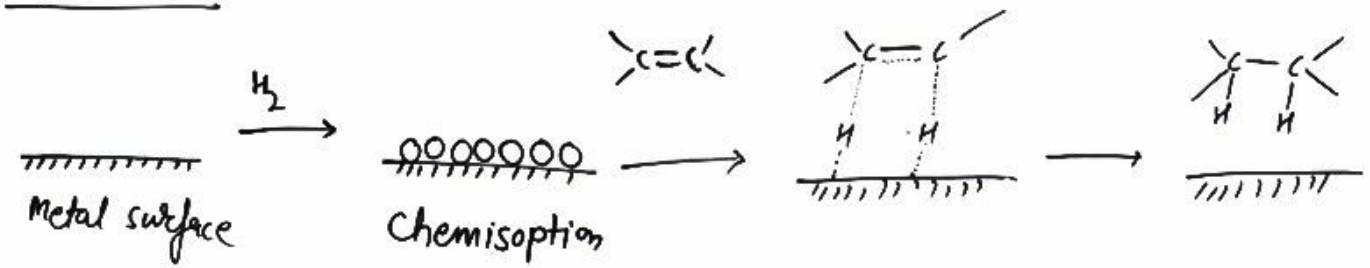
Reduction

(3) Metal Catalytic Hydrogenation:-

Ni boiled with
conc. NaOH (surface area of Ni is increased)



⇒ Mechanism



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REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION

(3) Metal Catalytic Hydrogenation:-

⇒ Hydrogenation = Syn addition (Addition of hydrogens from same plane)

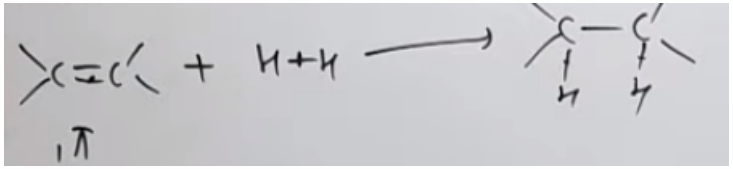
(Stereospecific hydrogenation reaction)

Cis → Syn → Erythro

Tran → Syn → Threo

⇒ Exothermic reaction ∴ Heat of hydrogenation $\propto$ no. of π bond in reactant

∴ ∴ ∴ ∴ ∴ $\propto \frac{1}{\text{Stability of alkene}}$ (for same no. of π bonds)



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REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION

(3) Metal Catalytic Hydrogenation:-

⇒ In presence of above metal catalyst complete hydrogenation take place on $-C\equiv C-$ of alkyne.

⇒ Reactivity of hydrogenation $\propto \frac{1}{\text{Steric hindrance on carbon-carbon multiple bond.}}$

eg: **Alkene < Alkyne** reactivity for hydrogenation.



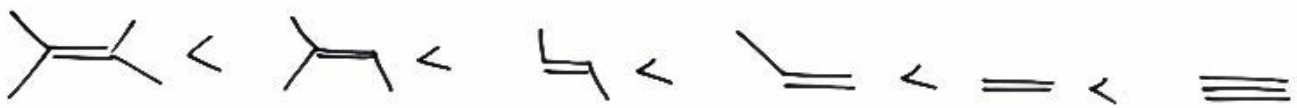
lesser steric hindrance by the two R groups of alkyne, compared to the 4 R groups on alkene

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REDUCTION, OXIDATION AND HYDROLYSIS

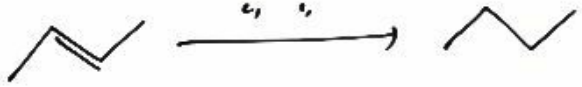
REDUCTION

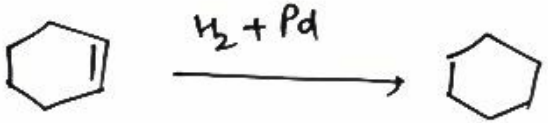
(3) Metal Catalytic Hydrogenation:-

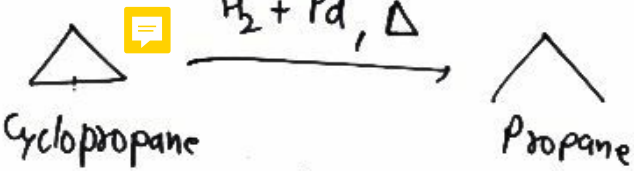
eg  Rate of hydrogenation

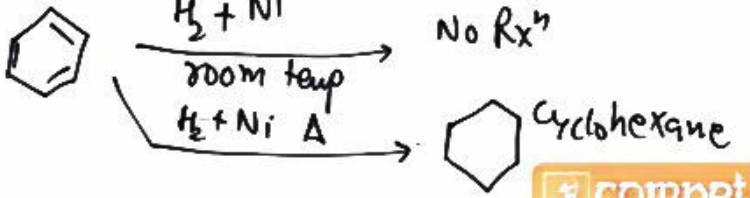
⇒ Hydrogenation of aromatic ring not possible at room temp (but possible on high temp.)

eg1 $CH_2=CH_2 \xrightarrow{H_2+Ni} CH_3-CH_3 + Heat$

eg2 

eg3 

*eg4 
Cyclopropane — Propane

eg5 
No Rx^n
Cyclohexane

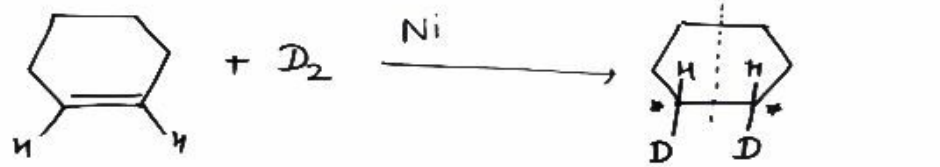
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REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION

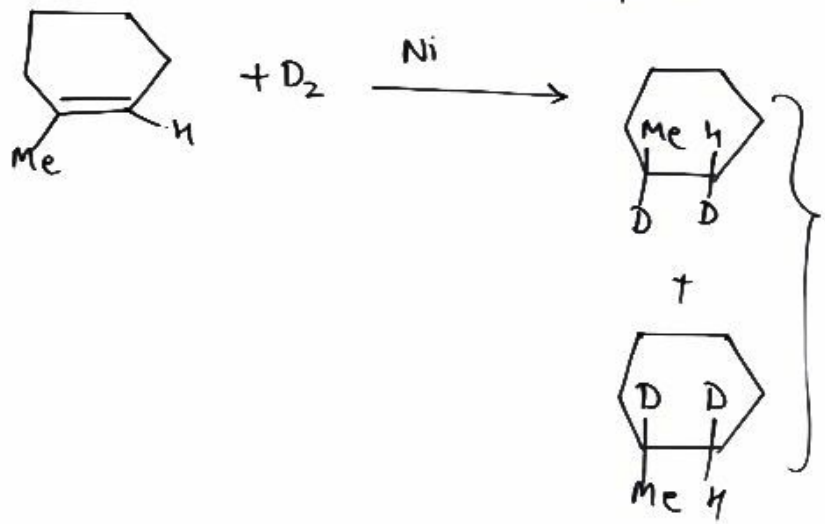
(3) Metal Catalytic Hydrogenation:-

eg 6



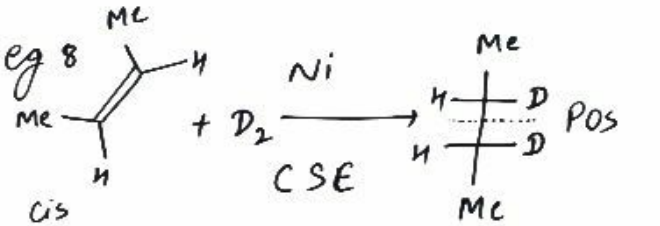
Cis / Meso
optical inactive

eg 7



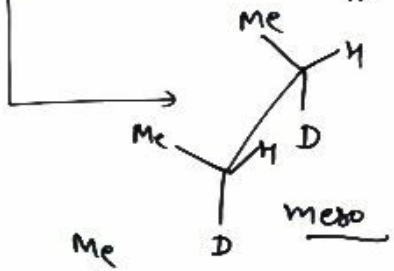
Racemic mix.

eg 8



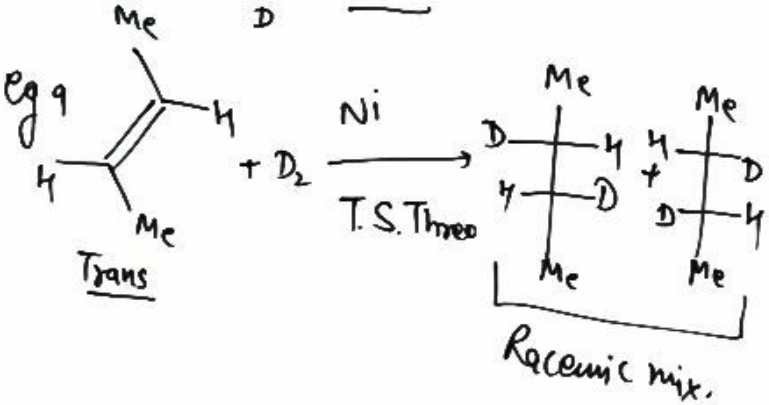
Meso / Erythro

"



meso

eg 9



Trans

Racemic mix.

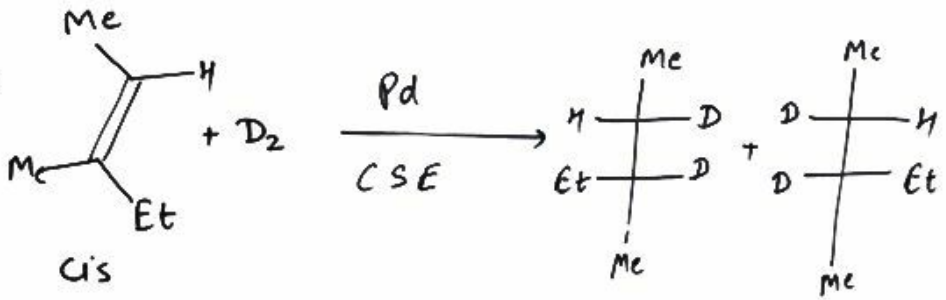
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REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION

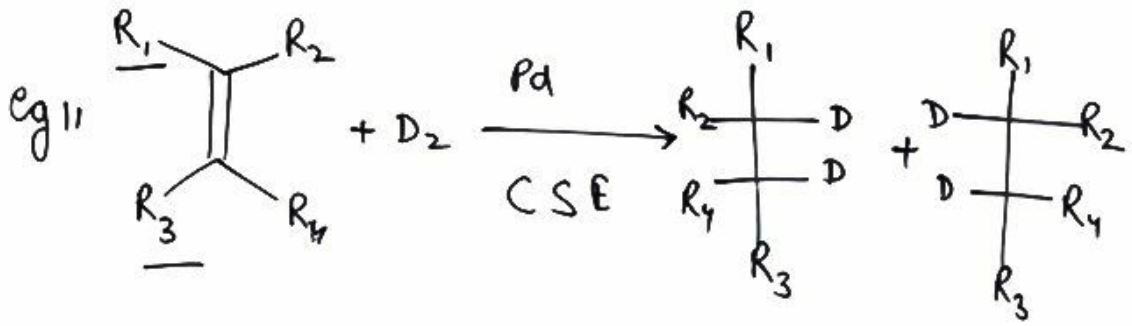
(3) Metal Catalytic Hydrogenation:-

eg 10

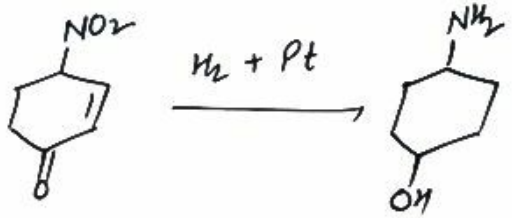


cis

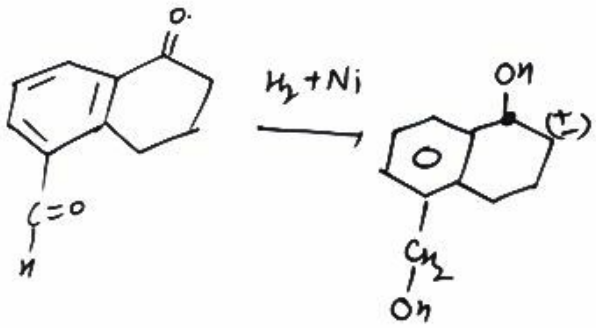
eg 11



eg 12



eg 13



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REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION

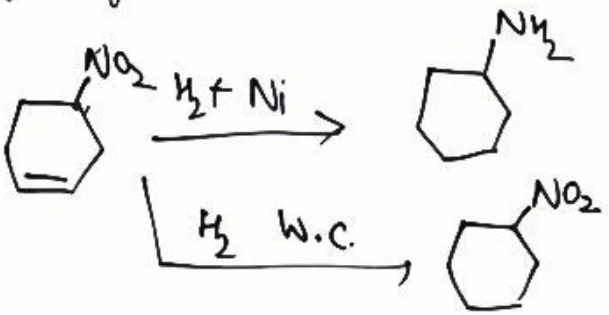
(3) Metal Catalytic Hydrogenation:-

⇒ In presence of metal catalyst hydrogenation = Heterogeneous hydrogenation.

⇒ Hydrogenation in presence of Will Kinson's catalyst - $[Rh(PPh_3)_3Cl]$ = Homogeneous hydrogenation.

⇒ In presence of Will Kinson's catalyst syn hydrogenation take place of only C=C & C≡C.

Eg



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REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION

(3) Metal Catalytic Hydrogenation:-

Partial Hydrogenation:-

In presence of poisond metal catalyst partial hydrogenation take place on alkyne & acid halide and form alkene and aldehyde respectively.

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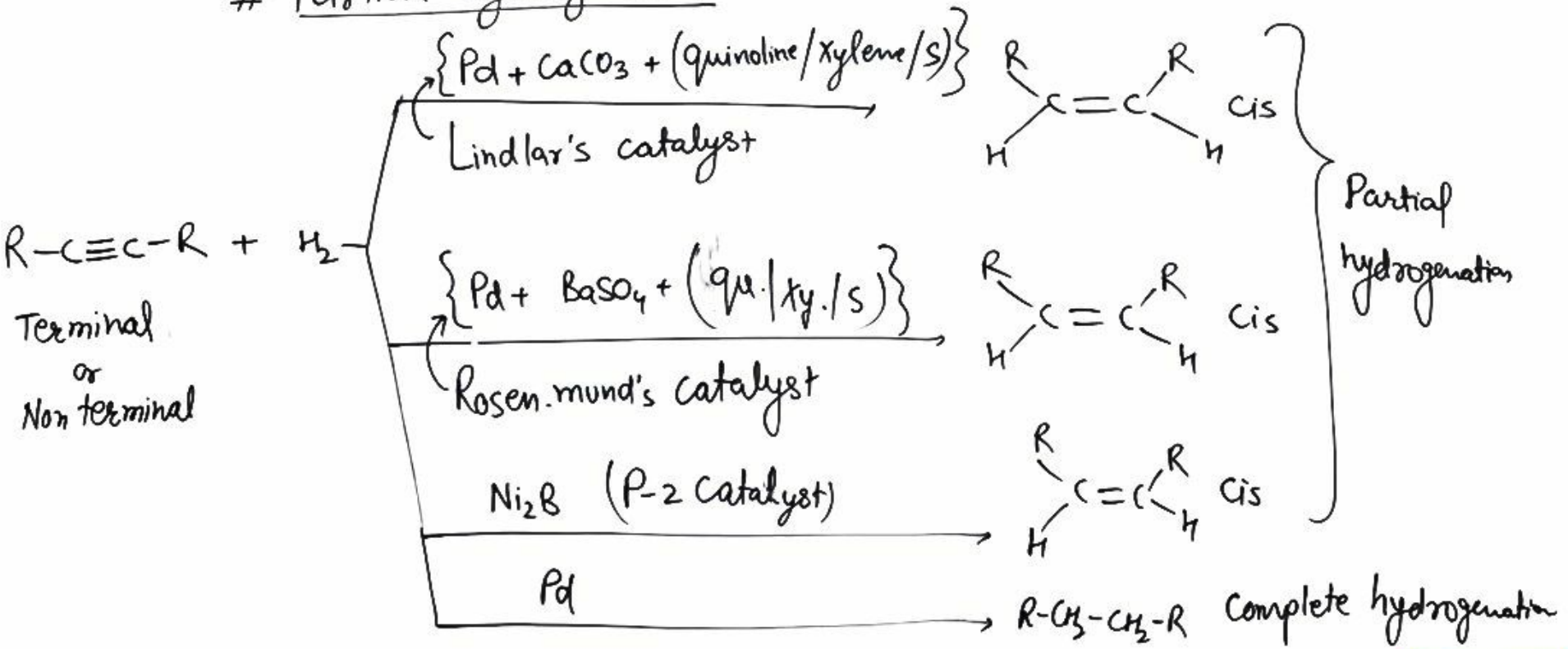
Reduction, Oxidation And Hydrolysis

Reduction

(3) Metal Catalytic Hydrogenation:-

Partial Hydrogenation:-

syn hydrogenation in these cases too

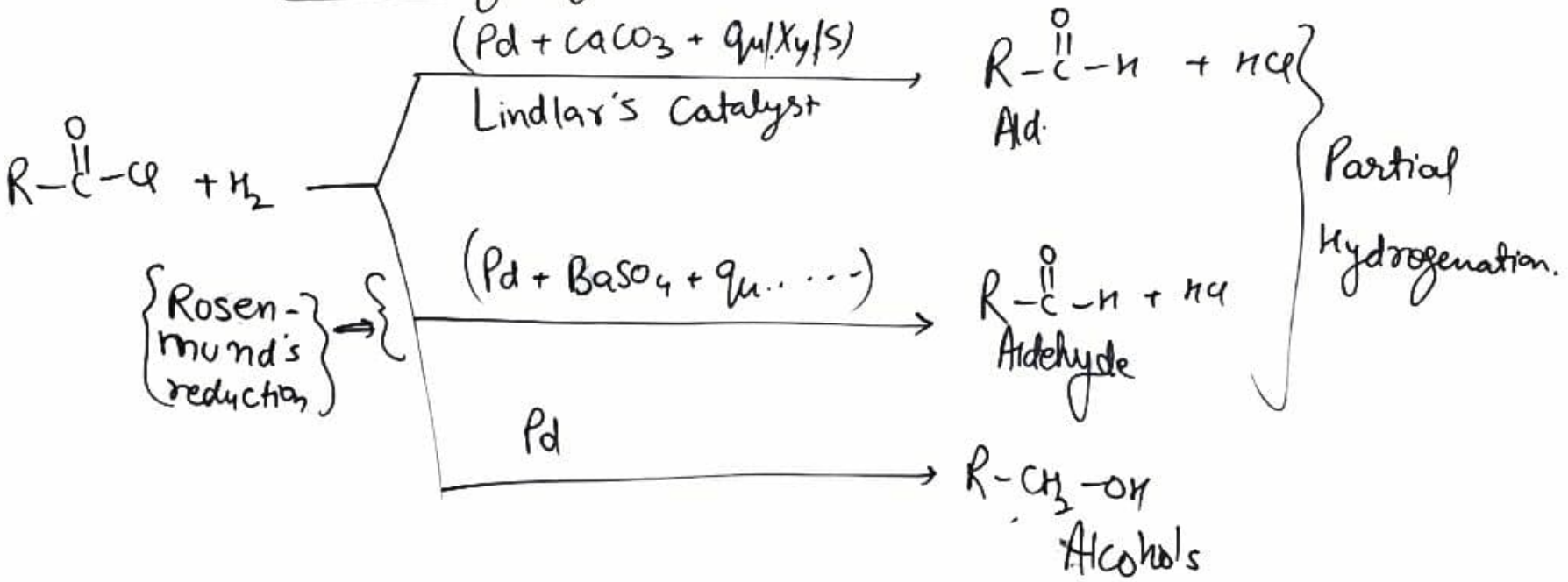


Reduction, Oxidation and Hydrolysis

Reduction

(3) Metal Catalytic Hydrogenation:-

Partial Hydrogenation:-

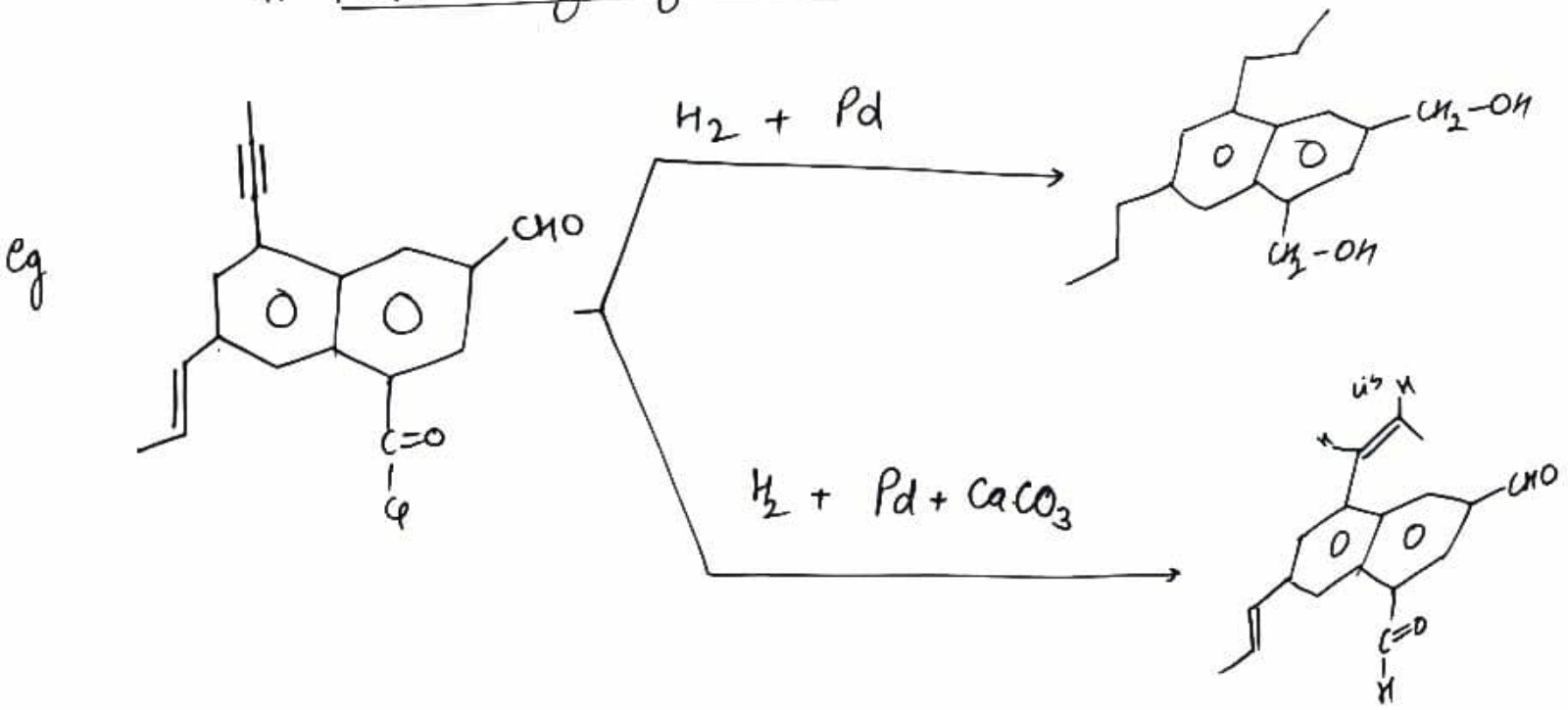


REDUCTION, OXIDATION AND HYDROLYSIS

REDUCTION

(3) Metal Catalytic Hydrogenation:-

Partial Hydrogenation:-

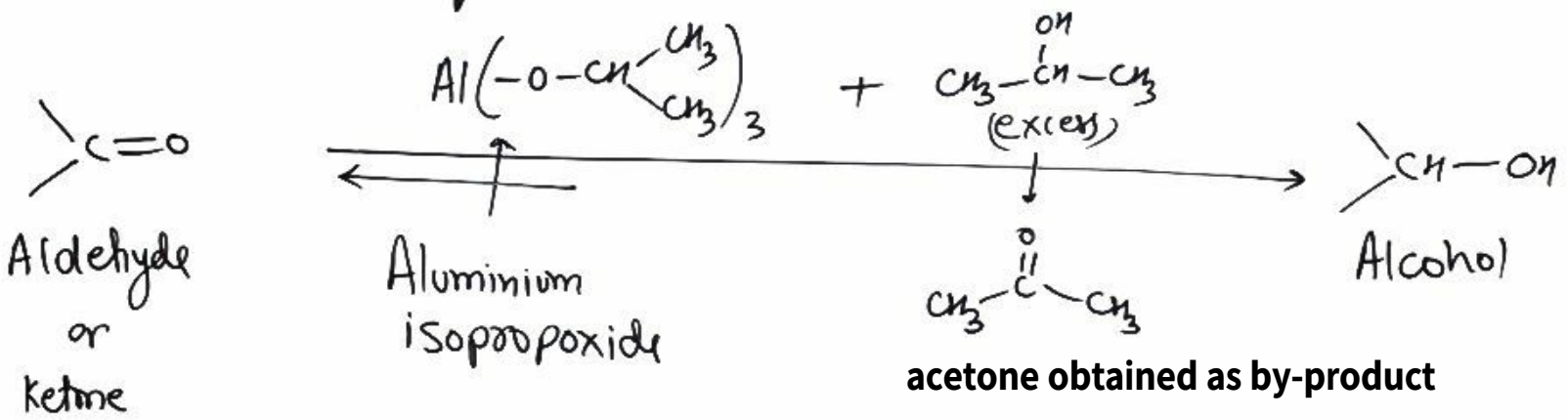


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Reduction, Oxidation And Hydrolysis

(I) Reduction :-

(4) Meerwein - Ponndorf - Verley (MPV) Reduction of carbonyl compounds



acetone obtained as by-product

⇒ This reaction is reversible reaction. Its reverse reaction is called Oppanauer oxidation.

⇒ Aldehyde and Ketones with high enol content, donot give this reduction.

like [β-Diketone, β-Ketoester OR etc]

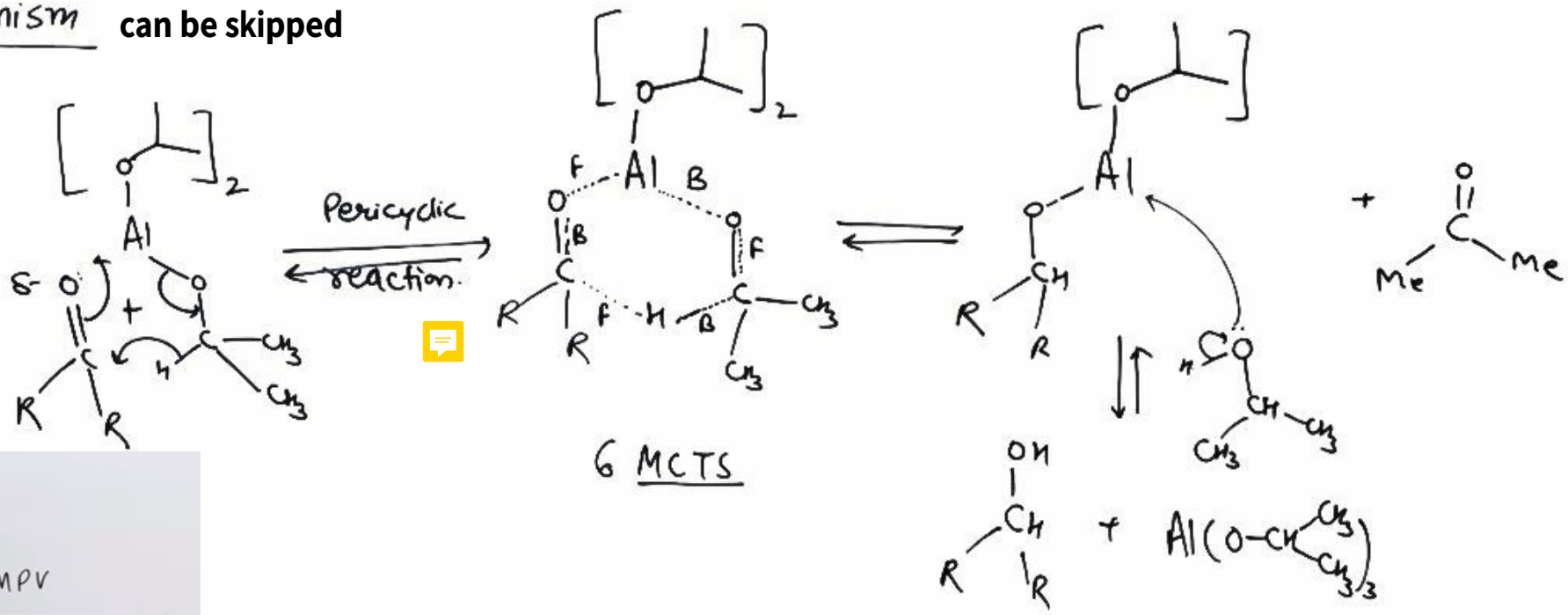
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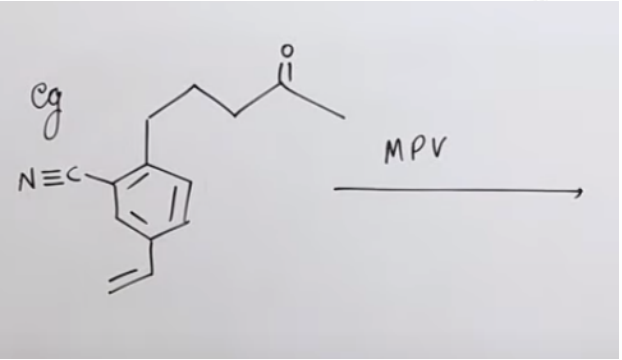
Reduction, Oxidation And Hydrolysis

(I) Reduction :-

(4) Meerwein - Ponndorf - Verley (MPV) Reduction: of carbonyl compounds

mechanism can be skipped





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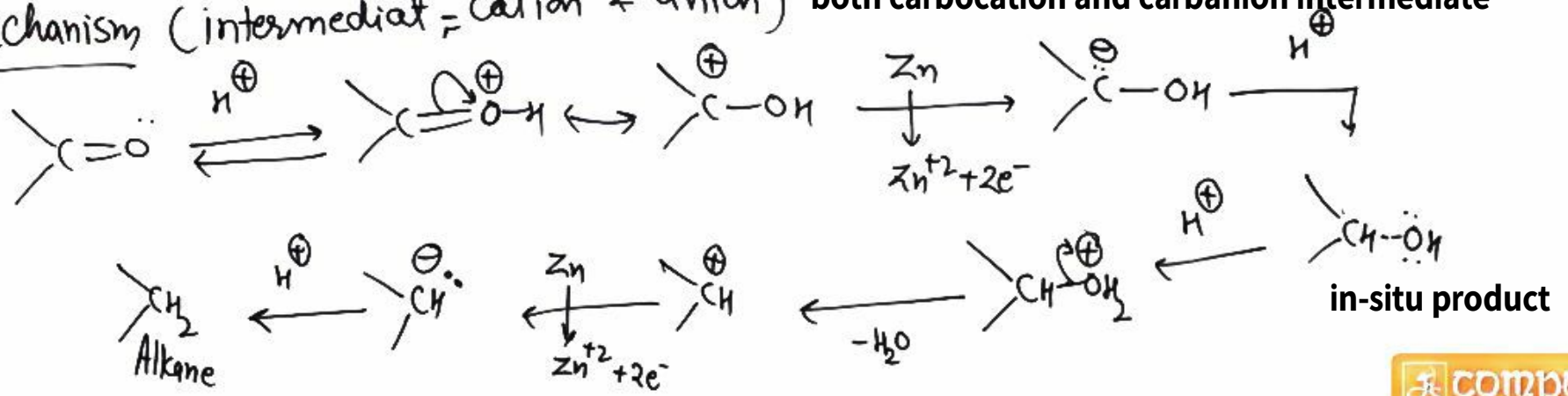
REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(5) Clemmensen's Reduction of carbonyl comp.



mechanism (intermediat = cation & anion) **both carbocation and carbanion intermediate**



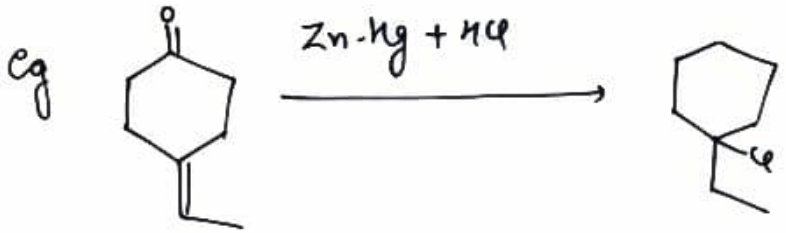
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REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(5) Clemmensen's Reduction of carbonyl comp.

⇒ Avoid clemmensen reduction in presence of acid sensitive groups like C=C, C≡C, alcohol, —OR ether etc.

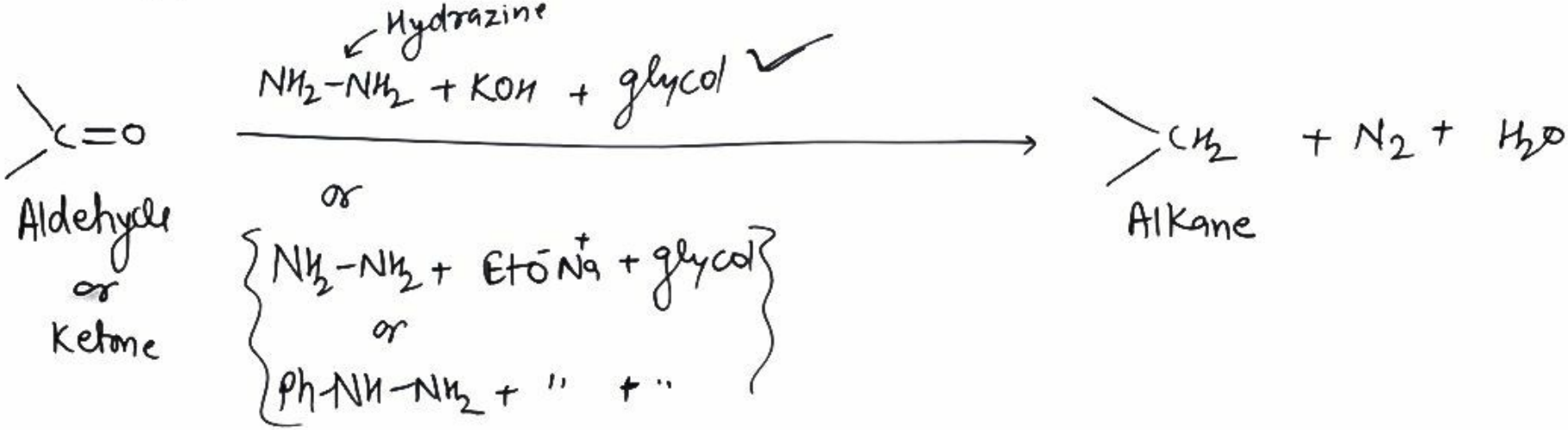


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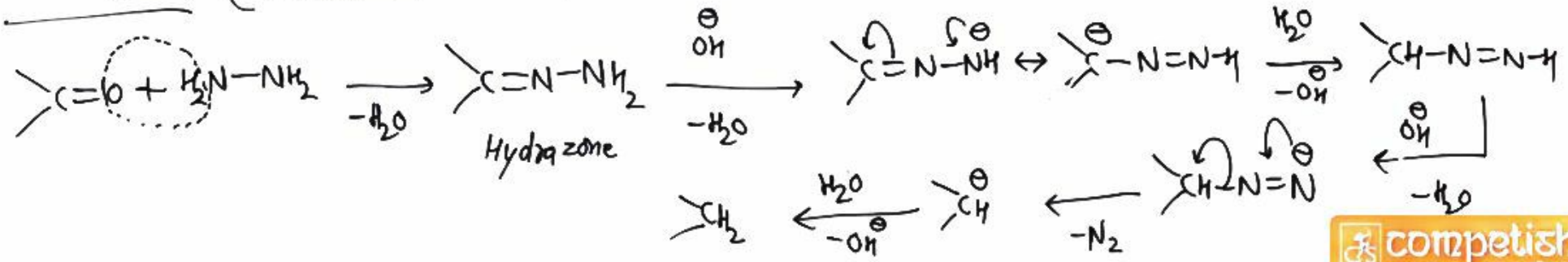
Reduction, Oxidation And Hydrolysis

(I) Reduction :-

(6) Wolff-Kishner Reduction:-



Mechanism (Intermediate = Carbanion)



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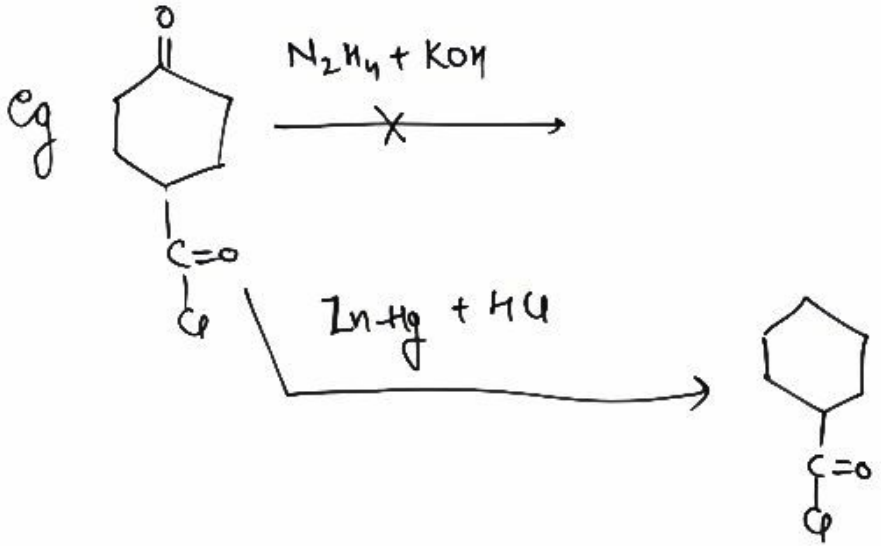
REDUCTION, OXIDATION AND HYDROLYSIS

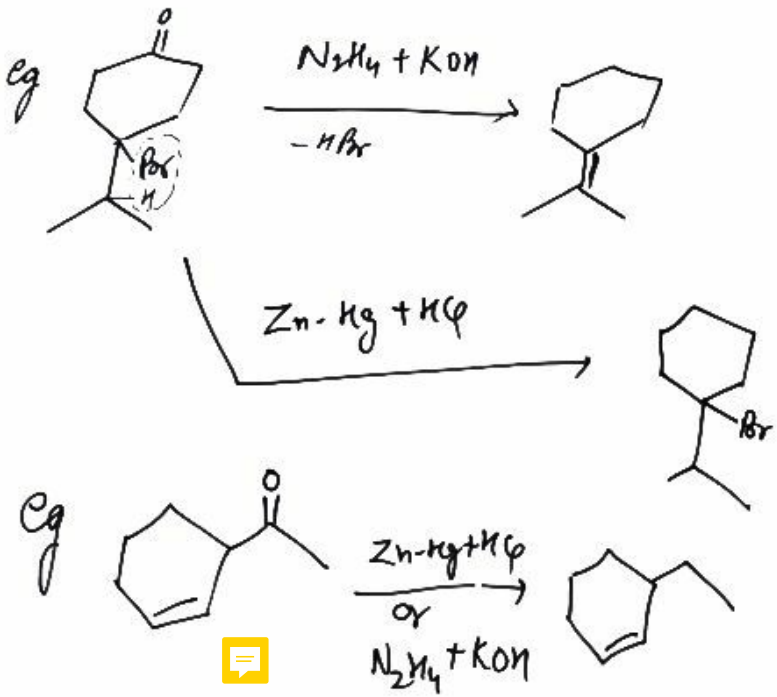
(I) REDUCTION :-

(6) Wolff-Kishner Reduction:-

⇒ Avoid Wolff-Kishner reduction in presence of base sensitive groups.

(like R-X, $R-\overset{O}{\overset{\|}{C}}-Cl$, epoxide etc.)





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REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(7) Birch Reduction

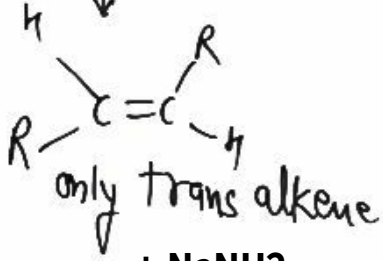
basic metal + liq. ammonia

⇒ Reagent $(Li/Na/K) + NH_{3(\ell)}$

(A) Birch Reduction of Non terminal alkyne

$R-C\equiv C-R$

$\downarrow Na + NH_{3(\ell)}$



only trans alkene

+ NaNH2

(B) Birch Red. of Terminal alkene

$R-CH=CH_2$

$\downarrow Na + NH_{3(\ell)}$

$R-CH_2-CH_3$

(C) B.R. of aromatic ring



$\downarrow Na + NH_{3(\ell)}$



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REDUCTION, OXIDATION AND HYDROLYSIS

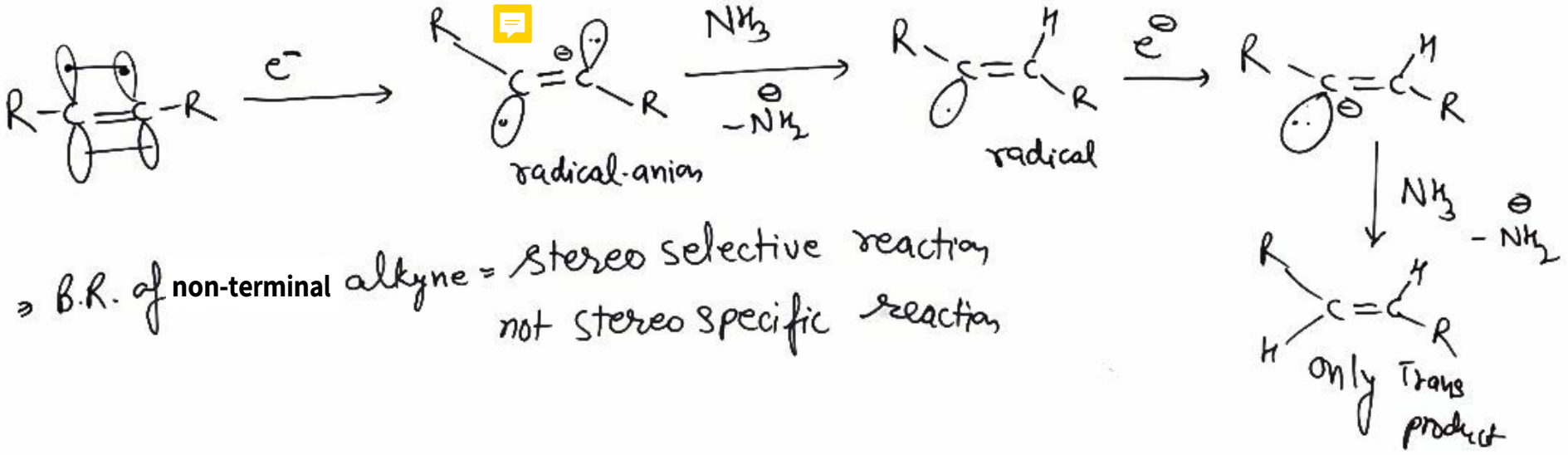
(I) REDUCTION :-

(7) Birch Reduction

for non terminal

(A) Alkyne (Mechanism)

$$2Na \longrightarrow 2Na^{\oplus} + 2e^{-}$$



⇒ B.R. of non-terminal alkyne = Stereo selective reaction
not Stereo specific reaction

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REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(7) Birch Reduction

eg $R-C\equiv \overset{sp}{C}-\overset{sp}{H}$ $\xrightarrow[\text{acid-base Rxn}]{Na}$ $R-C\equiv C^{\ominus}\,Na^{\oplus}$

Terminal alkyne

thus, terminal alkynes give acid-base rex. but not reduction rex. with Na/ liq. NH3

$>C=C^{\ominus}< + NH_3 \not\rightarrow >C=C< + NH_2^{\ominus}$

weaker acid than ammonia, so rex. goes forward

$-C\equiv C^{\ominus} + NH_3 \not\longrightarrow -C\equiv C-H + NH_2^{\ominus}$

stronger acid than ammonia
so rex. doesn't proceed in forward direction

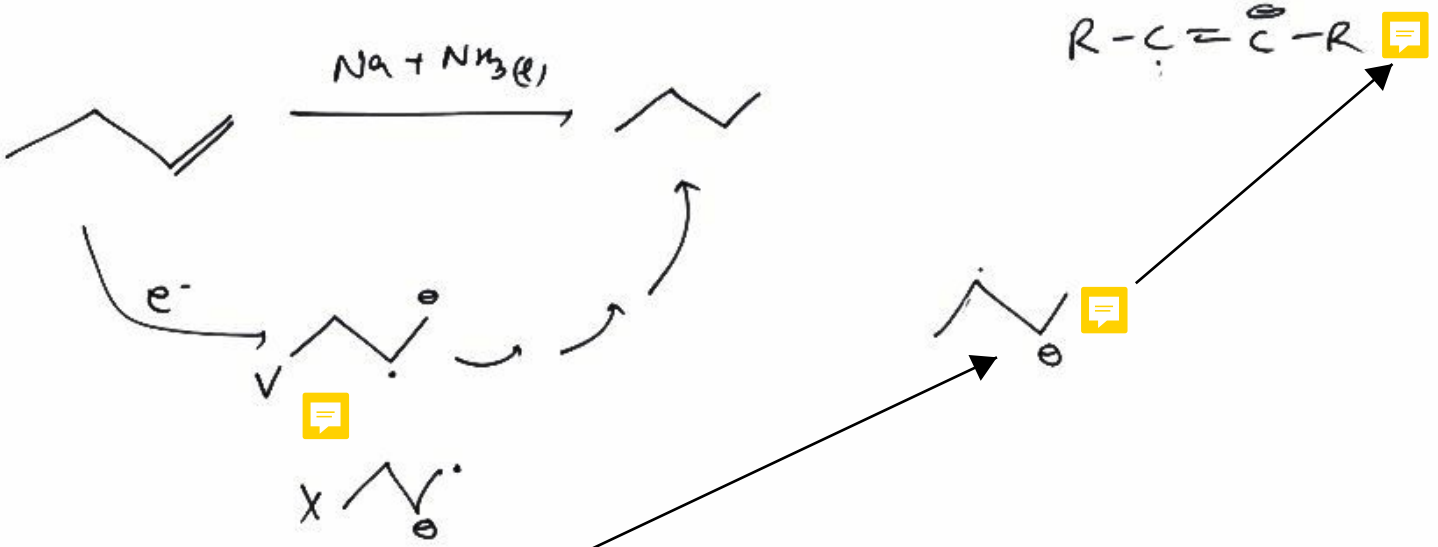
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Reduction, Oxidation And Hydrolysis

(I) Reduction :-

(7) Birch Reduction

(B) Terminal alkene



Na + NH3(l) No Rxn

Non terminal Alkene

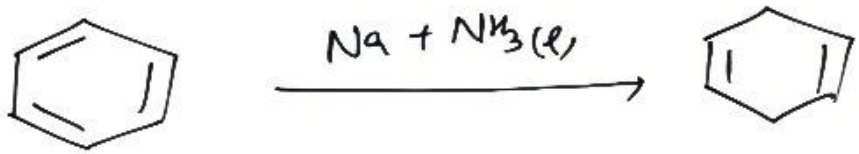
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REDUCTION, OXIDATION AND HYDROLYSIS

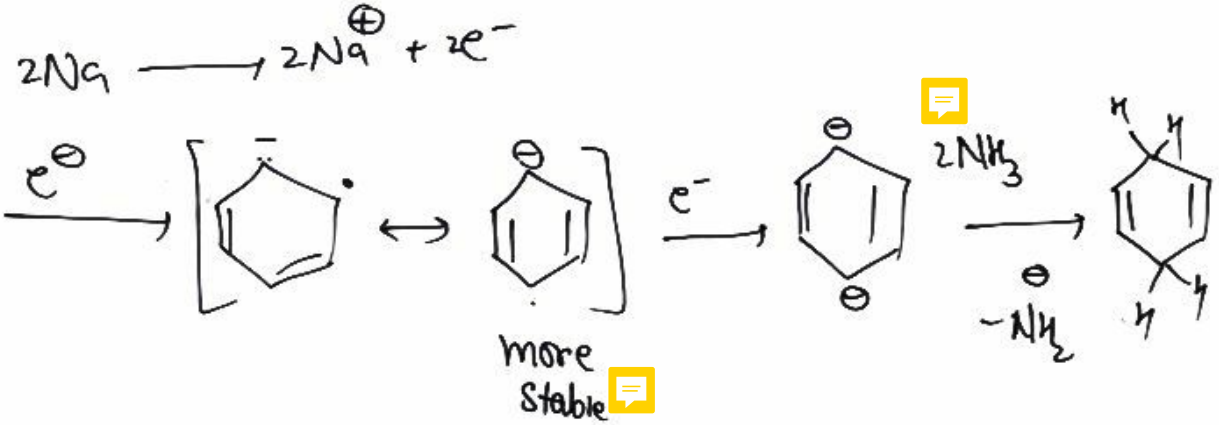
(I) REDUCTION :-

(7) Birch Reduction

[C] for aromatic rings



mechanism



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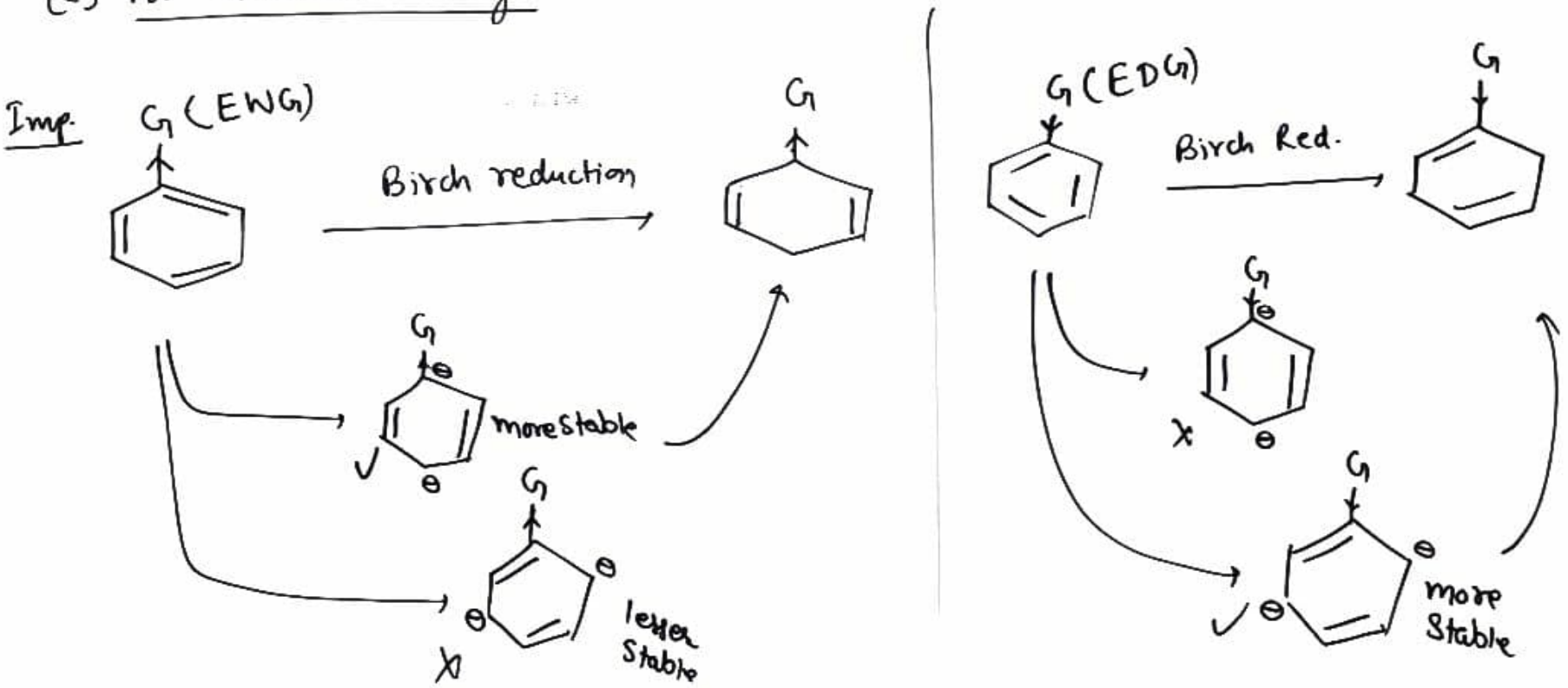
REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(7) Birch Reduction

[C] for aromatic rings

atleast one double bond will touch with the EDG

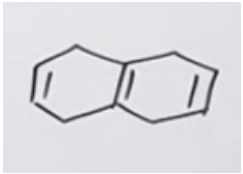


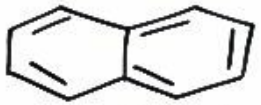
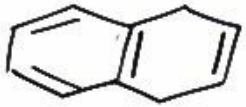
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Reduction, Oxidation And Hydrolysis

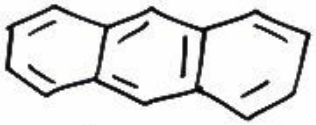
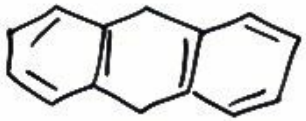
(I) Reduction :-

(7) Birch Reduction

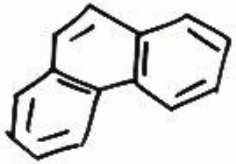
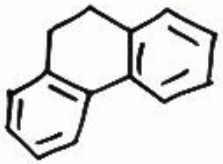


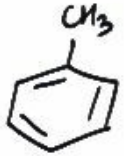
eg 1  $Na + NH_{3(l)}$ → 

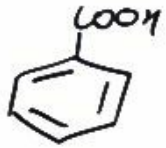
naphthalene — major

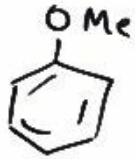
eg 2  " " → 

Anthracene

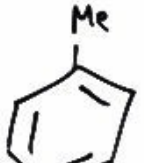
eg 3  " → 

eg 4  $Na + NH_{3(l)}$ →

eg 5  " " →

eg 6  " →

Anisole

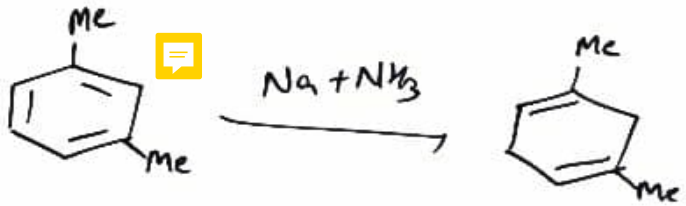
eg 7  " →

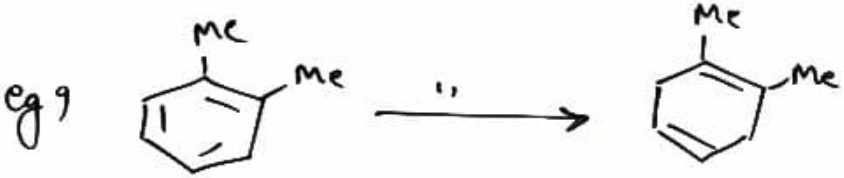
Me, Me — P. xylene

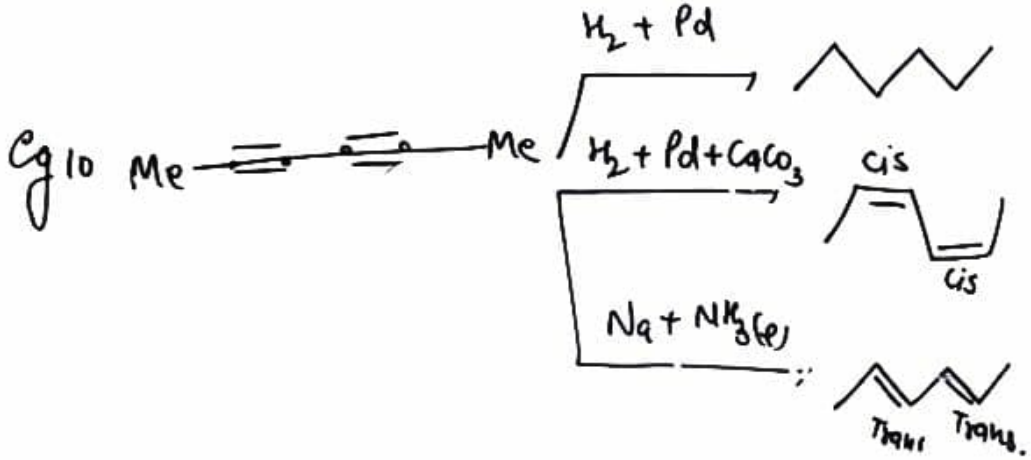
REDUCTION, OXIDATION AND HYDROLYSIS

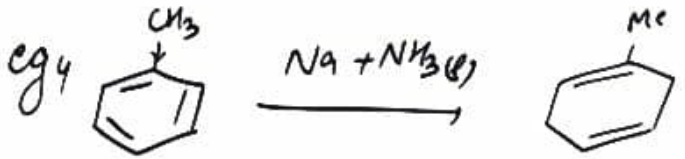
(I) REDUCTION :-

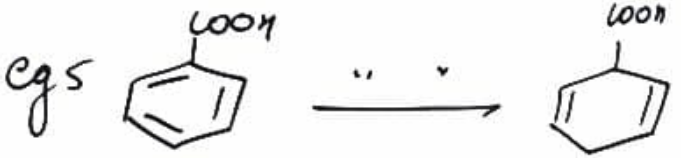
(7) Birch Reduction

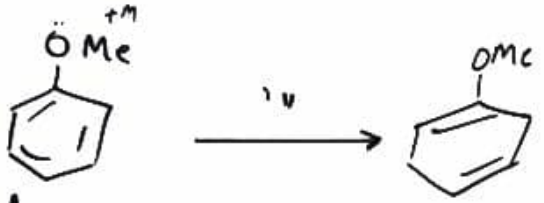
eg 8 

eg 9 

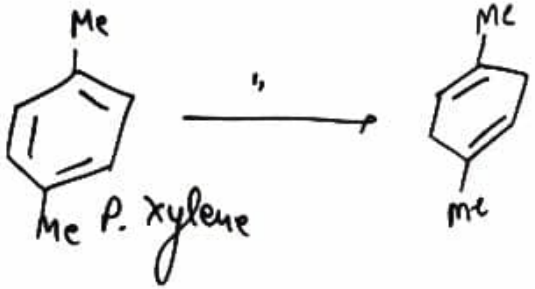
eg 10 

eg 4 

eg 5 

eg 6 

Anisole

eg 7 

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REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(8) Reduction with metal hydrides.

- (A) $LiAlH_4$
- (B) $NaBH_4$
- (C) Ph_3SnH — Triphenyl tin hydride
- (D) DIBAL-H — Di Iso Butyl-Aluminium Hydride

[8.A] $LiAlH_4$ in ether:

$$LiH + AlH_3 \longrightarrow LiAlH_4$$

$H^{\ominus}$ = $N_4^{\ominus}$

(a) Nitro $-NO_2$

$$R-NO_2 \xrightarrow{LiAlH_4} R-NH_2$$

Aliphatic

$$Ar-NO_2 \xrightarrow{LiAlH_4} Ar-N=N-Ar$$

Aromatic — Azo grene

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REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(8) Reduction with metal hydrides.

(8.A) $LiAlH_4$ in ether

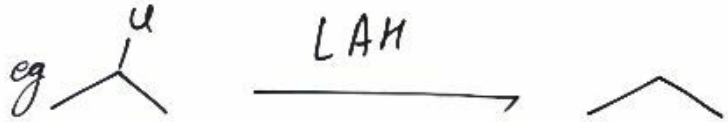
(b) Alkyl halide.

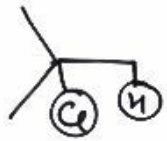
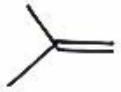
$H^{\ominus} = N_4^{\oplus}$

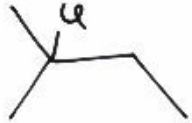
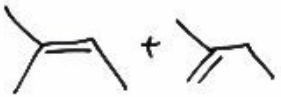
$R^{\delta+} \rightarrow X^{\delta-}$ (1°, 2°) $\xrightarrow{LiAlH_4}$ (S_N2) R–H Alkane

R–X (3°) $\xrightarrow{LiAlH_4}$ (E_2 elimination) Alkene

eg CH_3–Cl $\xrightarrow{LAH}$ CH_4

eg  $\xrightarrow{LAH}$

eg  $\xrightarrow[-HCl\ \text{elimination}]{LAH}$ 

eg  $\xrightarrow{LAH}$ 

major → more stable

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REDUCTION, OXIDATION AND HYDROLYSIS

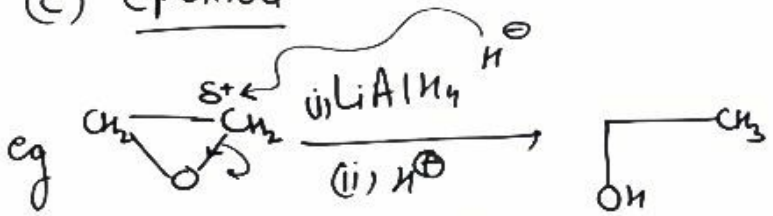
Ph-CH=CH-COOH Cinnamic acid

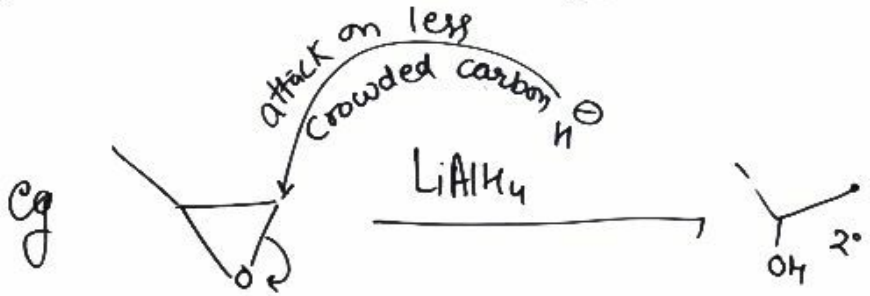
(I) REDUCTION :-

(8) Reduction with metal hydrides.

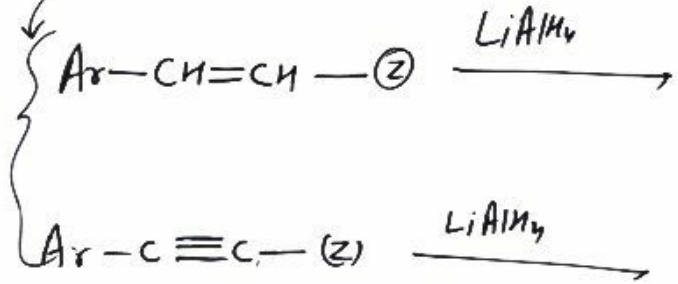
(8.A) $LiAlH_4$ in ether

(c) epoxide





(d) Cinnamyl carbon-carbon multiple bonds :-



[Z = -CHO, $-\overset{O}{\overset{\|}{C}}-R$, -COOH, -COOR, -COCl etc]

Z should be any group which can undergo reduction with LiAlH4

⇒ Generally No-reduction of C=C with C≡C $LiAlH_4$

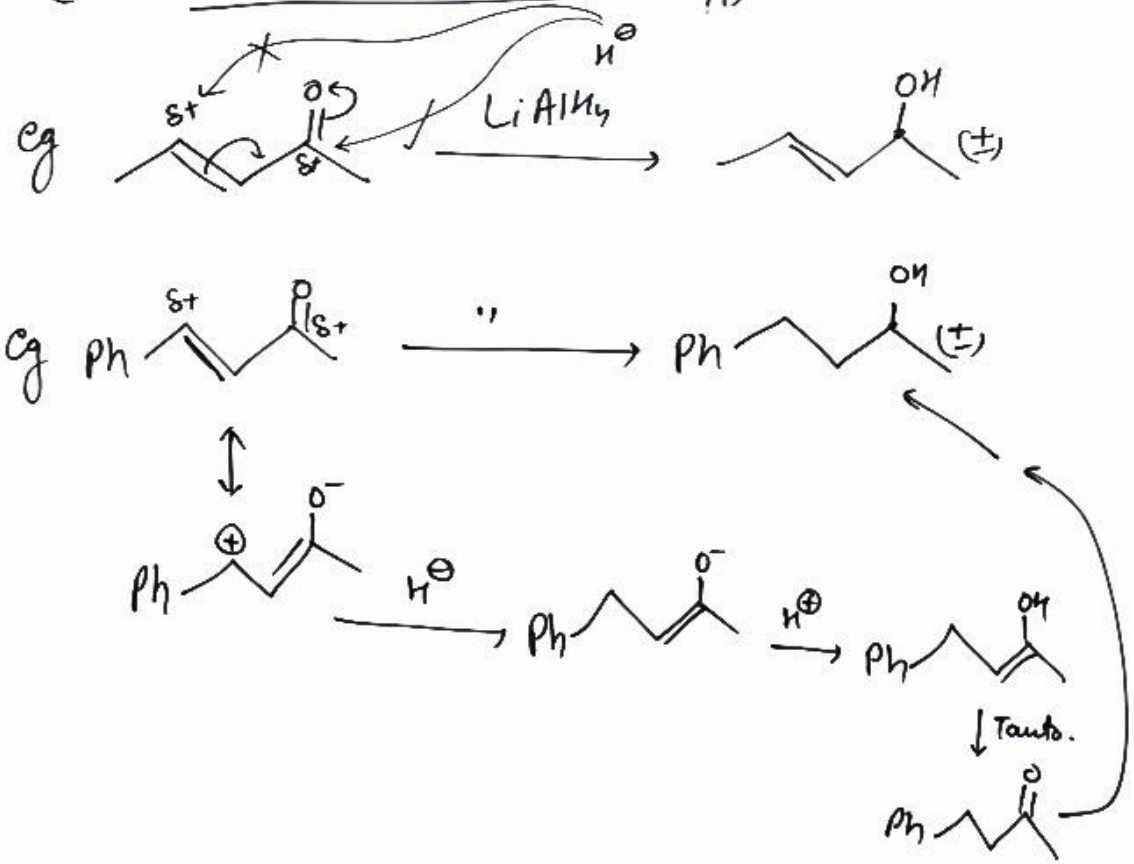
⇒ But reduction possible of cinnamyl C=C C≡C with LAH

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REDUCTION, OXIDATION AND HYDROLYSIS

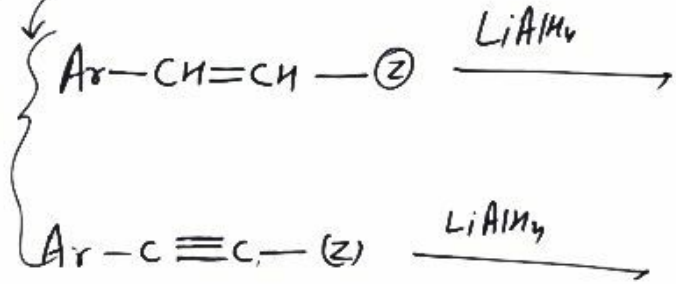
(I) REDUCTION :-

(8) Reduction with metal hydrides



Ph-CH=CH-COOH Cinnamic acid

(d) Cinnamyl carbon-carbon multiple bonds :-



[Z = -CHO, -C(=O)-R, -COOH, -COOR, -COCl etc]

⇒ Generally No-reduction of C=C with C≡C LiAlH4

⇒ But reduction possible of cinnamyl C=C C≡C with LAH

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REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(8) Reduction with metal hydrides.

(8.9) $LiAlH_4$:-

(e) $R-N=C=O$ isocyanate

$\xrightarrow{LiAlH_4}$ $R-NH-CH_3$

(f) $R-N_3 \xrightarrow{LiAlH_4} R-NH_2 + N_2$

8.b $NaBH_4$

$R-CHO \xrightarrow{NaBH_4} R-CH_2-OH$

$R-\overset{O}{\overset{\|}{C}}-R \xrightarrow{''} R-\overset{OH}{\overset{|}{C}}H-R$

$R-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow{'} R-CH_2-OH$

* $R-X \xrightarrow{NaBH_4} R-H$

$2^\circ, 3^\circ$

$R-X \xrightarrow{NaBH_4}$ No Rx^n.

1°

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REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(8) Reduction with metal hydrides.

(8.C) Ph_3SnH

→ Free radical Reaction

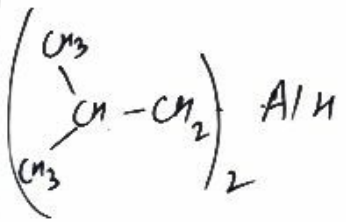
$$R-X \xrightarrow{Ph_3SnH} R-H$$

$1^\circ, 2^\circ, 3^\circ$

$$Ar-X \xrightarrow{Ph_3SnH} Ar-H$$

(8.D) DIBAL-H

Di Iso Butyl Aluminium Hydride



REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(8) Reduction with metal hydrides.

$R-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow[-70^\circ C]{DIBAL-H} R-\overset{O}{\overset{\|}{C}}-H$ Ald.

$R-\overset{O}{\overset{\|}{C}}+OR' \xrightarrow[-70^\circ]{DIBAL-H} R-\overset{O}{\overset{\|}{C}}-H + R'-OH$

$R-C\equiv N \xrightarrow[-70^\circ C]{DIBAL-H} [R-CH=NH]$ Imine $\xrightarrow{\text{Hydrolysis}}$ $R-CHO + NH_3$

$R-\overset{O}{\overset{\|}{C}}-OR' \xrightarrow{NaAlH4} R-\overset{O}{\overset{\|}{C}}-H + R'-OH$

sodium aluminum hydride

(8.D) DIBAL-H

Di Iso Butyl Aluminium Hydride

$\left(\begin{matrix} CH_3 \\ | \\ CH-CH_2 \\ | \\ CH_3 \end{matrix}\right)_2 AlH$

sterically crowded H, so lesser reactivity in comparison to LiAlH4

At low temp. (−70°c to −78°c) it gives partial reduction of R−COCl, R−COOR', R−CN give aldehyde.

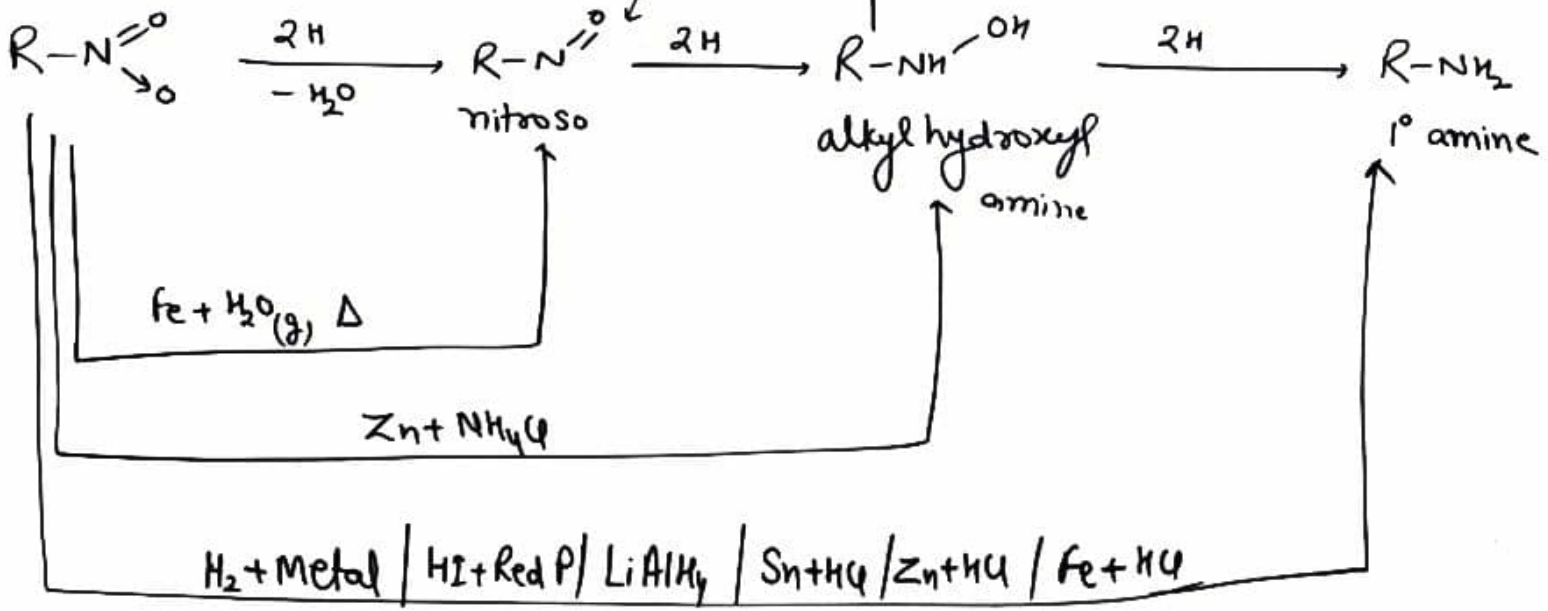
competishun

REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(9) Reduction of nitro group

(a) for aliphatic nitro compound :- Tollen's reagent



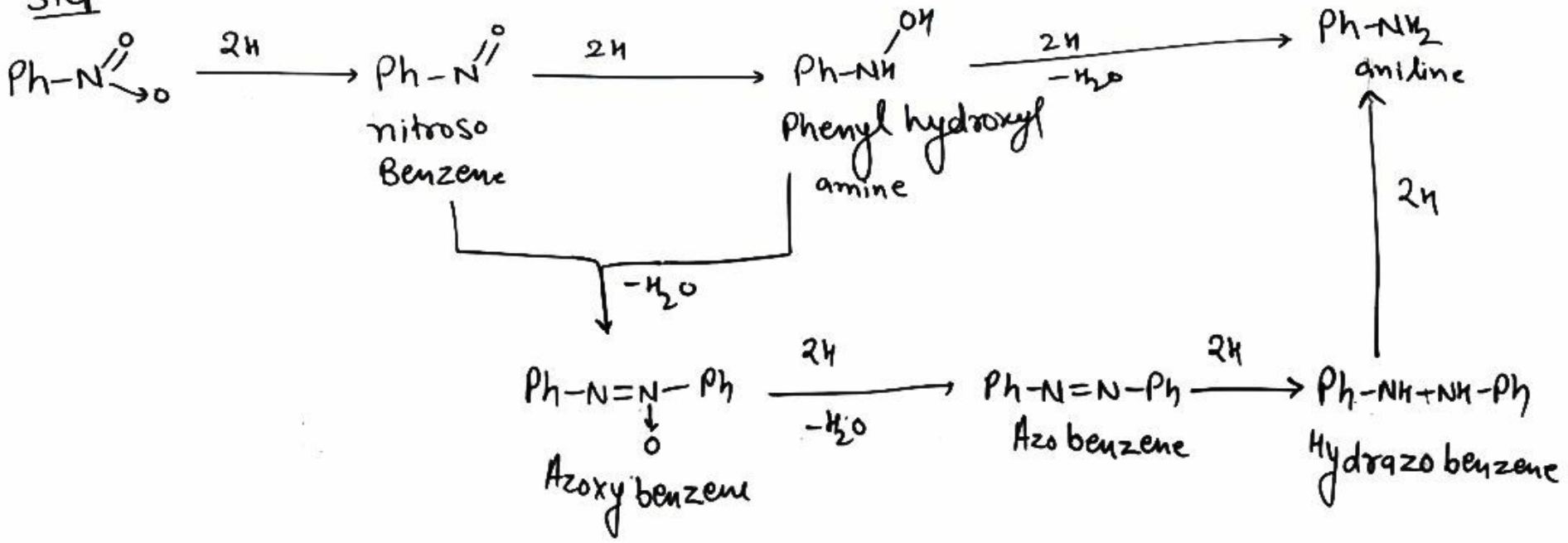
REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(9) Reduction of nitro group

(b) for aromatic nitro compound :-

Step

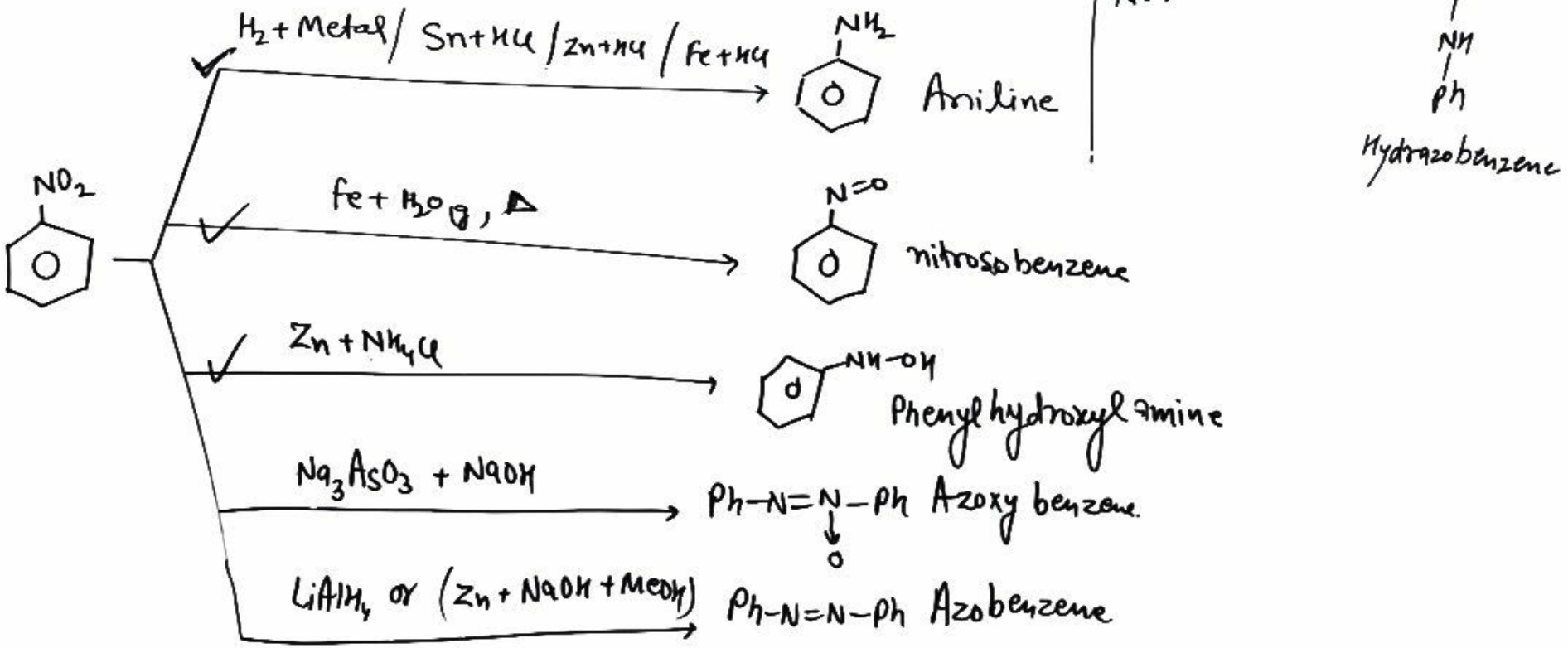


REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION :-

(a) Reduction of nitro group

(b) for aromatic nitro compound :-



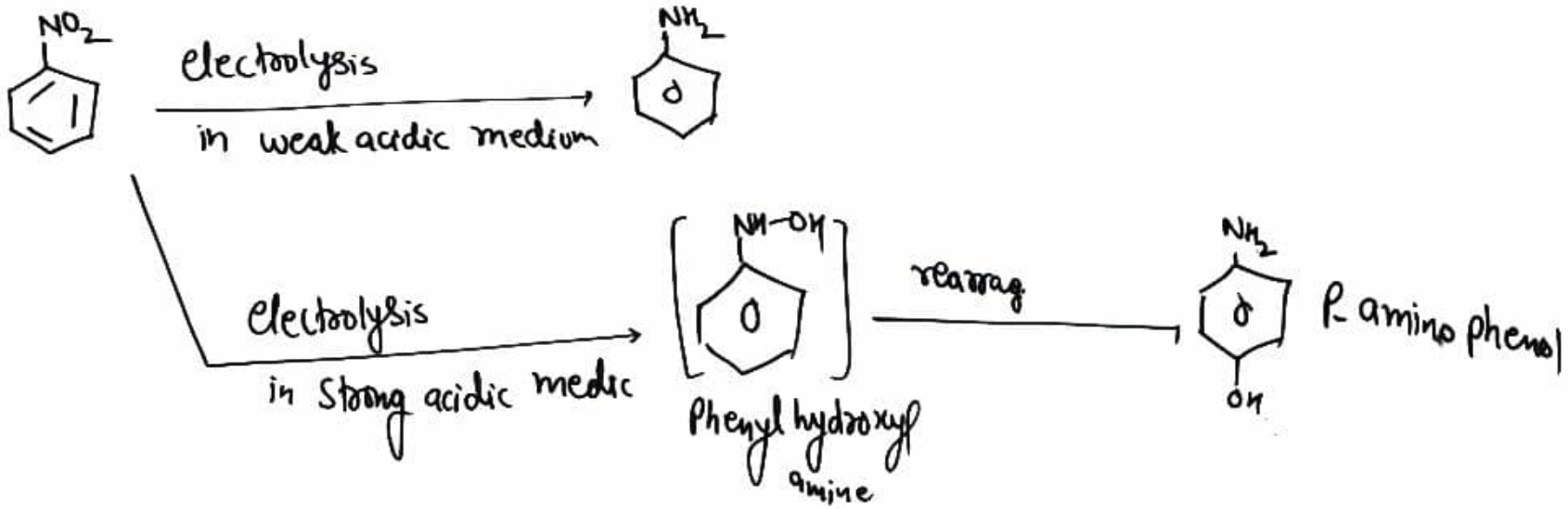
REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION:-

(a) Reduction of nitro group

(b) For aromatic nitro compound:-

→ Electrolysis (Reduction) of nitrobenzene.



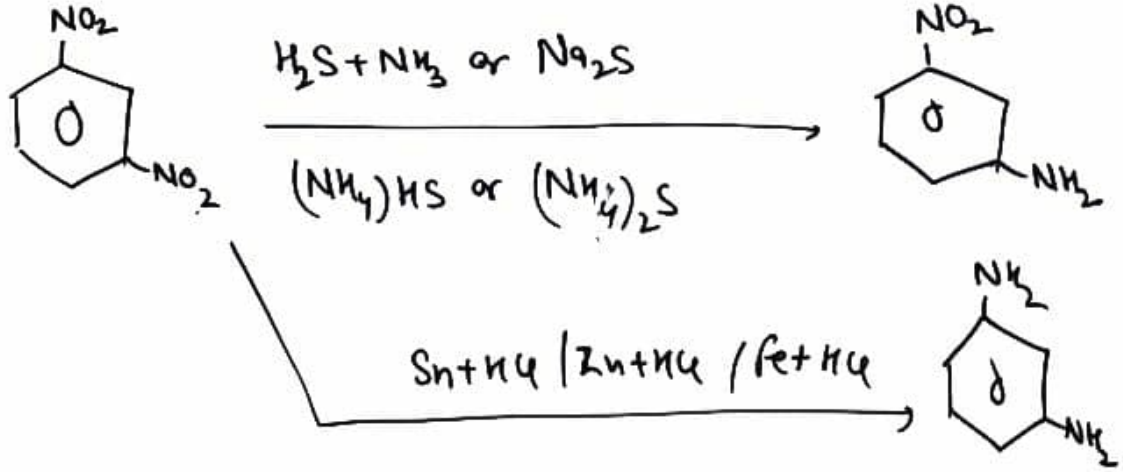
REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION:-

(a) Reduction of nitro group

(b) for aromatic nitro compound:-

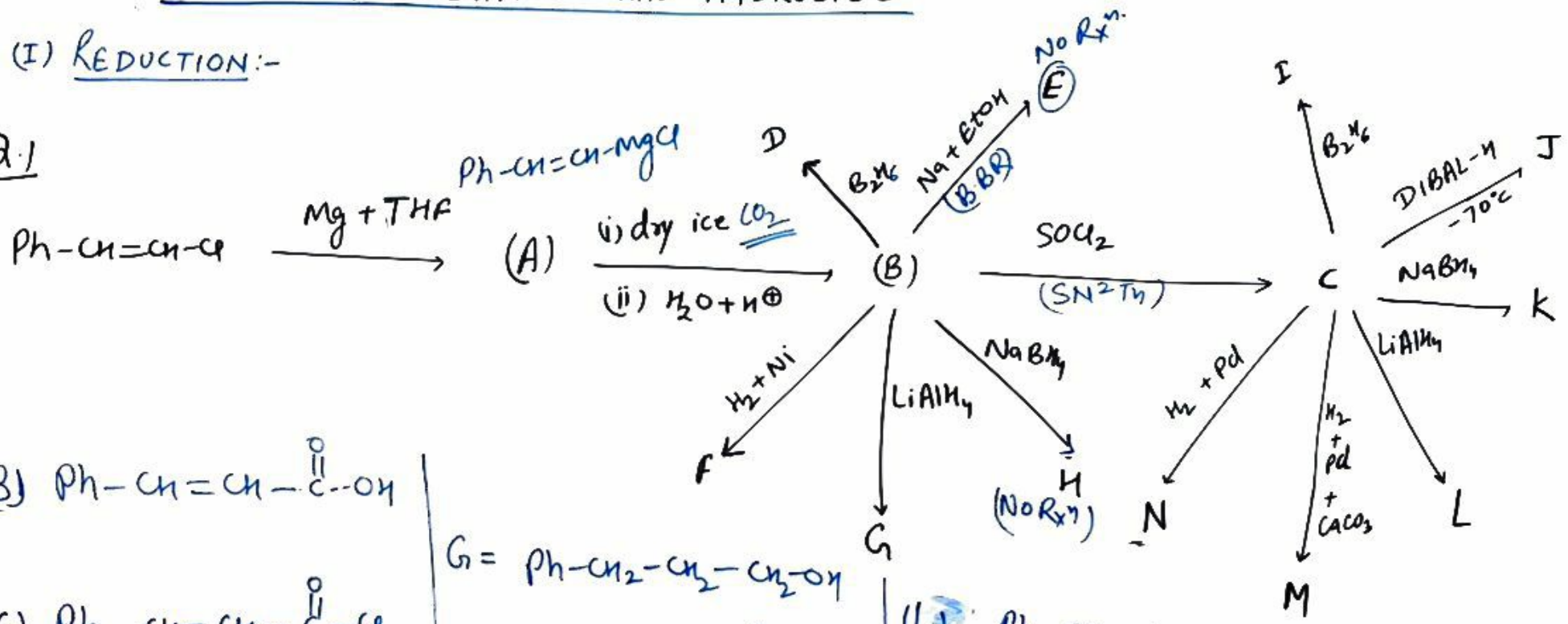
→ Zenith reaction (selective reduction)



REDUCTION, OXIDATION AND HYDROLYSIS

(I) REDUCTION:-

Q.1



(B) $Ph-CH=CH-\overset{O}{\overset{\|}{C}}-OH$

(C) $Ph-CH=CH-\overset{O}{\overset{\|}{C}}-Cl$

(D) $Ph-CH_2-CH_2-CH_2-OH$

(F) $Ph-CH_2-CH_2-COOH$

G = $Ph-CH_2-CH_2-CH_2-OH$

(I) $Ph-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-Cl$

(J) $Ph-CH=CH-\overset{O}{\overset{\|}{C}}-H$

(K) $Ph-CH=CH-CH_2-OH$

(L) $Ph-CH_2-CH_2-CH_2-OH$

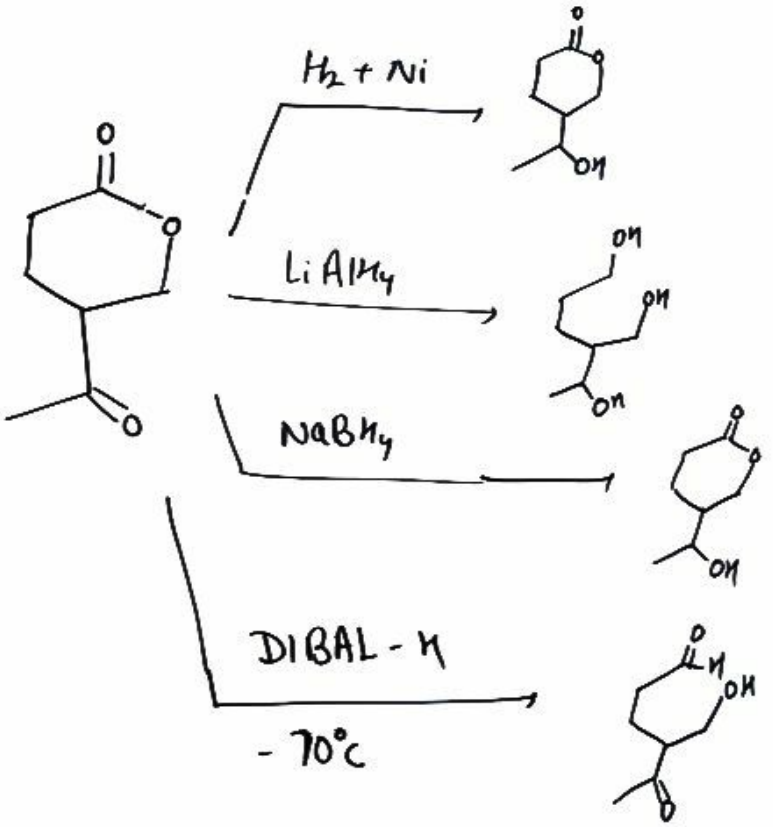
(M) $Ph-CH=CH-\overset{O}{\overset{\|}{C}}-H$

(N) $Ph-CH_2-CH_2-CH_2-OH$

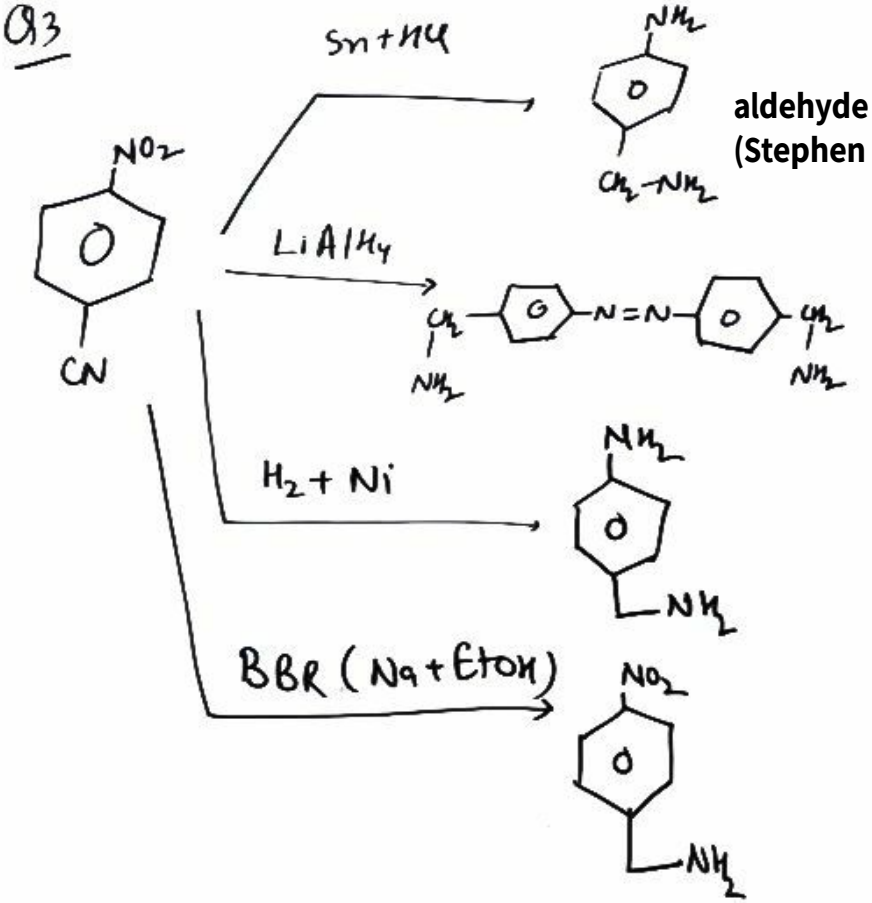
Reduction, Oxidation And Hydrolysis

(I) Reduction :-

Q.2



Q3



aldehyde here
(Stephen red.)

REDUCTION, OXIDATION AND HYDROLYSIS

(II) Oxidation:-

Reactant	Strong oxidising agent (SOA)	Moderate oxi. Agent (MOA)	Weak oxi. Agents (WOA)	
	⇒ $KMnO_4 + H^{\oplus}/OH^{-}/H_2O + \Delta$ ⇒ $K_2Cr_2O_7 + H^{\oplus} + \Delta$ ⇒ $CrO_3 + H^{\oplus}, \Delta$ ⇒ H_2CrO_4, Δ	Jone's Reagent (CrO_3 + Acetone + H_2O)	⇒ PCC ⇒ PDC ⇒ Collin's Reagent ⇒ Oppanaurs oxi.	$Cu, 300^{\circ}\Delta$
① $R-CH_2-OH$ (1°) Oxidation →	$R-\overset{O}{\overset{\|\|}{C}}-OH$	$R-\overset{O}{\overset{\|\|}{C}}-OH$	$R-\overset{O}{\overset{\|\|}{C}}-H$	$R-\overset{O}{\overset{\|\|}{C}}-H + H_2$ (Dehydrogenation)
② $R-\overset{OH}{\underset{\|}{CH}}-R$ (2°) Oxi →	$R-\overset{O}{\overset{\|\|}{C}}-R$ (Generally)	$R-\overset{O}{\overset{\|\|}{C}}-R$	$R-\overset{O}{\overset{\|\|}{C}}-R$	$R-\overset{O}{\overset{\|\|}{C}}-R + H_2$ (Dehydrogenation)
③ $R-\overset{OH}{\underset{\|}{C}}-R$ (3°) Oxi →	No oxidation	NO Oxi.	NO Oxi.	Alkene (Dehydration)

REDUCTION, OXIDATION AND HYDROLYSIS

(II) Oxidation:-

Reactant	Strong oxidising agents (SOA) ⇒ $KMnO_4 + H^{\oplus}/OH^{-}/H_2O + \Delta$ ⇒ $K_2Cr_2O_7 + H^{\oplus} + \Delta$ ⇒ $CrO_3 + H^{\oplus}, \Delta$ ⇒ H_2CrO_4, Δ	Moderate oxi. Agent (MOA) Jone's Reagent (CrO_3 + Acetone + H_2O)	Weak oxi. Agents (WOA) ⇒ PCC ⇒ PDC ⇒ Collin's Reagent ⇒ Oppanaurs oxi.	$Cu, 300^{\circ}\Delta$
(4) $R-\overset{O}{\overset{\|}{C}}-H \xrightarrow{\text{oxi.}}$	$R-\overset{O}{\overset{\|}{C}}-OH$	$R-\overset{O}{\overset{\|}{C}}-OH$	X	X
(5) $R-\overset{O}{\overset{\|}{C}}-R \xrightarrow{\text{oxi.}}$	mix. of carboxylic acid with lesser no. of C atom (only in specific condition)	X	X	X

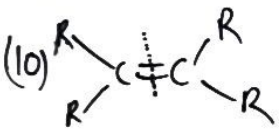
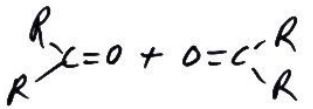
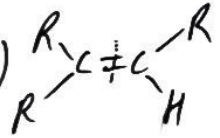
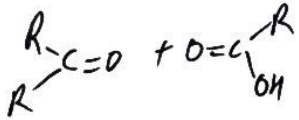
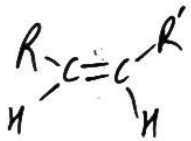
REDUCTION, OXIDATION AND HYDROLYSIS

(II) Oxidation:-

Reactant	Strong oxidising agent (SOA)	Moderate oxi. Agent (MOA)	Weak oxi. Agents (WOA)	
	⇒ $KMnO_4 + H^{\oplus}/OH^{-}/H_2O + \Delta$ ⇒ $K_2Cr_2O_7 + H^{\oplus} + \Delta$ ⇒ $CrO_3 + H^{\oplus}, \Delta$ ⇒ H_2CrO_4, Δ	Jone's Reagent (CrO_3 + Acetone + H_2O)	⇒ PCC ⇒ PDC ⇒ Collin's Reagent ⇒ Oppanaurs oxi.	$Cu, 300^{\circ}\Delta$
(6) $R-COOH \xrightarrow{oxi.}$	X	X	X	X
(7) $H-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{oxi}$ formic	$(H-O-\overset{O}{\overset{\|}{C}}-OH) \rightarrow CO_2 + H_2O$ carbonic acid	—	X	X
(8) $H-\overset{O}{\overset{\|}{C}}-H \xrightarrow{oxi}$ formaldehyde	$CO_2 + H_2O$	$(H-COOH)$	X	X
(9) $CH_3-OH \xrightarrow{oxi.}$	$CO_2 + H_2O$	$(H-COOH)$	$H-\overset{O}{\overset{\|}{C}}-H$	$H-\overset{O}{\overset{\|}{C}}-H$

REDUCTION, OXIDATION AND HYDROLYSIS

(II) Oxidation:-

Reactant	Strong oxidising agent (SOA)	Moderate oxi. Agent (MOA)	Weak oxi. Agents (WOA)	
	⇒ $KMnO_4 + H^{\oplus}/OH^{-}/H_2O + \Delta$ ⇒ $K_2Cr_2O_7 + H^{\oplus} + \Delta$ ⇒ $CrO_3 + H^{\oplus}, \Delta$ ⇒ H_2CrO_4, Δ	Jone's Reagent (CrO_3 + Acetone + H_2O)	⇒ PCC ⇒ PDC ⇒ Collin's Reagent ⇒ Oppanaurs oxi.	$Cu, 300^{\circ}\Delta$
(10)  Oxi. ⟶		X	X	X
(11)  ⟶		X	X	X
(12)  ⟶	$R\text{-}COOH + R'\text{-}COOH$	X	X	X

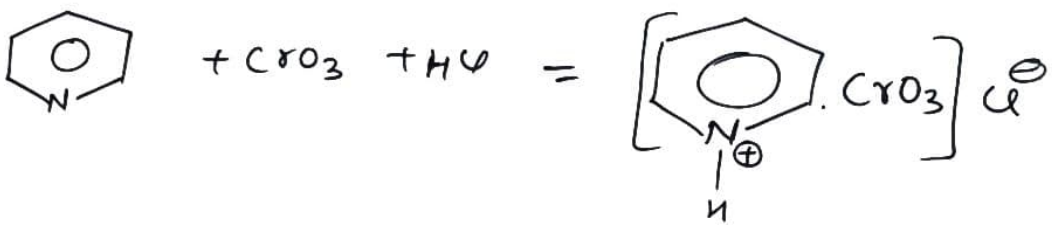
REDUCTION, OXIDATION AND HYDROLYSIS

(II) Oxidation:-

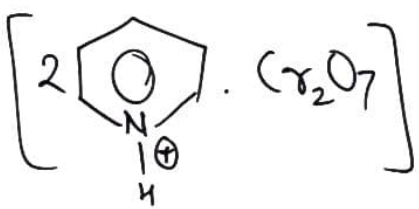
Reactant	Strong oxidising agents (SOA)	Moderate oxi. Agent (MOA)	Weak oxi. Agents (WOA)	
	⇒ $KMnO_4 + H^{\oplus}/OH^{-}/H_2O + \Delta$ ⇒ $K_2Cr_2O_7 + H^{\oplus} + \Delta$ ⇒ $CrO_3 + H^{\oplus}, \Delta$ ⇒ H_2CrO_4, Δ	Jone's Reagent (CrO_3 + Acetone + H_2O)	⇒ PCC ⇒ PDC ⇒ Collin's Reagent ⇒ Oppanaurs oxi.	$Cu, 300^{\circ}\Delta$
(13) $R-CH=CH_2$ (Terminal alkene) $\xrightarrow{oxi}$	$R-COOH + CO_2$	×	×	×
(14) $R-C\equiv C-R' \xrightarrow{oxi.}$	$R-COOH + R'-COOH$	×	×	×
(15) $R-C\equiv CH \xrightarrow{oxi}$	$R-COOH + CO_2 + H_2O$	×	×	×

REDUCTION, OXIDATION AND HYDROLYSIS

⇒ PCC = Pyridinium chloro cromate



⇒ PDC = Pyridinium dicromate



⇒ Collin's reagent =  $+ CrO_3 + CH_2Cl_2$

⇒ Oppanaur's oxidation = $Al(-O-C(CH_3)_3)_3$ + Acetone

aluminum tertiary butoxide

REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION:-

[A] Oxidation of alkane

[B] " " " alkene and alkyne

[C] " " " alcohols

[D] " " " Aldehydes and Ketones

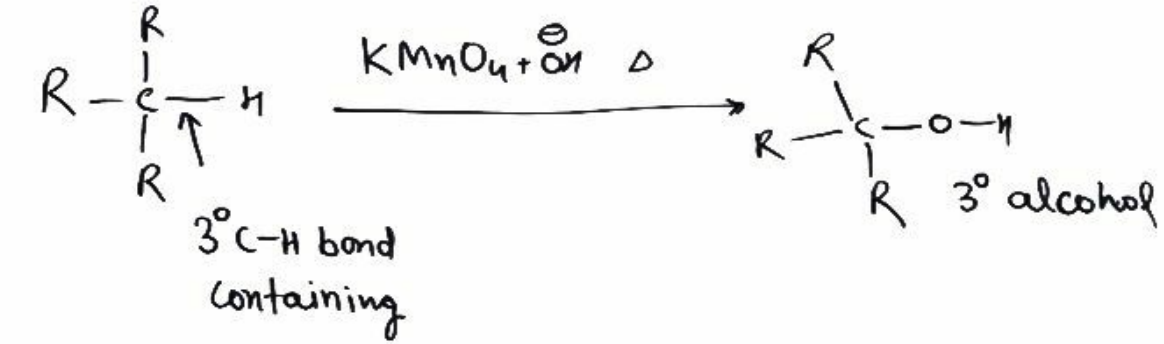
[E] " " " aromatic comp.

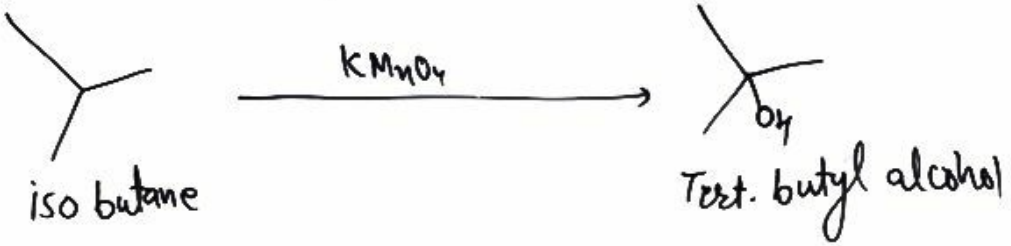
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION:-

[A] Oxidation of alkane ⇒ Difficult

n-Alkane $\xrightarrow{KMnO_4 / K_2Cr_2O_7}$ No oxidation

* 

eg 

REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION:-

[A] Oxidation of alkane ⇒ Difficult

Oxidation of methane with O_2

⇒ $CH_4 + O_2 \xrightarrow[\text{Cu tube}]{200^\circ C \;\; 100 \text{ atm}} CH_3-OH$

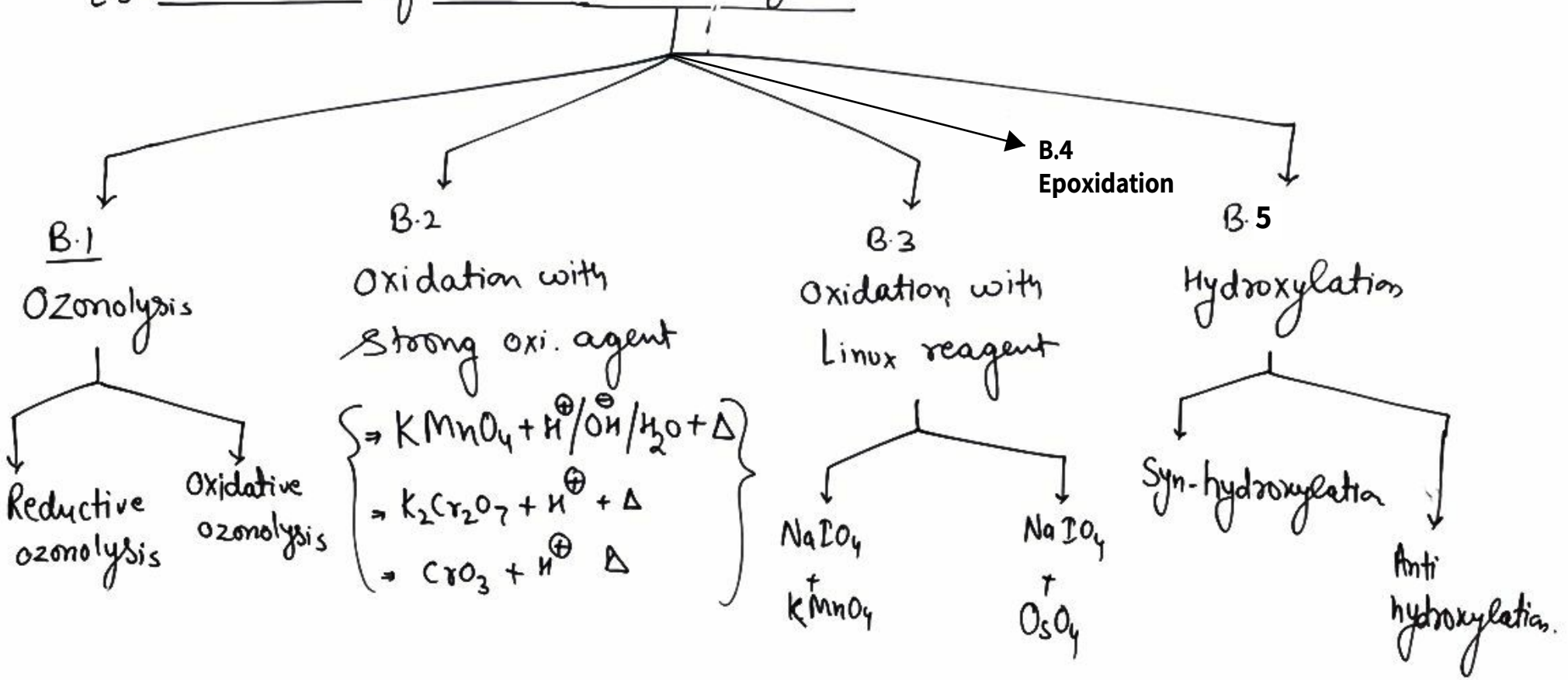
⇒ $CH_4 + O_2 \xrightarrow{MoO, \Delta \text{ Pressure}} CH_2=O + H_2O$

⇒ $CH_4 + O_2 \xrightarrow{1000^\circ C} H-C\equiv C-H + CO + H_2$

REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION:-

[B] Oxidation of alkenes and alkyne :-



REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION:-

[B] Oxidation of alkenes and alkyne :-

B.1 Ozonolysis of C=C & C≡C :- dissociation with ozone



Reductive ozonolysis

Reagent:-

(i) O_3

(ii) $Zn + H_2O$ / Me_2S / $H_2 + Ni_{(Raney)}$

Oxidative ozonolysis

Reagent

(i) O_3

(ii) H_2O | H_2O_2 | HCOOOH

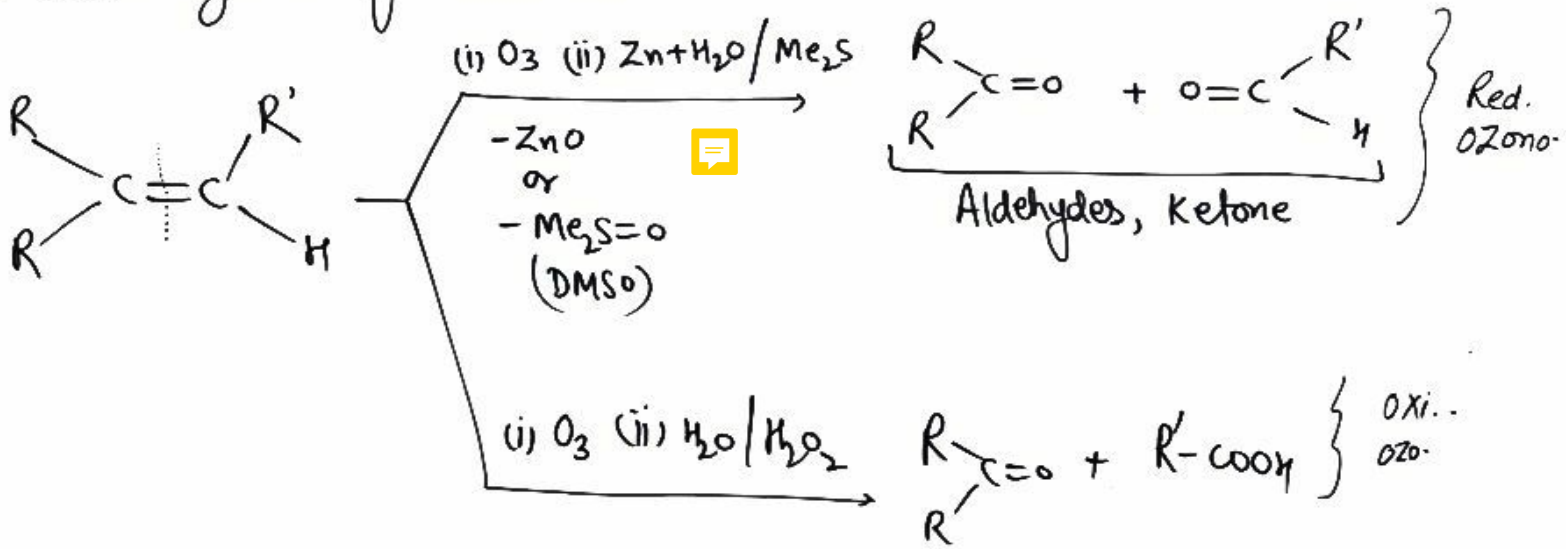
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION:-

[B] Oxidation of alkenes and alkyne :-

B·1 Ozonolysis of C=C & C≡C :- dissociation with ozone

⇒ Ozonolysis of alkene:-



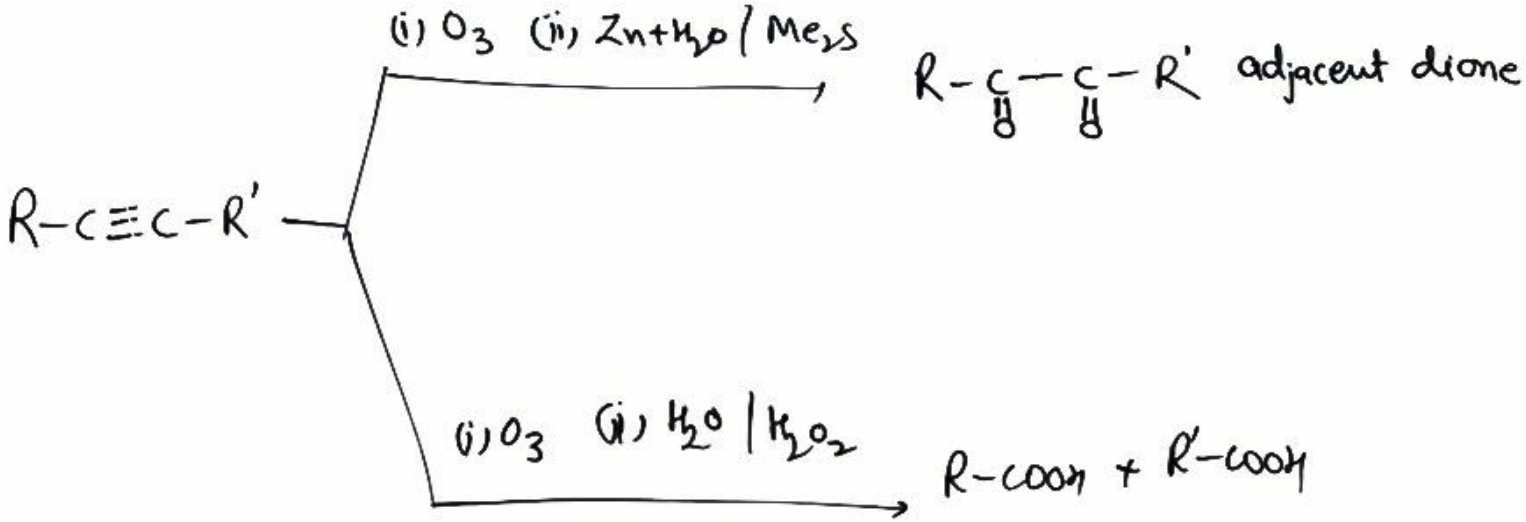
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION:-

[B] Oxidation of alkenes and alkyne :-

B.1 Ozonolysis of C=C & C≡C :- dissociation with ozone

→ Ozonolysis of alkynes.



REDUCTION, OXIDATION AND HYDROLYSIS

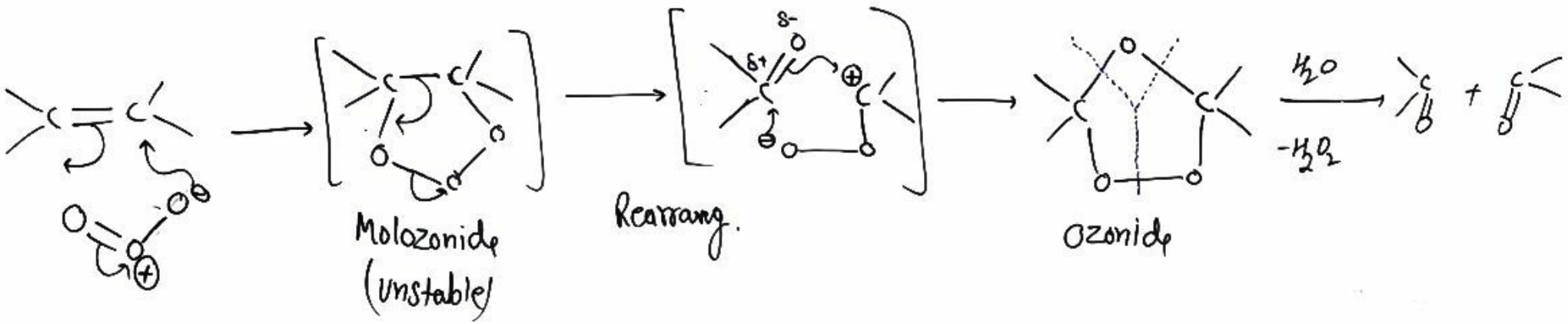
(II) OXIDATION:-

[B] Oxidation of alkenes and alkyne :-

B.1 Ozonolysis of C=C & C≡C :- dissociation with ozone

mechanism of ozonolysis

for >C=C<



REDUCTION, OXIDATION AND HYDROLYSIS

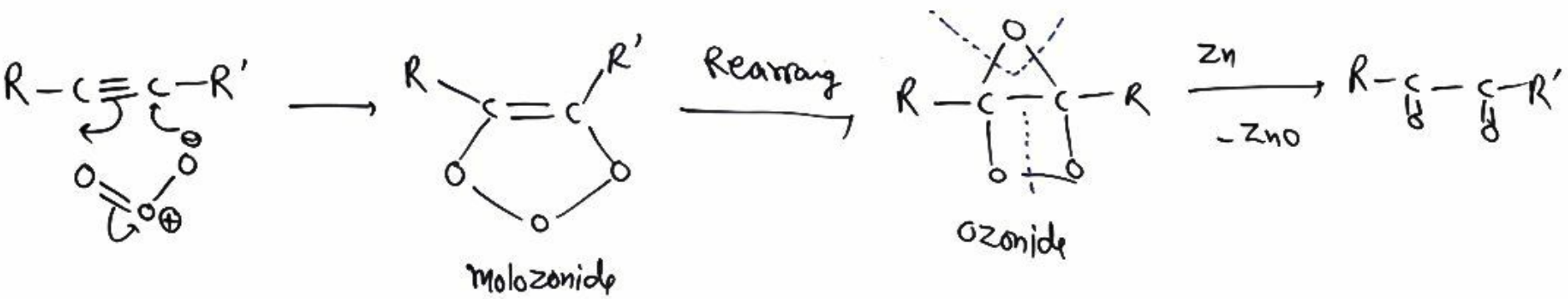
(II) OXIDATION:-

[B] Oxidation of alkenes and alkyne :-

B.1 Ozonolysis of C=C & C≡C :- dissociation with ozone

mechanism of ozonolysis

for C≡C



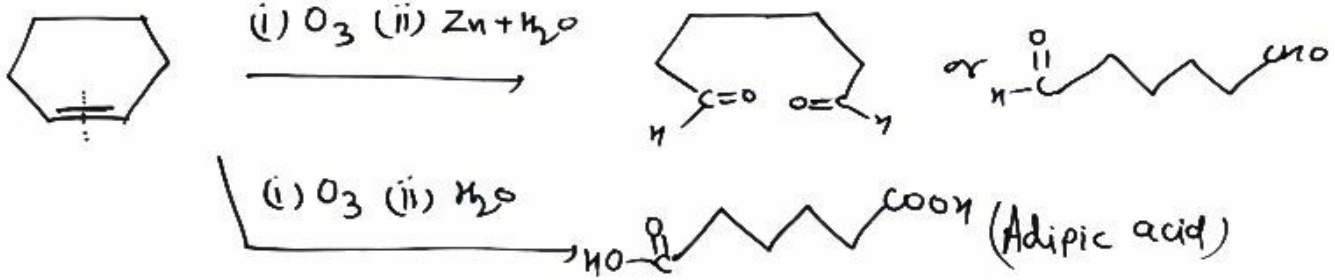
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION:-

[B] Oxidation of alkenes and alkyne :-

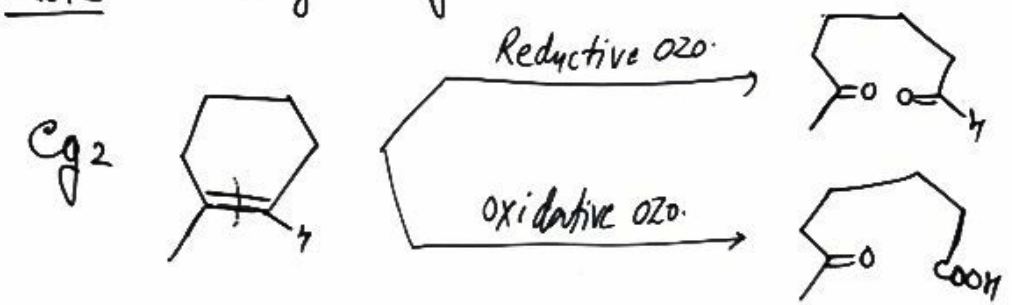
B-1 Ozonolysis of C=C & C≡C :- dissociation with ozone

Eg 1



Note ozonolysis of aromatic possible on high temp. (not at room temp.)

Eg 2

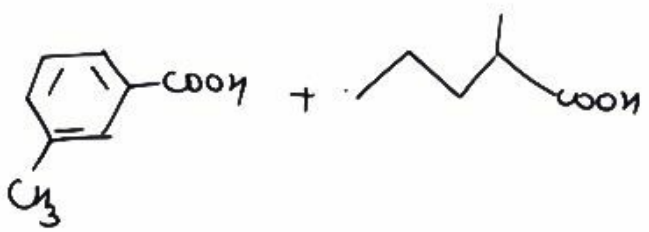


REDUCTION, OXIDATION AND HYDROLYSIS

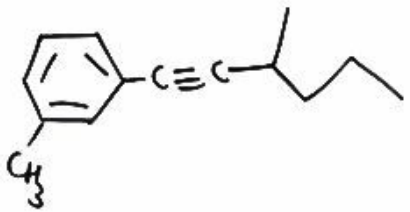
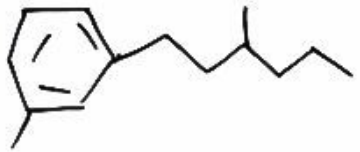
(II) OXIDATION:-

[B] Oxidation of alkenes and alkyne :-

B.1 Ozonolysis of C=C & C≡C :- dissociation with ozone

Q Alkyne ? —Ozonolysis→  +

Alkyne ? —H_2 + Ni, room Temp→ Product ?

Sol. Alkyne  , Product = 

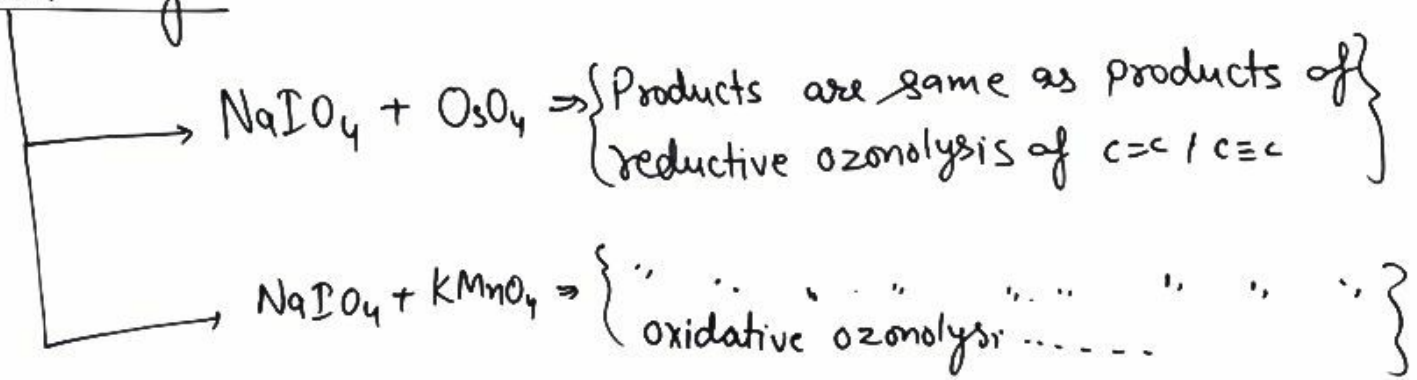
REDUCTION, OXIDATION AND HYDROLYSIS

(II) **OXIDATION:-** ⇒ [B] Oxidation of alkene and alkyne

[B.2] Oxidation with strong oxidising agent

⇒ On oxidation of alkene and alkyne with S.O.A., Products are same as Products of oxidative ozonolysis

[B.3] Oxidation with Linux reagent:- **Lemieux reagent**



REDUCTION, OXIDATION AND HYDROLYSIS

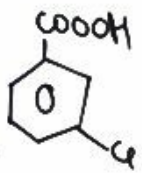
(II) **OXIDATION:-** ⇒ [B] Oxidation of alkene and alkyne

[B.4] Epoxidation of alkene:



Reagents

① → $R-\overset{O}{\overset{\|}{C}}-O-OH$ Per acid

- $H-\overset{O}{\overset{\|}{C}}-O-OH$ (Performic acid, PFA)
- $CH_3-\overset{O}{\overset{\|}{C}}-O-OH$ (Peracetic acid; PAA)
- $Ph-\overset{O}{\overset{\|}{C}}-O-OH$ (Perbenzoic acid, PBA)
-  m. Chloro perbenzoic acid (MCPBA)

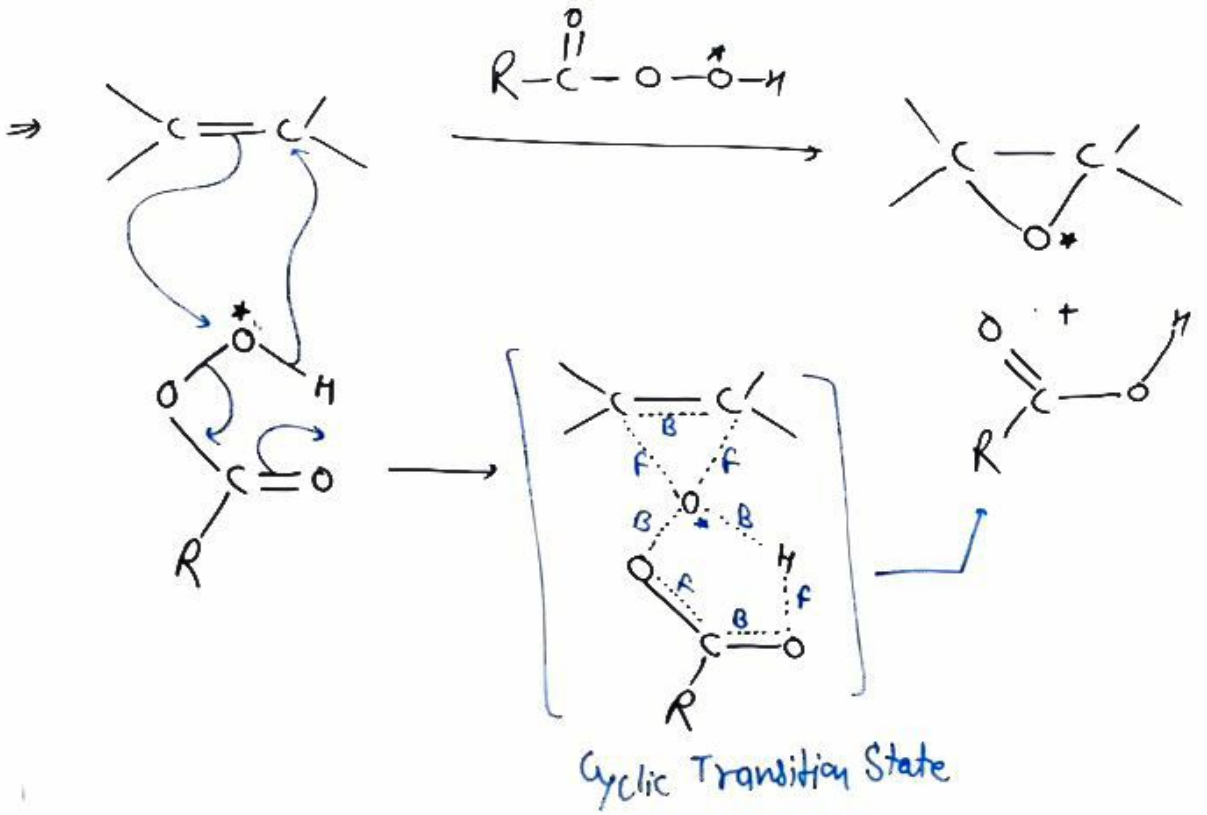
② → $Ag + O_2$ (air) Δ

(indirect method) ③ → (i) HOBr (ii) NaOH (Base)

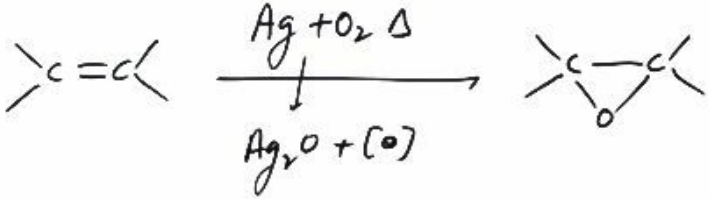
Reduction, Oxidation And Hydrolysis

(II) **Oxidation:-** ⇒ [B] Oxidation of alkene and alkyne

[B.4] Epoxidation of alkene:-



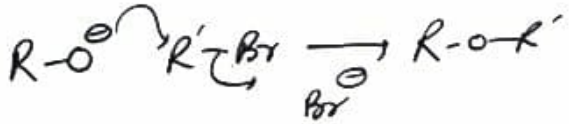
⇒ $2Ag + O_2 \xrightarrow{h\nu/\Delta} Ag_2O + [O]$



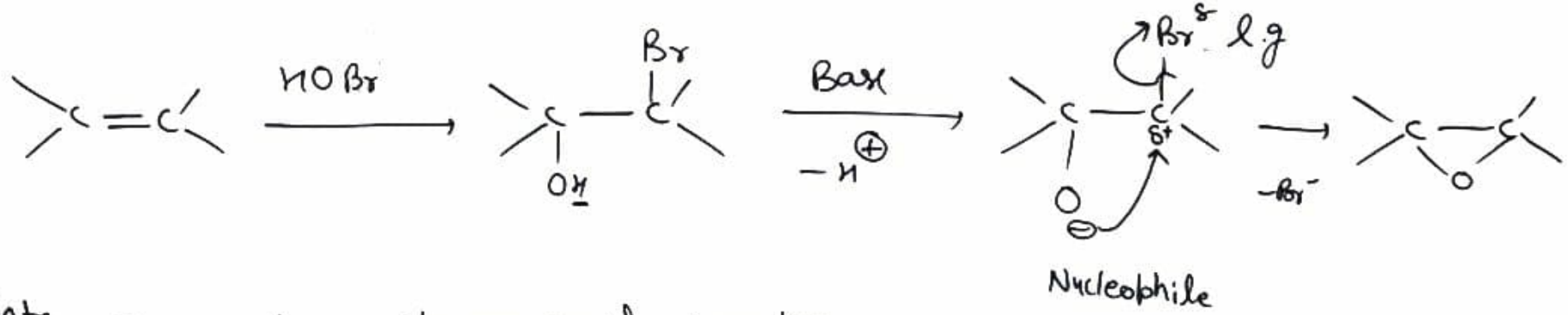
REDUCTION, OXIDATION AND HYDROLYSIS

(II) **OXIDATION:-** ⇒ [B] Oxidation of alkene and alkyne

[B.4] Epoxidation of alkene:



⇒ Indirect epoxidation



Note Epoxidation ⇒ Stereo specific reaction

REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION:- ⇒ [B] Oxidation of alkene and alkyne

[B.4] Epoxidation of alkene:

eg1



CH_3COOOH

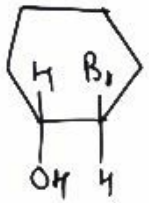


+ CH_3COOH

$Ag + O_2$ Δ

" + Ag_2O

$HOBr$

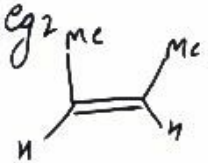


Trans

Base

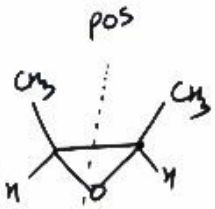


eg2



cis

R-COOOH
or $Ag + O_2$ Δ
or
$HOBr$ then Base



cis
meso

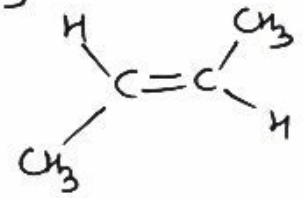
Reduction, Oxidation And Hydrolysis

(II) **Oxidation:-** ⇒ [B] Oxidation of alkene and alkyne

[B.4] Epoxidation of alkene:

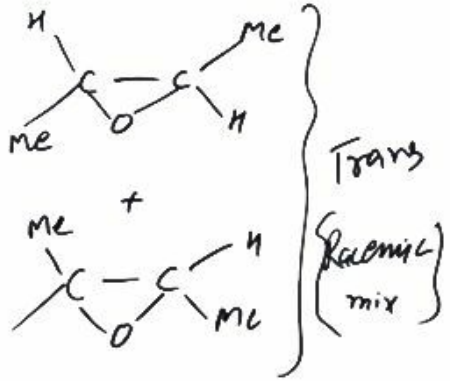
if you go by mechanism in case of HOBr + NaOH, it may be confusing
Hence, simply remember that
cis forms cis
trans forms trans

eg 3



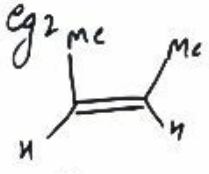
Trans

R-COOOH
or
Ag + O₂
or
(HOBr + NaOH)



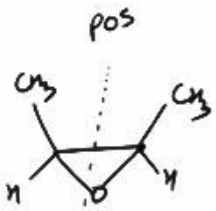
Trans (Racemic mix)

eg 2



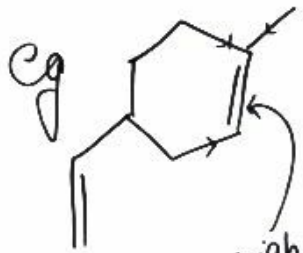
cis

R-COOOH
or Ag + O₂ Δ
or
HOBr then Base

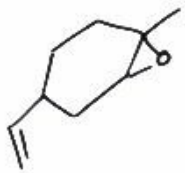


cis meso

eg



1 mole R-COOOH

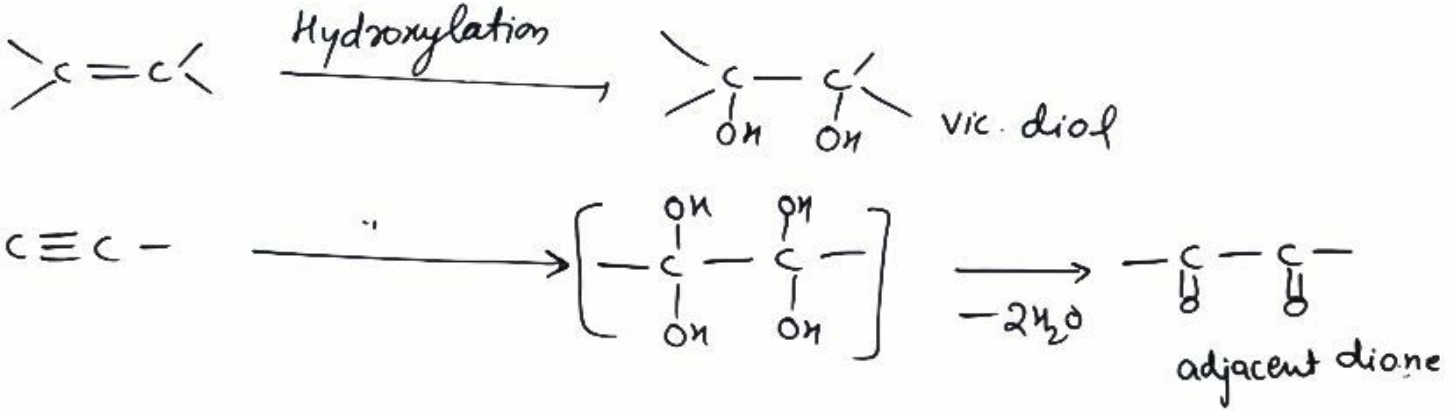


higher density

REDUCTION, OXIDATION AND HYDROLYSIS

(II) **OXIDATION:-** ⇒ [B] Oxidation of alkene and alkyne

B.5 Hydroxylation of C=C and C≡C



REDUCTION, OXIDATION AND HYDROLYSIS

(II) **OXIDATION:-** ⇒ [B] Oxidation of alkene and alkyne

B.5 Hydroxylation of C=C and C≡C

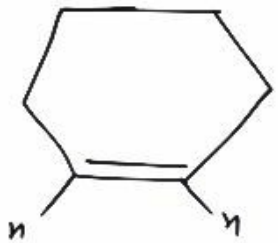
- **Syn hydroxylation**
  - → Syn Addition (CSE / TST)
  - → C=C, C≡C
  - Reagent
    - ① → Cold. dil. $KMnO_4 + \overset{\ominus}{O}H$ (Baeyer's reagent) (Purple colour)
      (rest of alkene and alkyne)
    - ② → (i) OsO_4 (ii) $NaHSO_3$ / H_2O_2
- **Anti hydroxylation**
  - → Anti addition (CAT / TAE)
  - → only C=C
  - Reagent
    - Epoxidation then hydrolysis

Reduction, Oxidation And Hydrolysis

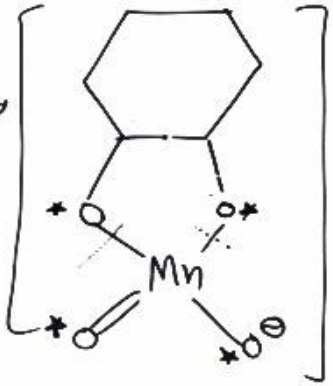
(II) **Oxidation:-** ⇒ [B] Oxidation of alkene and alkyne

B.5 Hydroxylation of C=C and C≡C

eg1

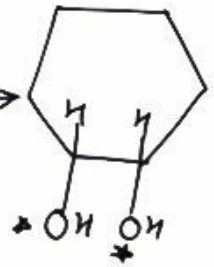


Cold: dil. $KMnO_4$ $^{+7}$ $^{\ominus}OH$



Cyclic intermediate

$2H_2O$

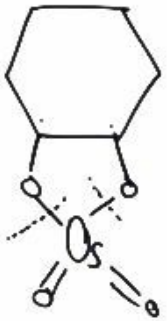


only cis

(Meso)

(i) OsO_4

(ii) $NaHSO_3$ aq



$2H_2O$

H_2OsO_4



cis/Meso

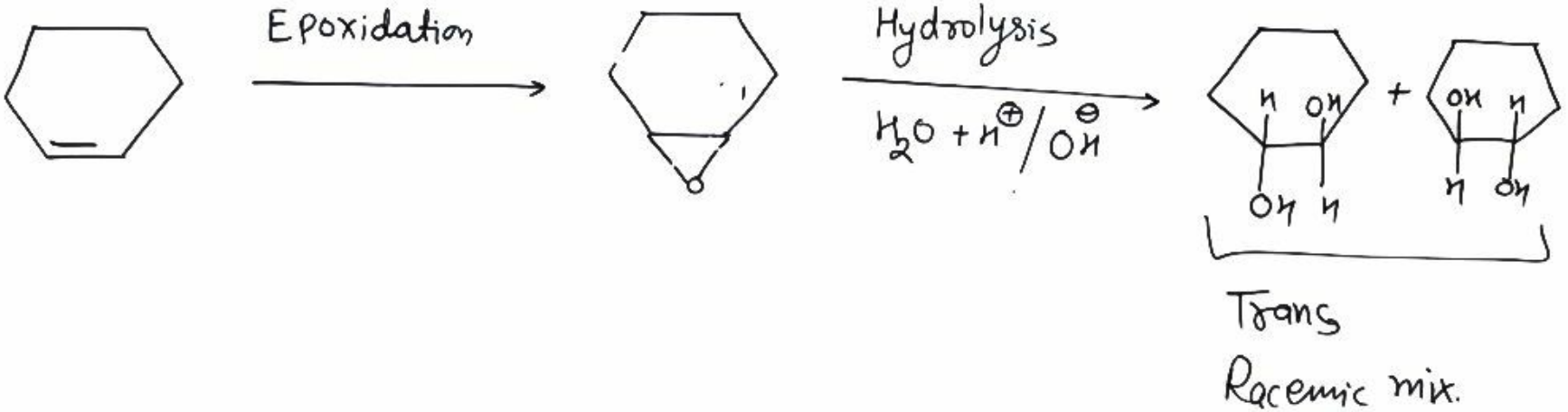
REDUCTION, OXIDATION AND HYDROLYSIS

(II) **OXIDATION:-** ⇒ [B] Oxidation of alkene and alkyne

B.5 Hydroxylation of C=C and C≡C

mechanism covered in detail in ORM-3

eg 2



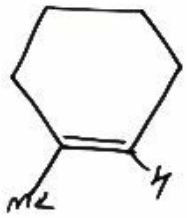
Trans

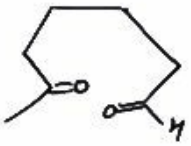
Racemic mix.

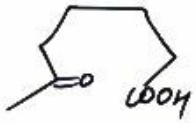
REDUCTION, OXIDATION AND HYDROLYSIS

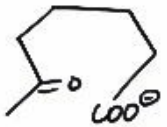
(II) OXIDATION :-

(B) Oxidation of alkenes and alkynes.

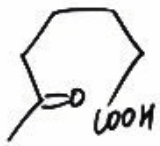
eg 

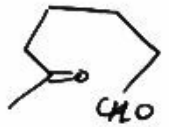
① $O_3 + Zn + H_2O$ → 

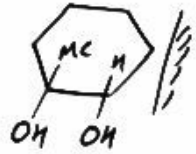
② (i) O_3 (ii) H_2O → 

③ $KMnO_4 + OH^{\ominus}$ Δ → 

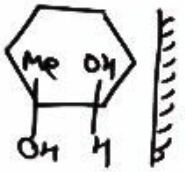
④ cold dil. $KMnO_4$ → 

⑤ $NaIO_4 + KMnO_4$ → 

⑥ $NaIO_4 + OsO_4$ → 

⑦ (i) OsO_4 (ii) $NaHSO_3$ → 

⑨ MCPBA → 

⑩ (i) $Ag + O_2$ Δ (ii) $H_3O^{\oplus}$ → 

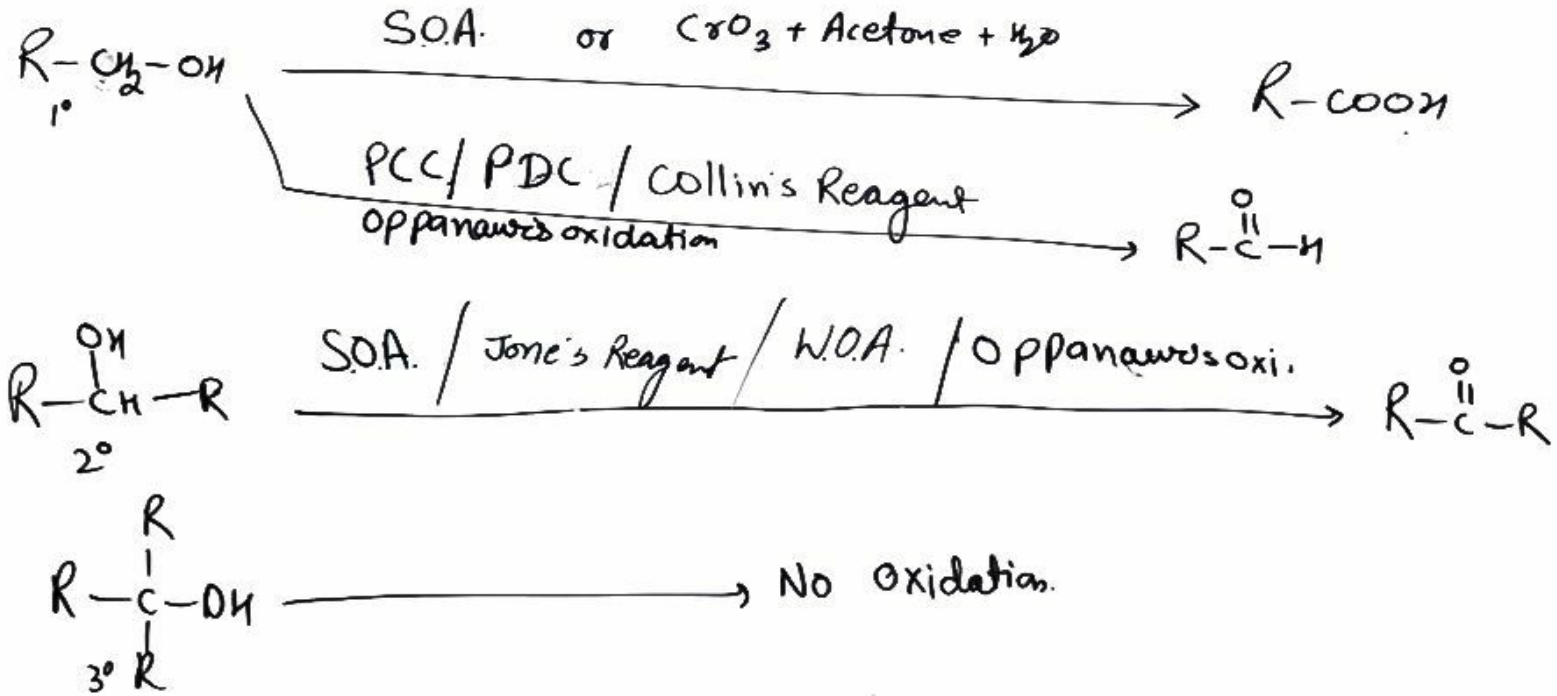


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Reduction, Oxidation and Hydrolysis

(II) Oxidation :-

(c) Oxidation of alcohols :-



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REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

(c) Oxidation of alcohols :-

(i) $Cu + 300^{\circ}C$

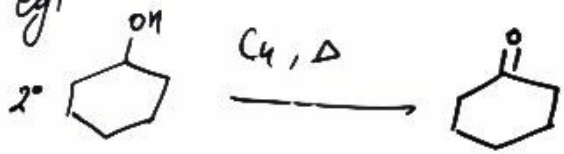
$$R-\underset{1^{\circ}}{CH_2}-OH \xrightarrow{Cu,\ \Delta} R-CH=O$$

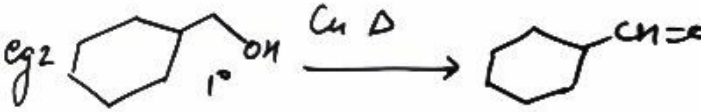
$$R-\underset{2^{\circ}}{\overset{OH}{\overset{|}{C}H}}-R \xrightarrow{Cu,\ \Delta} R-\overset{O}{\overset{||}{C}}-R$$

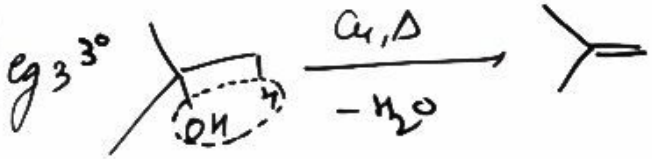
} Dehydrogenation

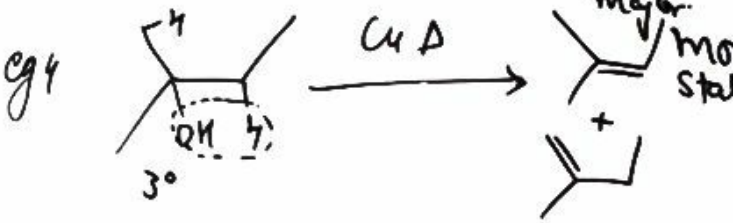
$$R-\overset{R}{\overset{|}{\underset{k}{C}}}-OH \xrightarrow[-H_2O]{Cu,\ \Delta} \text{Alkene}$$

dehydration

eg1 

eg2 

eg3 

eg4 

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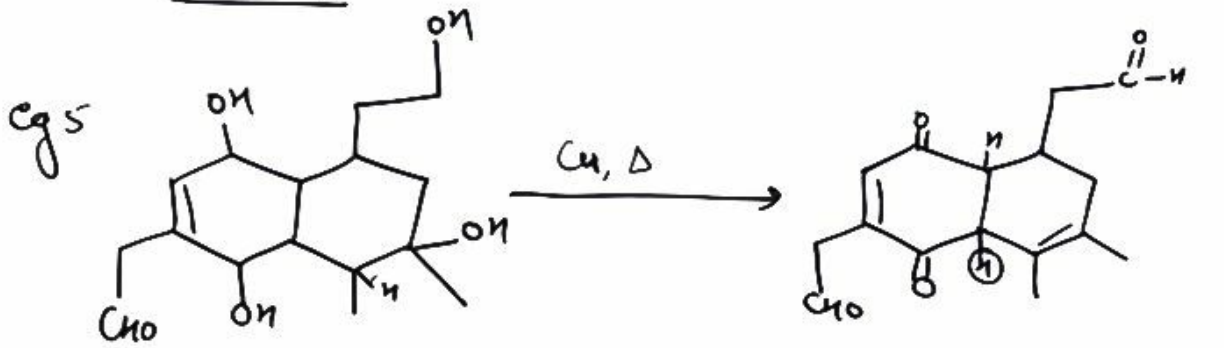
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

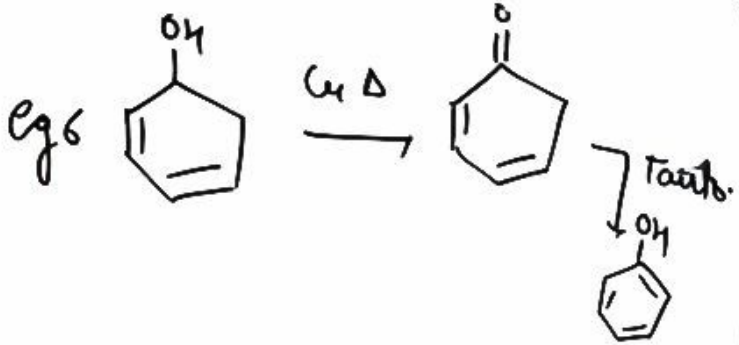
(C) Oxidation of alcohols :-

(i) Cu + 300°C

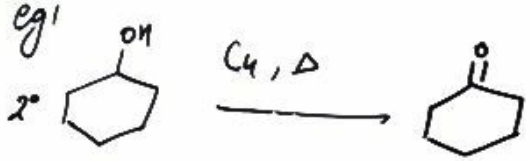
eg 5



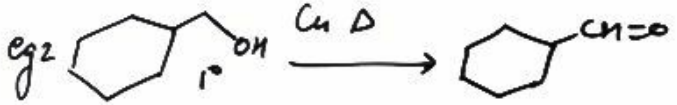
eg 6



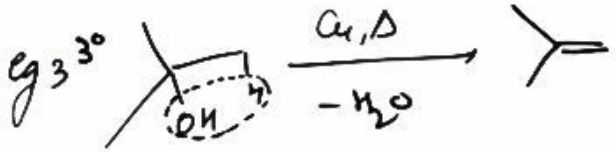
eg 1



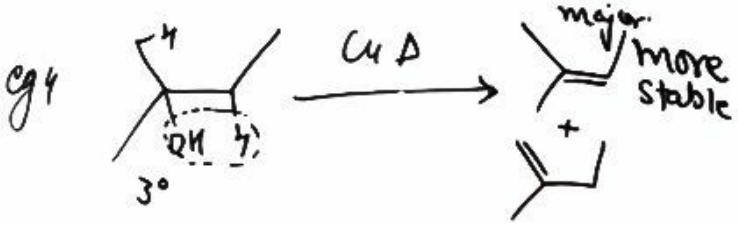
eg 2



eg 3 3°



eg 4



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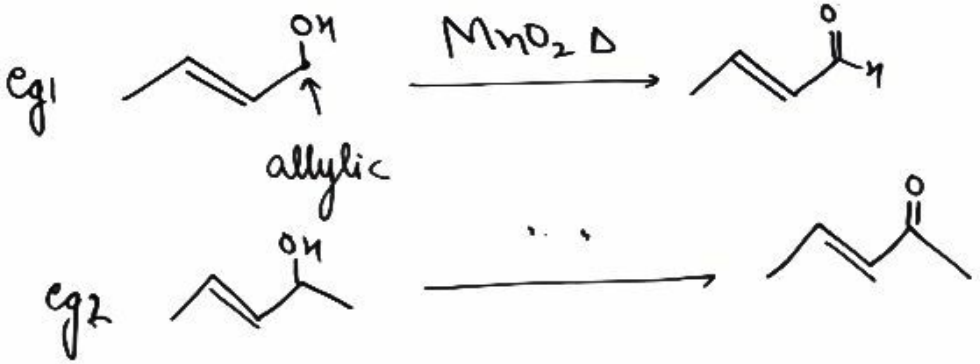
REDUCTION, OXIDATION AND HYDROLYSIS

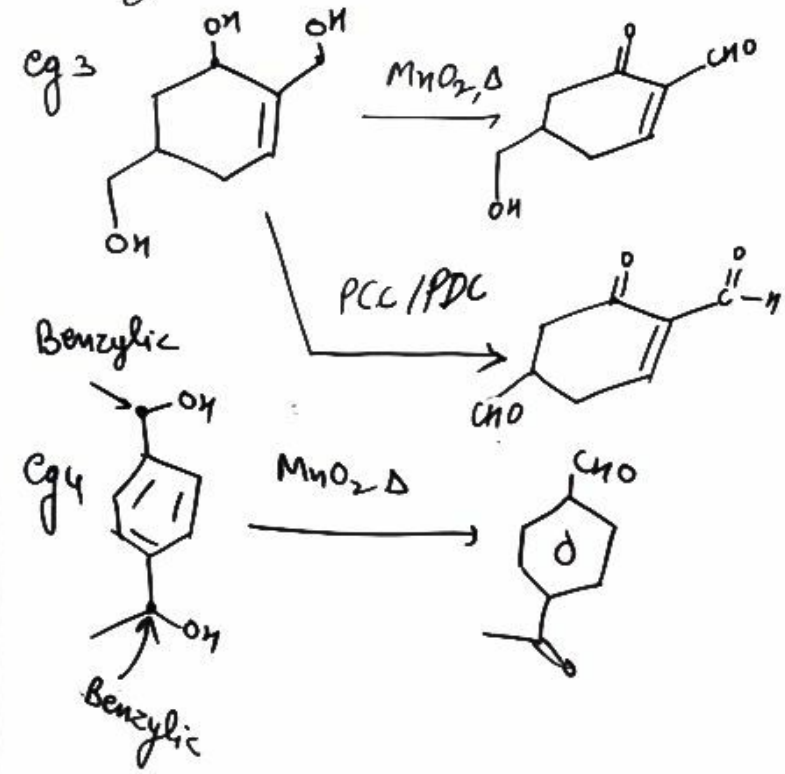
(II) OXIDATION :-

(C) Oxidation of alcohols :-

(ii) MnO_2, Δ : Selective oxidation of allylic and benzylic alcohols (1°, 2°)

- Weak oxidising agent
- Products = Aldehyde, Ketone.





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REDUCTION, OXIDATION AND HYDROLYSIS

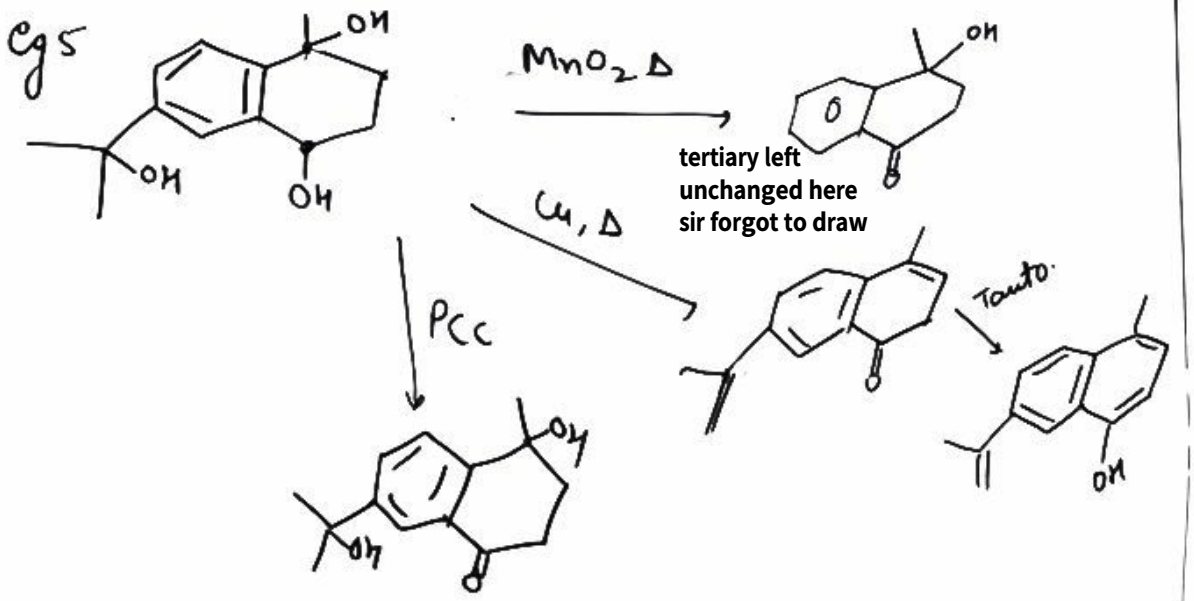
(II) OXIDATION :-

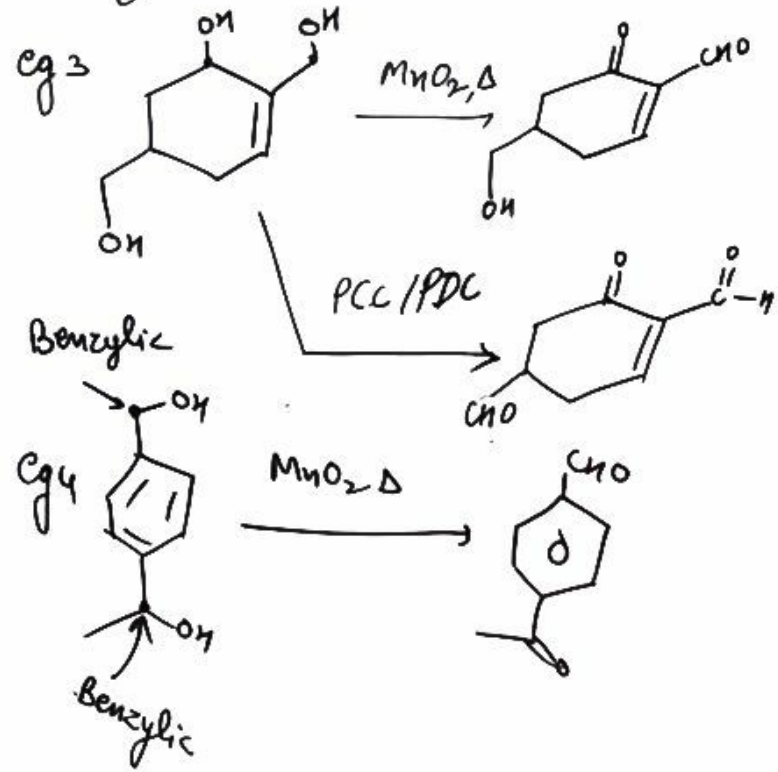
(C) Oxidation of alcohols :-

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» Weak oxidising agent

» Products = Aldehyde, Ketone.





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REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

(C) Oxidation of alcohols :-

(iii) dil. HNO_3

$$R-CH_2-OH \xrightarrow{\text{dil } HNO_3} R-COOH$$

1°

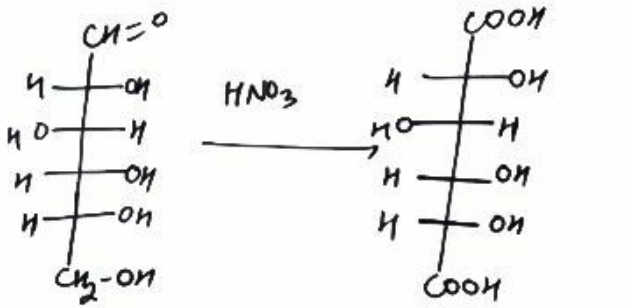
$$R-OH \xrightarrow{\text{dil } HNO_3} \text{No oxidation}$$

2° or 3°

$$R-CH=O \xrightarrow{\text{dil } HNO_3} R-COOH$$

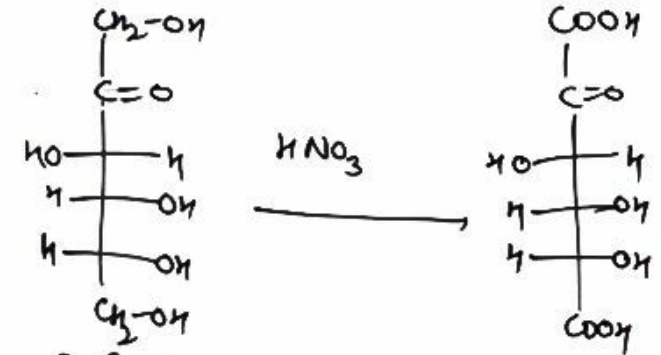
$$\text{Ketone} \xrightarrow{\quad\quad} \text{No Rx}^n.$$

eg1



D-Glucose → Gluconic acid

eg2



D. Fructose

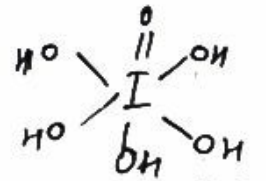
more sweeter than glucose

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REDUCTION, OXIDATION AND HYDROLYSIS

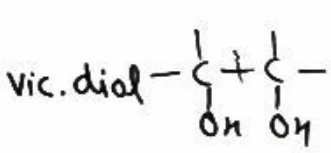
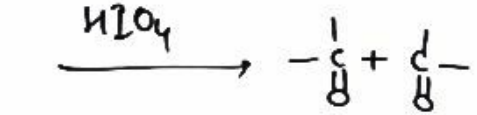
(II) OXIDATION :-

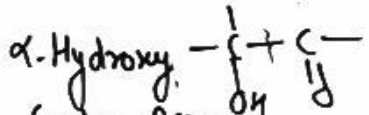
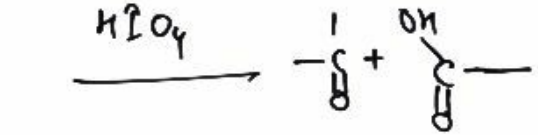
(C) Oxidation of alcohols :-

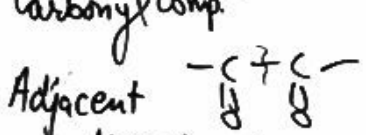
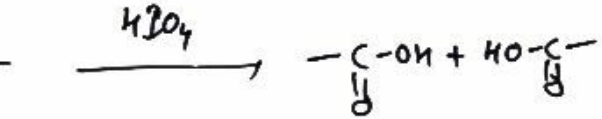
(IV) { HIO_4 (Per iodic acid) or $(CH_3COO)_4Pb$ (Lead tetra acetate) } $HIO_4 + 2H_2O \longrightarrow H_5IO_6$ = 

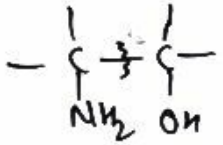
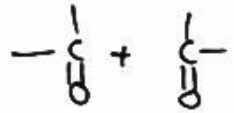
sp^3d^2

⇒ Oxidative dissociation of -- **to dissociate one such bond, one HIO4 is required**

vic.dial  $\xrightarrow{HIO_4}$ 

α-Hydroxy Carbonyl comp  $\xrightarrow{HIO_4}$ 

Adjacent di carbonyl  $\xrightarrow{HIO_4}$ 

 $\xrightarrow{HIO_4}$ 

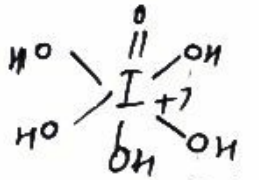
product similar to case of vic-diol

competishun
Power of real gurus

REDUCTION, OXIDATION AND HYDROLYSIS

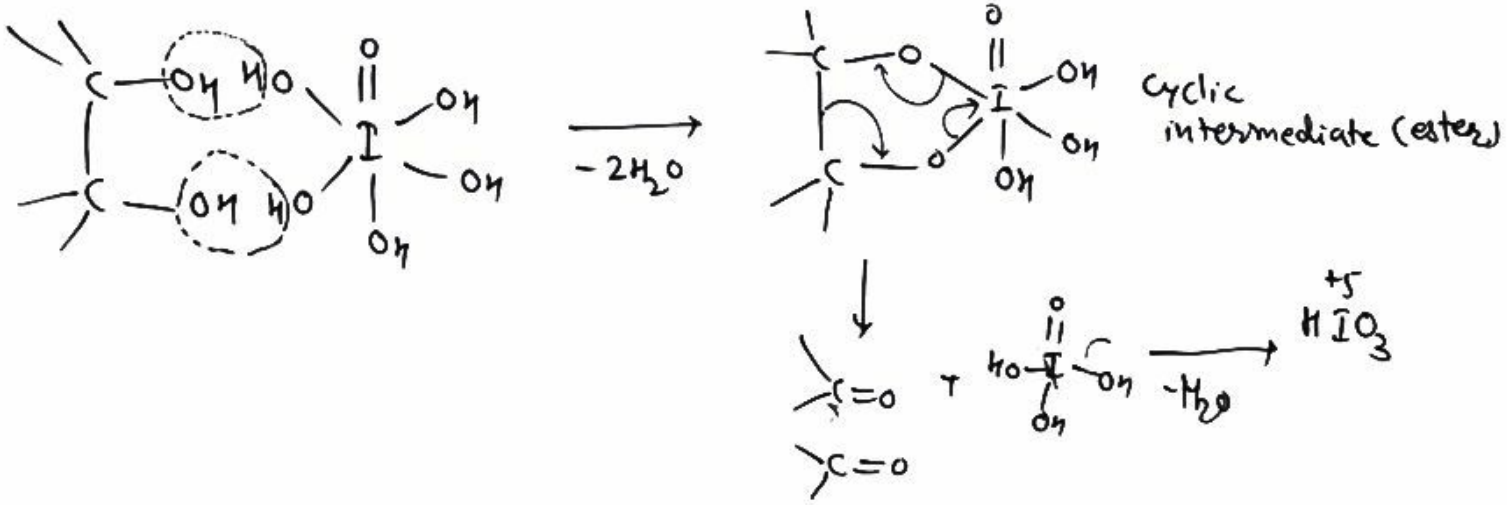
(II) OXIDATION :-

(C) Oxidation of alcohols :-

(IV) { HIO_4 (Per iodic acid) or $(CH_3COO)_4Pb$ (Lead tetra acetate) } $HIO_4 + 2H_2O \longrightarrow H_5IO_6$ = 

sp^3d^2

mechanism



competishun
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REDUCTION, OXIDATION AND HYDROLYSIS

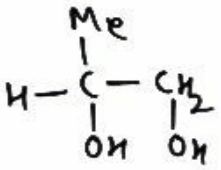
(II) OXIDATION :-

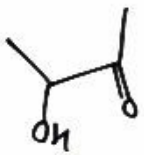
(C) Oxidation of alcohols :-

(iv) { HIO_4 (Per iodic acid) or $(CH_3COO)_4Pb$ (Lead tetra acetate) }

product depends on the no. of bonds which HIO4 is able to cleave

Hint Alcohol ⟶ carbonyl comp. ⟶ carboxylic acid ⟶ CO_2

eg 1  $\xrightarrow{HIO_4}$

eg 2  $\xrightarrow{HIO_4}$

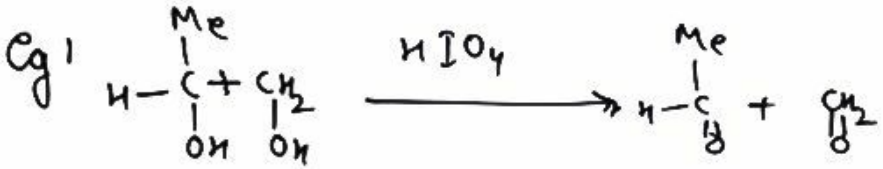
competishun
Power of real gurus

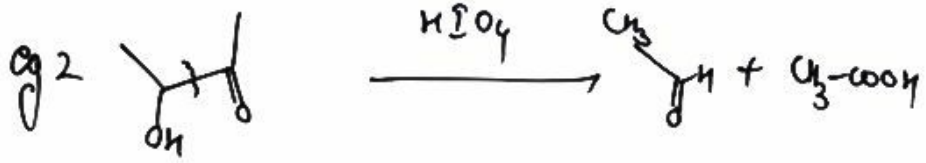
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

(C) Oxidation of alcohols :-

(iv) { HIO_4 (Per iodic acid) or $(CH_3COO)_4Pb$ (Lead tetra acetate) }

eg 1 

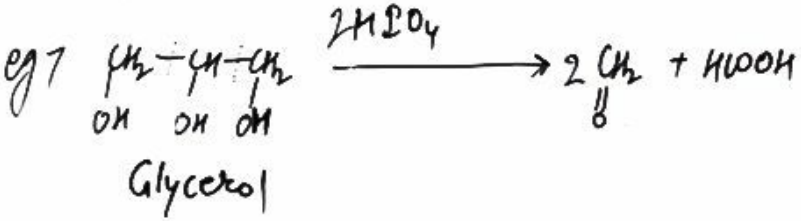
eg 2 

eg 3 

eg 4 

eg 5 

eg 6 

eg 7 

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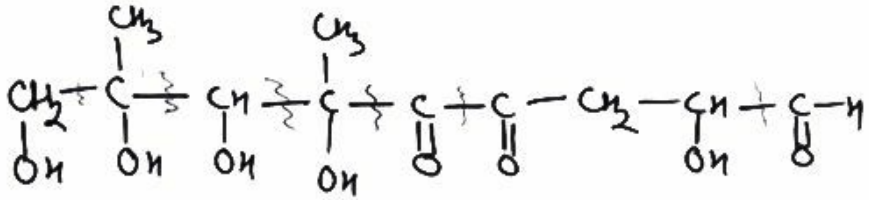
Reduction, Oxidation And Hydrolysis

(II) Oxidation :-

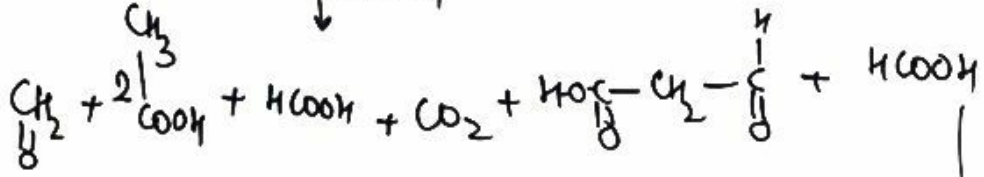
(c) Oxidation of alcohols :-

(iv) { HIO_4 (Per iodic acid) or $(CH_3COO)_4Pb$ (Lead tetra acetate) }

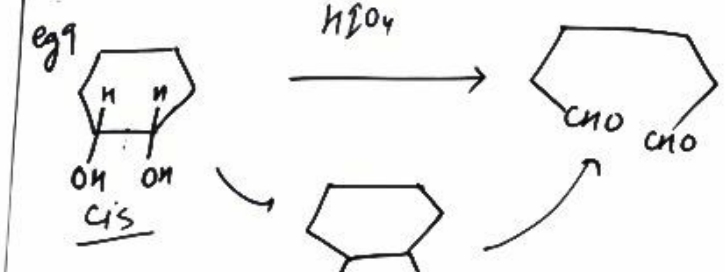
eg 8



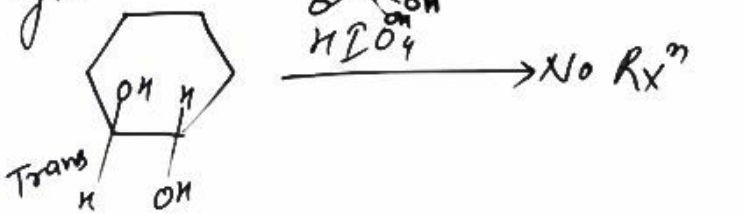
$\xrightarrow{6HIO_4}$



eg 9



eg 10



No Cyclic intermediate not possible

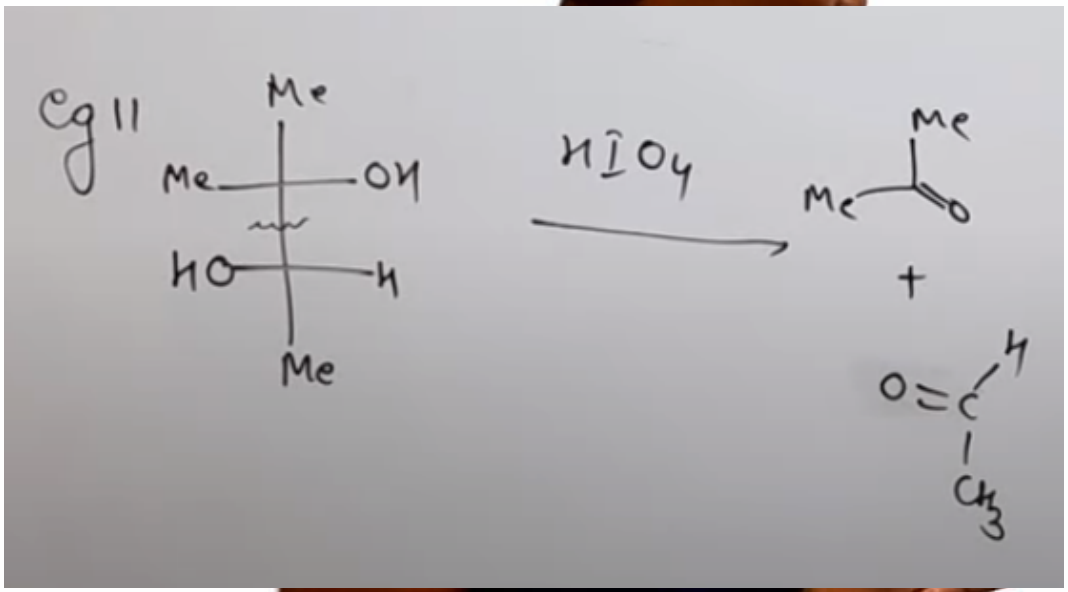
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

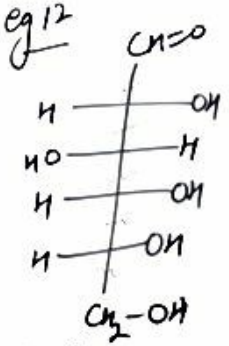
(C) Oxidation of alcohols :-

(iv) { HIO_4 (Per iodic
or
$(CH_3COO)_4Pb$

eg 11



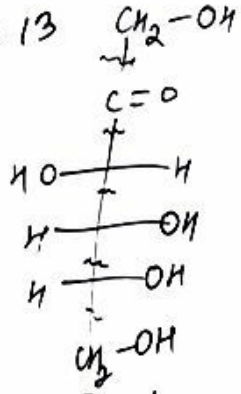
eg 12



D. Glucose

$5\,HIO_4 \longrightarrow 5\,HCOOH + CH_2=O$

eg 13



D. Fructose

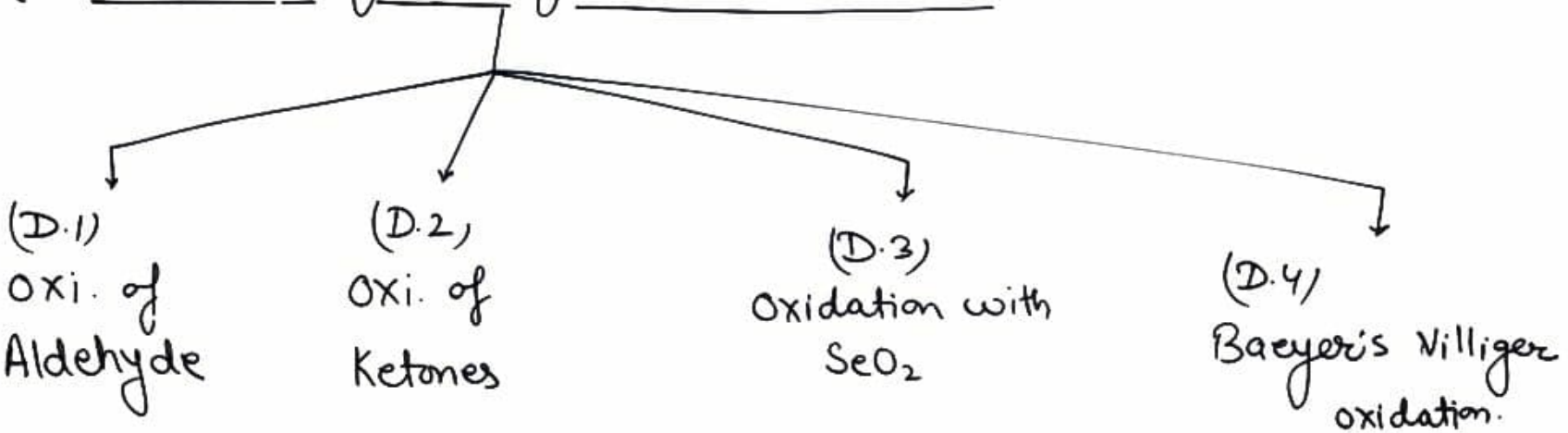
$5\,HIO_4 \longrightarrow 2\,CH_2=O + CO_2 + 3\,HCOOH$

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REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

(D) Oxidation of aldehydes and ketones :-

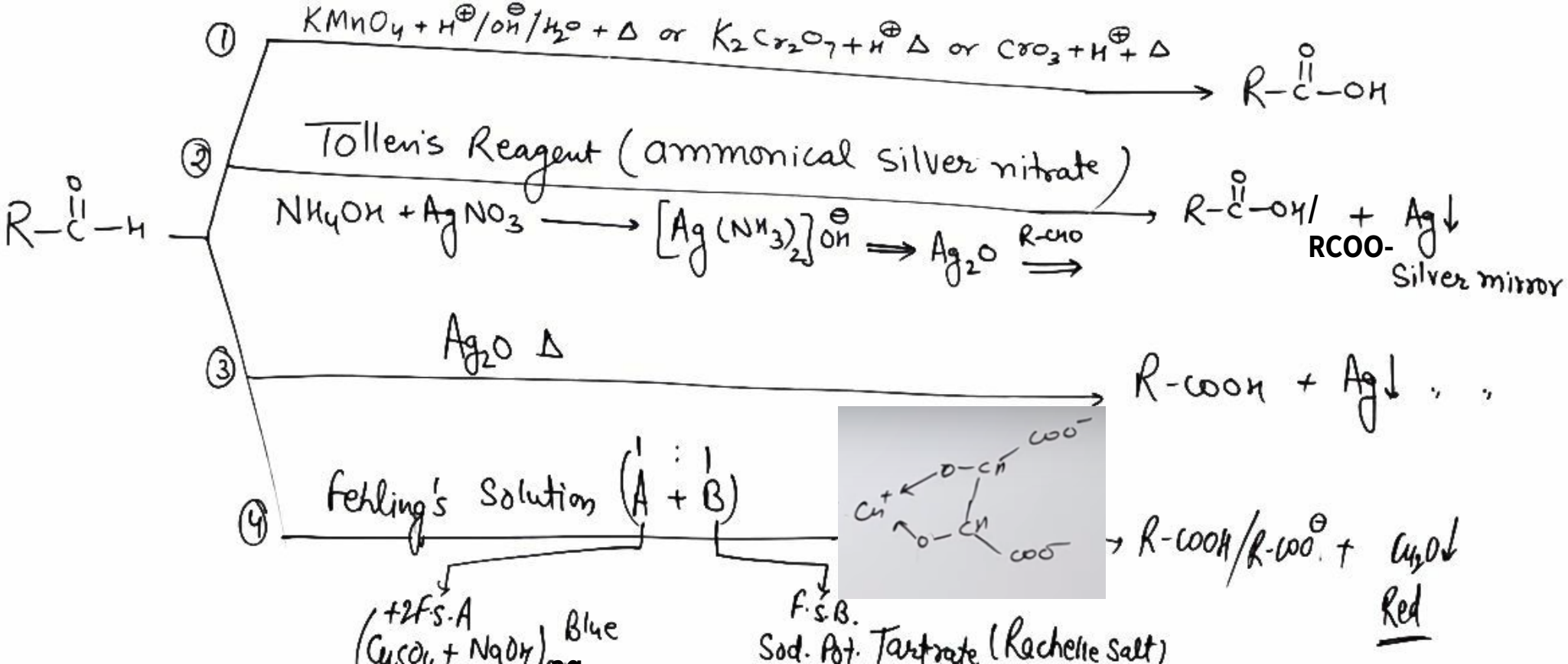


Reduction, Oxidation and Hydrolysis

(II) Oxidation :-

(D) Oxidation of aldehydes and ketones :-

(D1) Oxidation of aldehydes :-

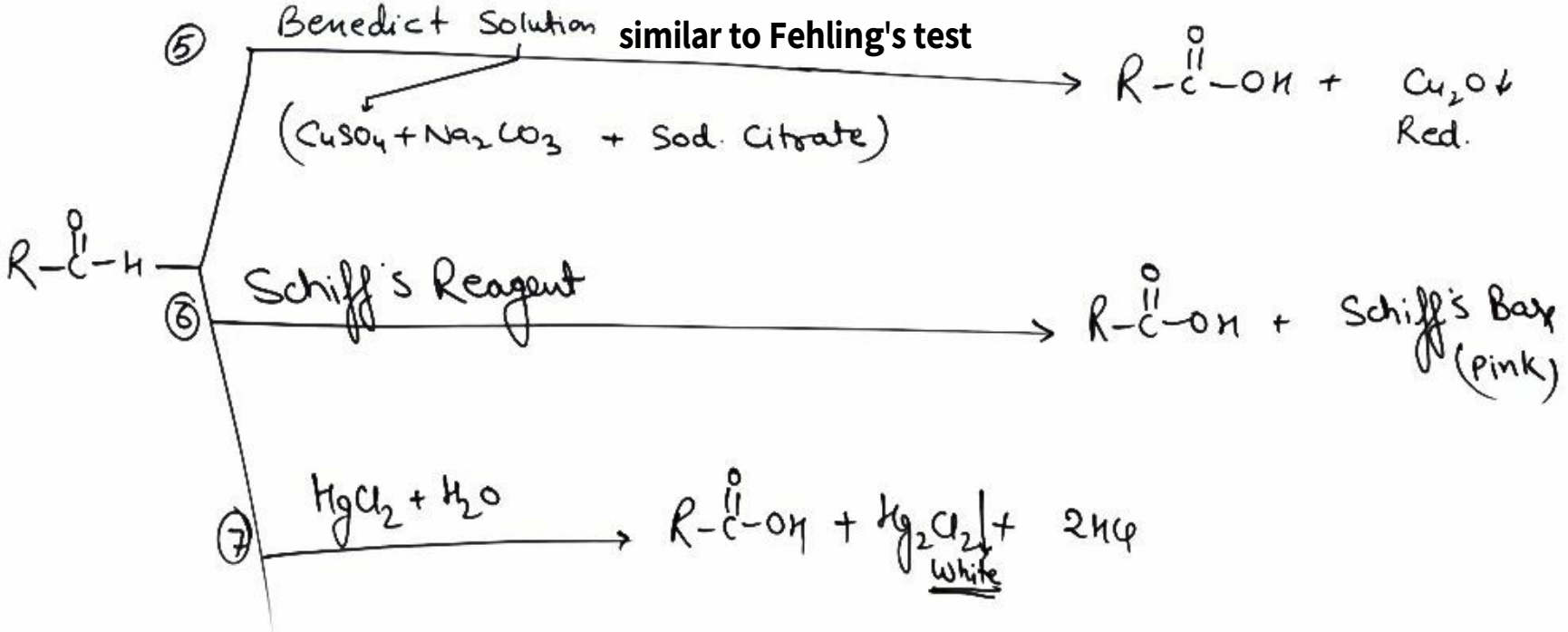


Reduction, Oxidation And Hydrolysis

(II) Oxidation :-

(D) Oxidation of aldehydes and ketones :-

(D1) Oxidation of aldehydes :-



Reduction, Oxidation and Hydrolysis

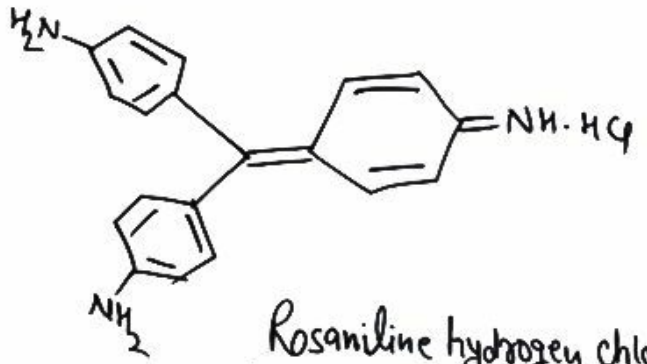
(II) Oxidation :-

(D) Oxidation of aldehydes and ketones :-

(D1) Oxidation of aldehydes :-

Note ① above reaction ② to ⑦ all are test of aldehydes.

② Schiff reagent :-



Rosaniline hydrogen chloride
(Red)

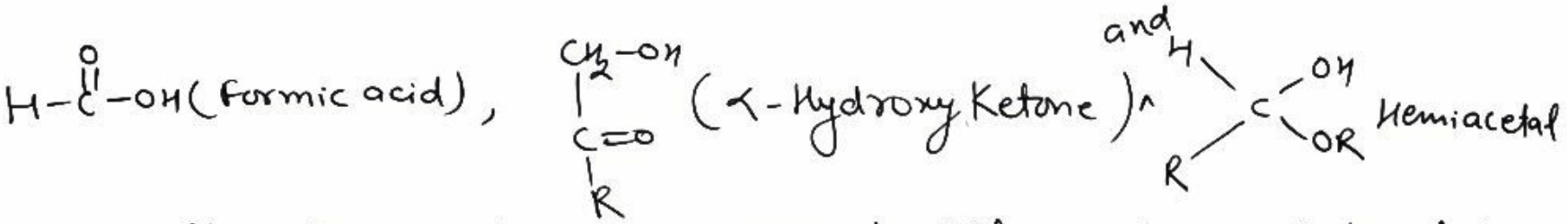
$\xrightarrow{H_2SO_3/SO_2}$ Schiff's reagent (Colourless) $\xrightarrow{R-CHO}$ Schiff base (Pink)

Reduction, Oxidation And Hydrolysis

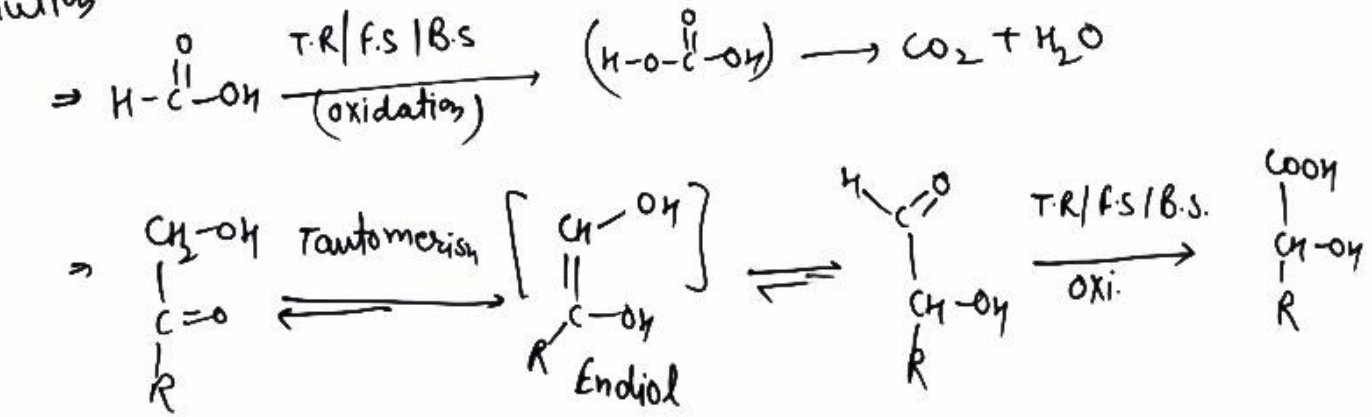
(II) Oxidation :-

(D) Oxidation of aldehydes and Ketones :-

(D1) Oxidation of aldehydes :-

Note (3) 

Can give positive test with tollen's reagent, fehling solution and benedict Solution



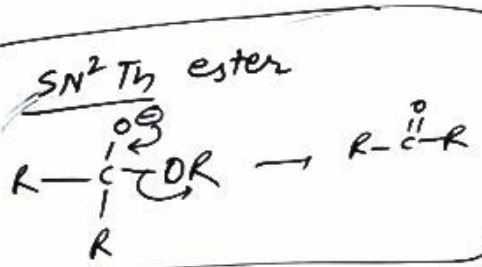
Reduction, Oxidation and Hydrolysis

(II) Oxidation :-

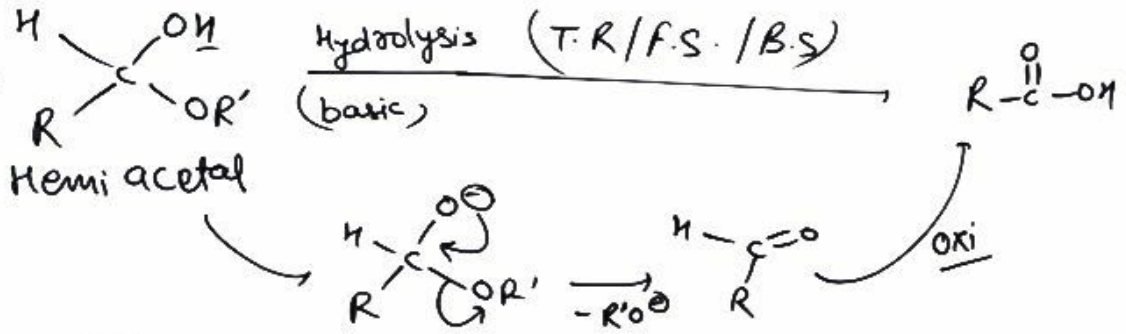
(D) Oxidation of aldehydes and Ketones :-

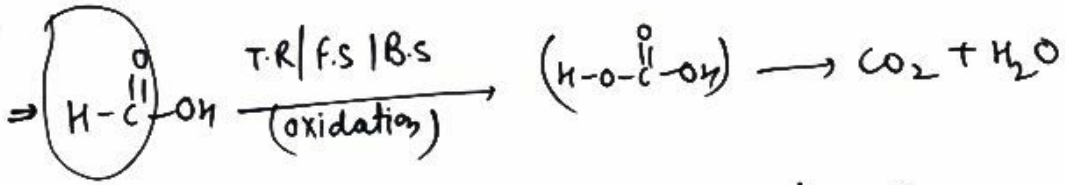
(DI) Oxidation of aldehydes :-

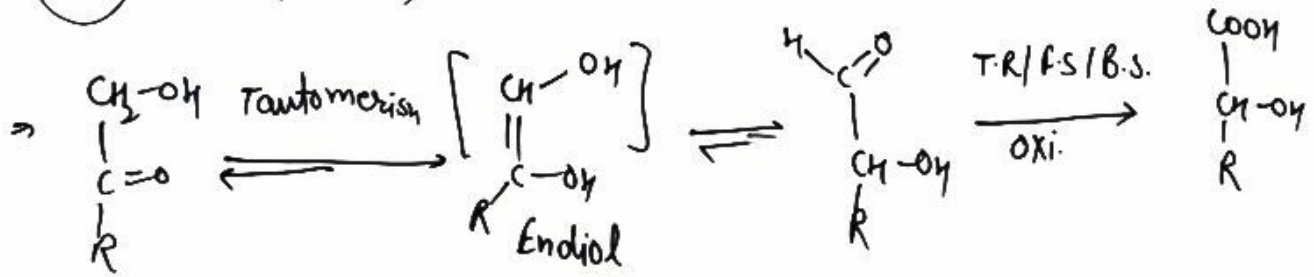
S_N2 Th ester



Note (3)







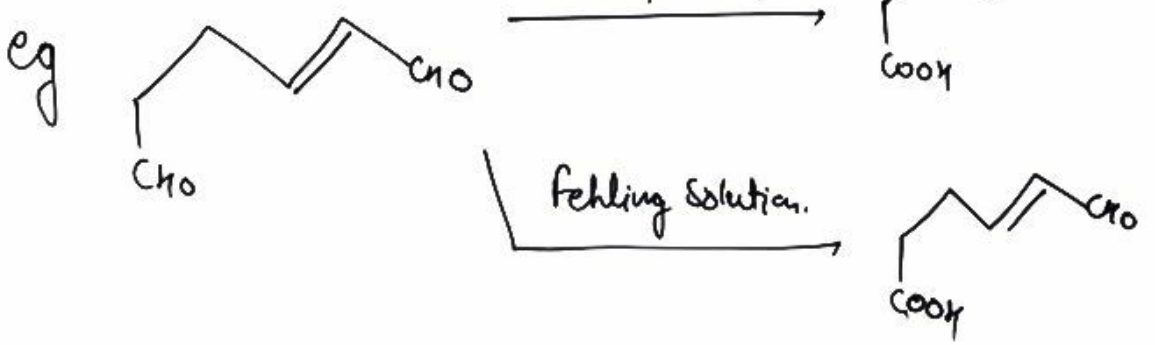
Reduction, Oxidation And Hydrolysis

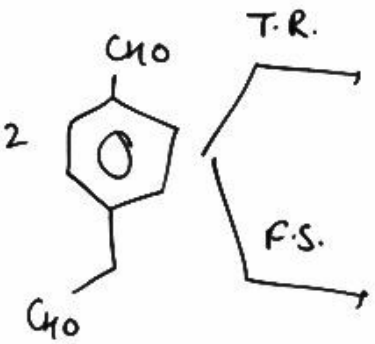
(II) Oxidation :-

(D) Oxidation of aldehydes and ketones :-

(D1) Oxidation of aldehydes :-

Note : (4) $\begin{bmatrix} Ar-CHO \\ CH_2=CH-CHO \end{bmatrix}$ Negative test with Fehling, Benedict solution & Schiff's test give Tollen's however

eg 

eg 2 

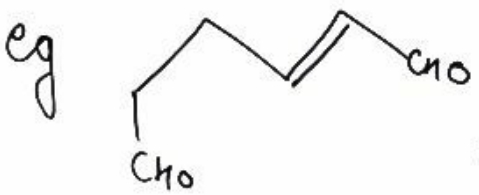
Reduction, Oxidation and Hydrolysis

(II) Oxidation :-

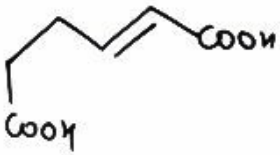
(D) Oxidation of aldehydes and ketones :-

(D1) Oxidation of aldehydes :-

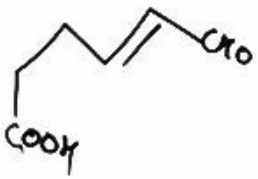
Note : (4) $\left[\begin{array}{l} Ar-CHO \\ CH_2=CH-CHO \end{array}\right]$ Negative test with Fehling, Benedict solution & Schiff's test

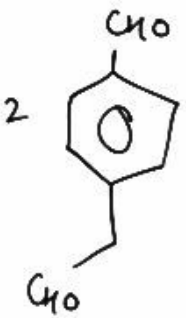
eg 

Tollen's reagent →



Fehling Solution →



eg 2 

T.R. →



F.S. →



Reduction, Oxidation And Hydrolysis

(II) Oxidation :-

CH_3 - methyl

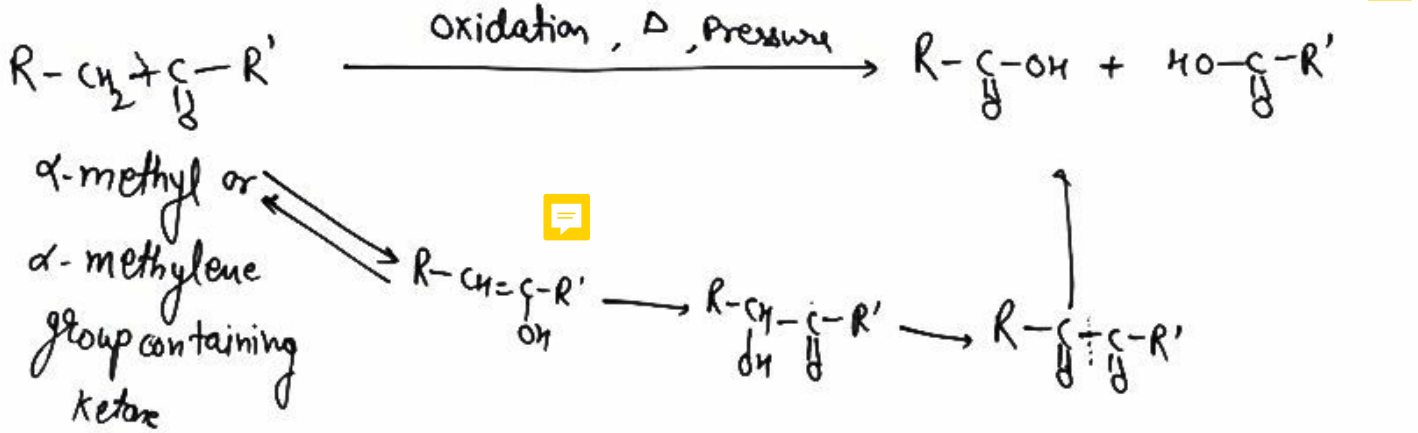
$-CH_2-$ methylene

(D) Oxidation of aldehyde & Ketones:-

(D.2) Oxidation of Ketones:-

Oxidation of ketones is possible only in difficult conditions in presence of strong oxidising agent like conc. $\overset{\Delta}{HNO_3}$, conc. $KMnO_4, \Delta$, conc. $K_2Cr_2O_7$ etc. and form lesser carbon containing carboxylic acid(s).

only ketones having methyl/methylene in adjacent position undergo oxidation

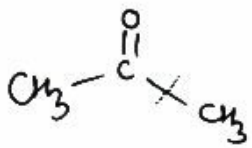


Reduction, Oxidation And Hydrolysis

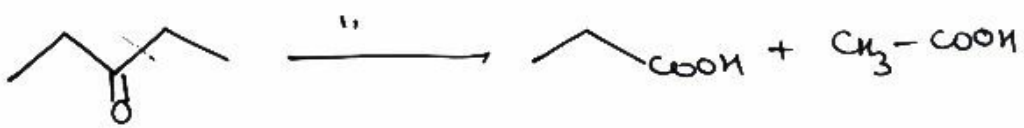
(II) Oxidation :-

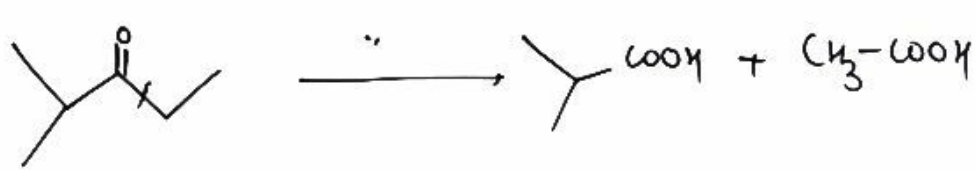
(D) Oxidation of aldehyde & Ketones:-

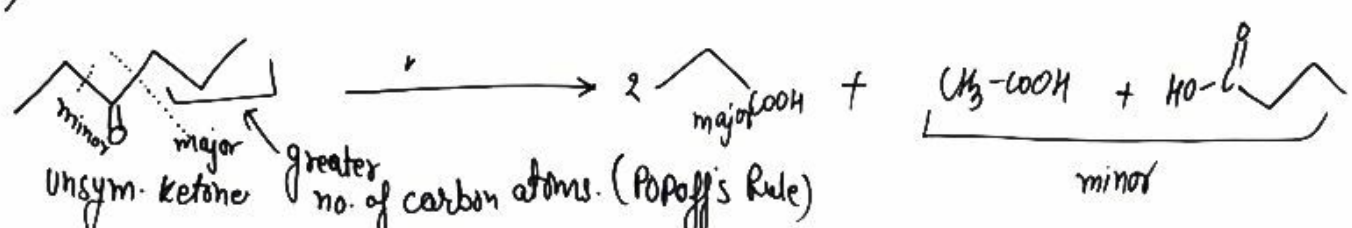
(D·2) Oxidation of Ketones:-

Eg 1 

formic acid gets further oxidised to carbon dioxide

Eg 2 

Eg 3 

Eg 4 

REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

(D) Oxidation of aldehyde & Ketones:-

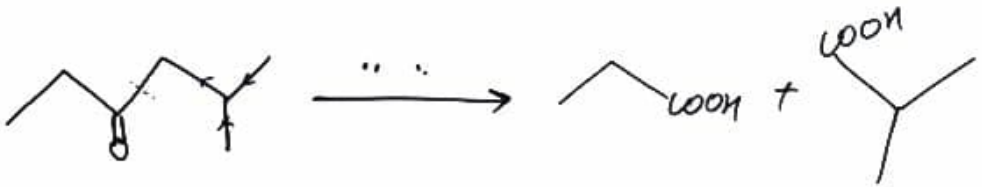
(D·2) Oxidation of Ketones:-

Eg 5

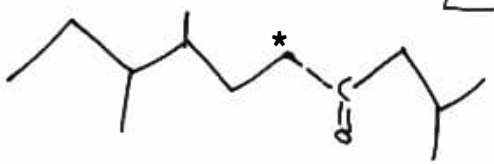


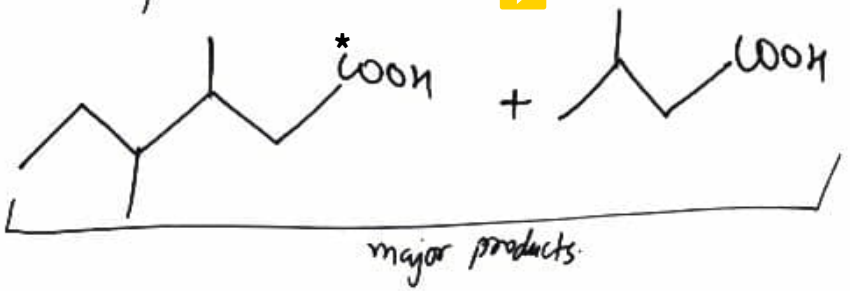
Adipic acid

Eg 6



Q. Ketone (X) ?





major products.

Reduction, Oxidation And Hydrolysis

(II) Oxidation :-

(D) Oxidation of aldehyde & Ketones:-

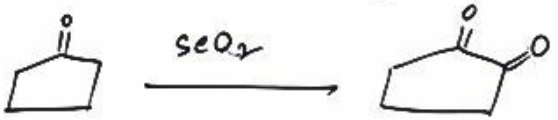
D.3 Oxidation with SeO_2, Δ ⇒ α-carbonylation on α-methyl/α-methylene group of carbonyl comp. (Aldehyde/Ketone)

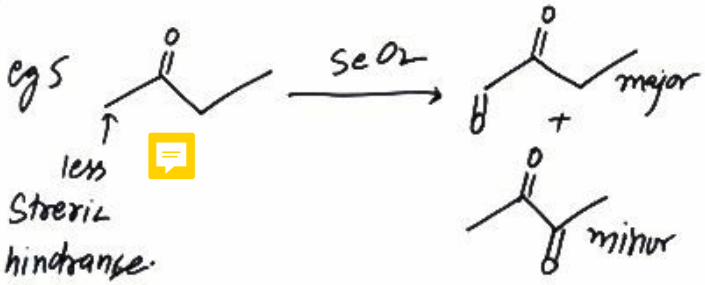
cyclic transition state
pericyclic mechanism (not in syllabus)

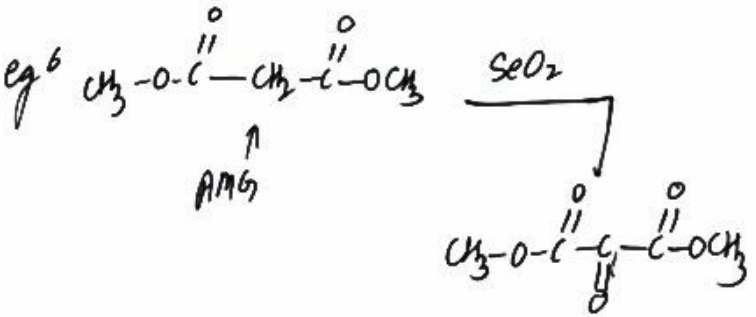
eg 1 

eg 2 

eg 3 

eg 4 

eg 5 

eg 6 

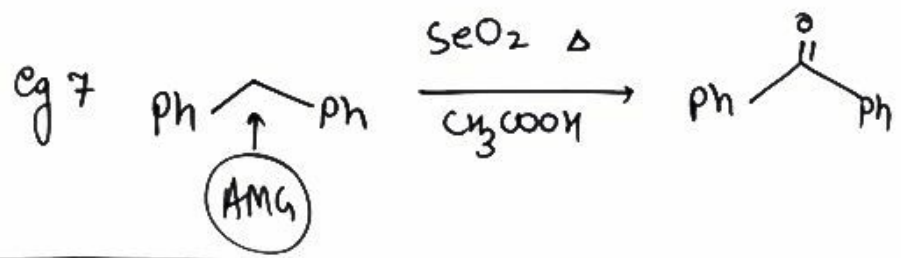
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

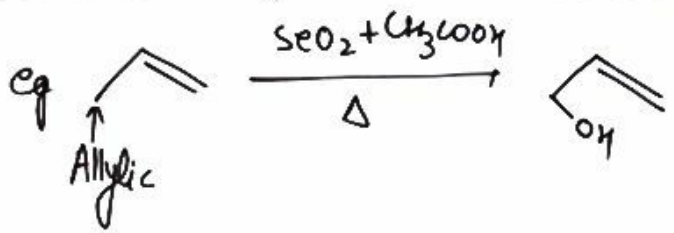
(D) Oxidation of aldehyde & Ketones:-

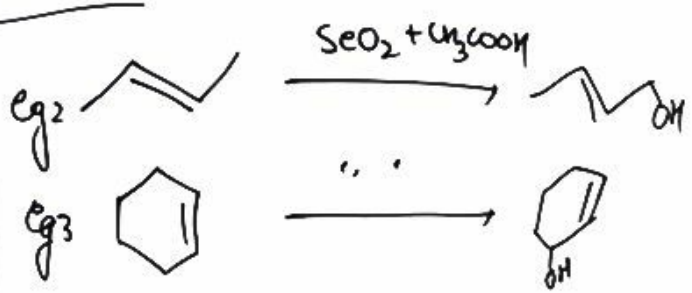
D.3 Oxidation with SeO_2, Δ ⇒ α-carbonylation on α-methyl/α-methylene group of carbonyl comp. (Aldehyde/Ketone)

Eg 7



Allylic hydroxylation with SeO_2 of alkenes:





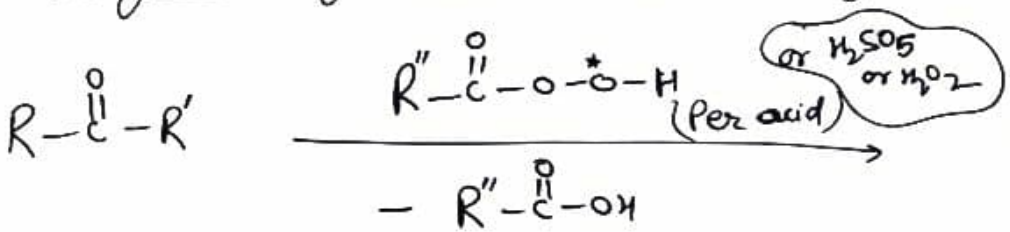
kisi bhi ek jagah laga do

REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

(D) Oxidation of aldehyde & Ketones:-

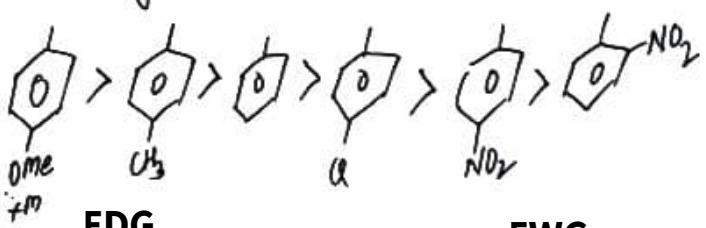
D.4 Baeyer - Villiger oxidation (Rearrangement)



R' > R ← migratory aptitude (Relative order of migration of a group in rearrangement)

⇒ H > Phenyl > Alkyl

Alkyl ⇒ 3° > 2° > 1°

Aryl ⇒ 

EDG

EWG

REDUCTION, OXIDATION AND HYDROLYSIS

Fries Rearr.

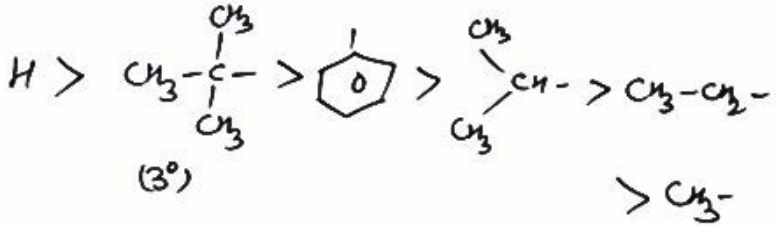


(II) OXIDATION :-

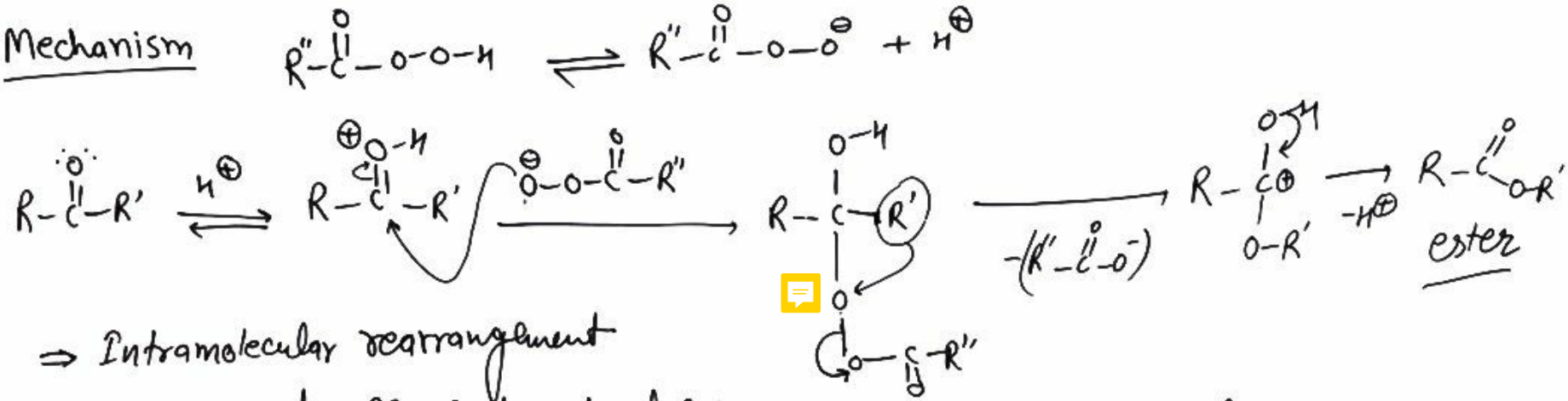
(D) Oxidation of aldehyde & Ketones:-

D.4 Baeyer - Villiger oxidation (Rearrangement)

Final migratory abtitude for B.V. oxidation.



Mechanism



⇒ Intramolecular rearrangement

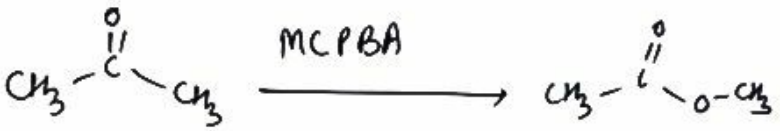
⇒ When migratory group has chiral carbon, retension take plane of chiral carbon after rearr.

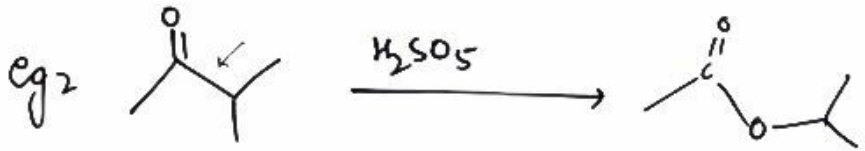
REDUCTION, OXIDATION AND HYDROLYSIS

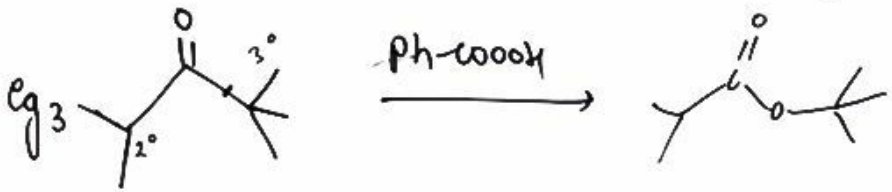
(II) OXIDATION :-

(D) Oxidation of aldehyde & Ketones:-

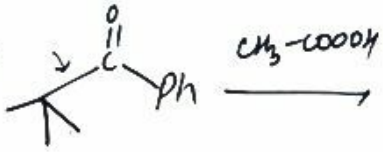
D.4 Baeyer - Villiger oxidation (Rearrangement)

eg 1 

eg 2 

eg 3 

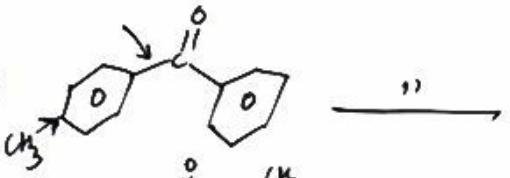
eg 4 

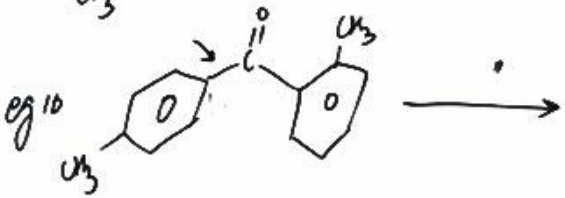
eg 5 

★ eg 6 

eg 7 

eg 8 

eg 9 

eg 10 

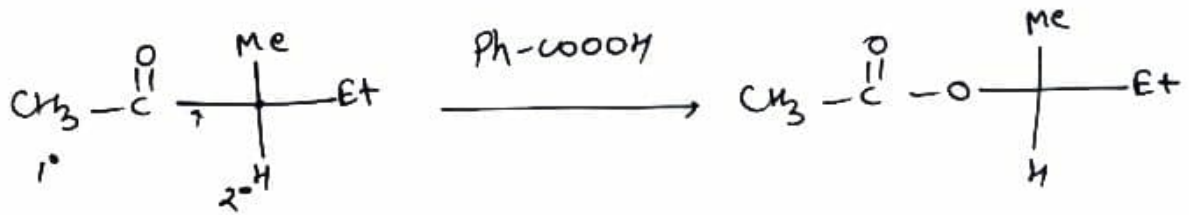
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

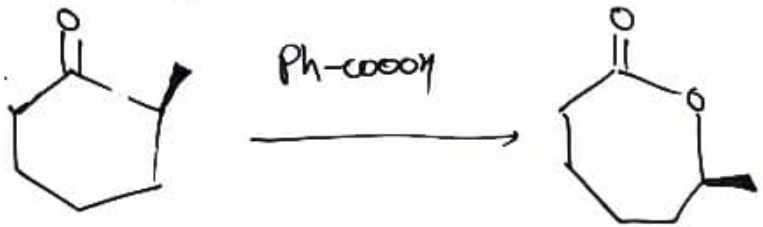
(D) Oxidation of aldehyde & Ketones:-

D.4 Baeyer - Villiger oxidation (Rearrangement)

eg 11



eg 12

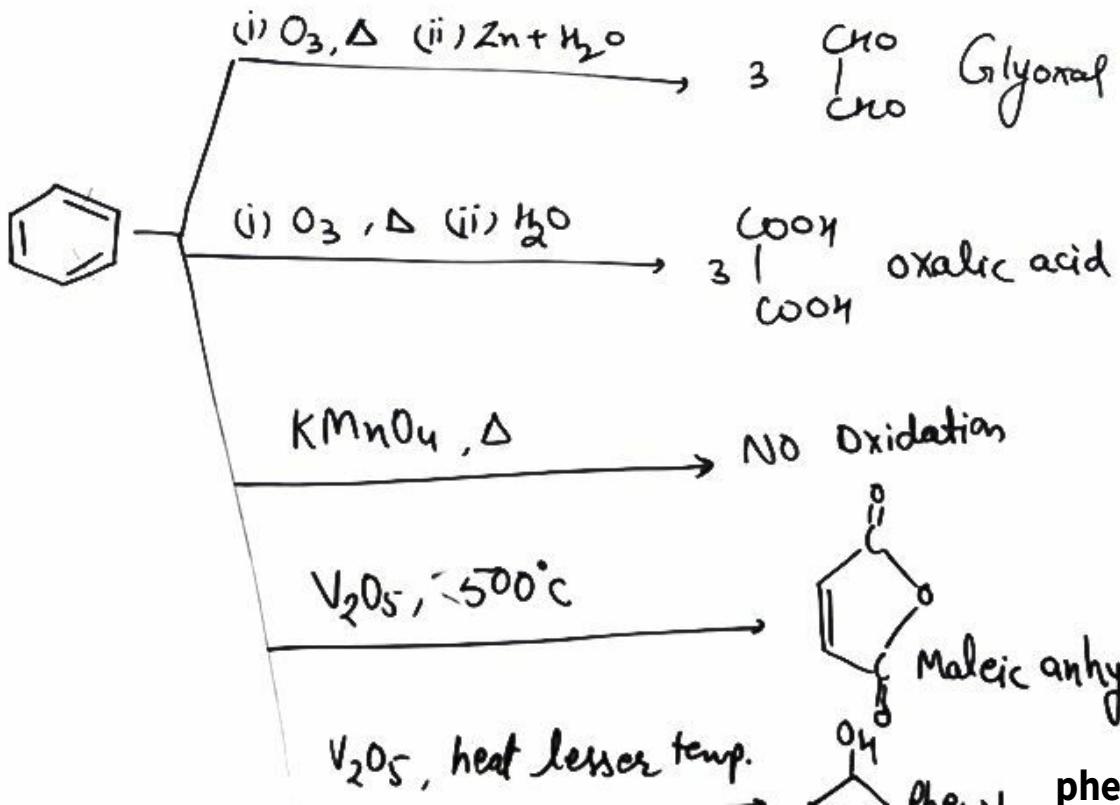


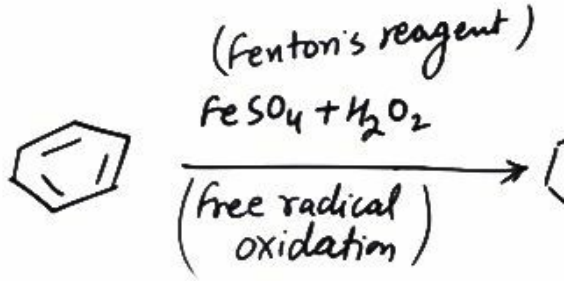
Reduction, Oxidation And Hydrolysis

(II) Oxidation :-

(E) Oxidation of aromatic compounds.

① oxidation of benzene





product phenol

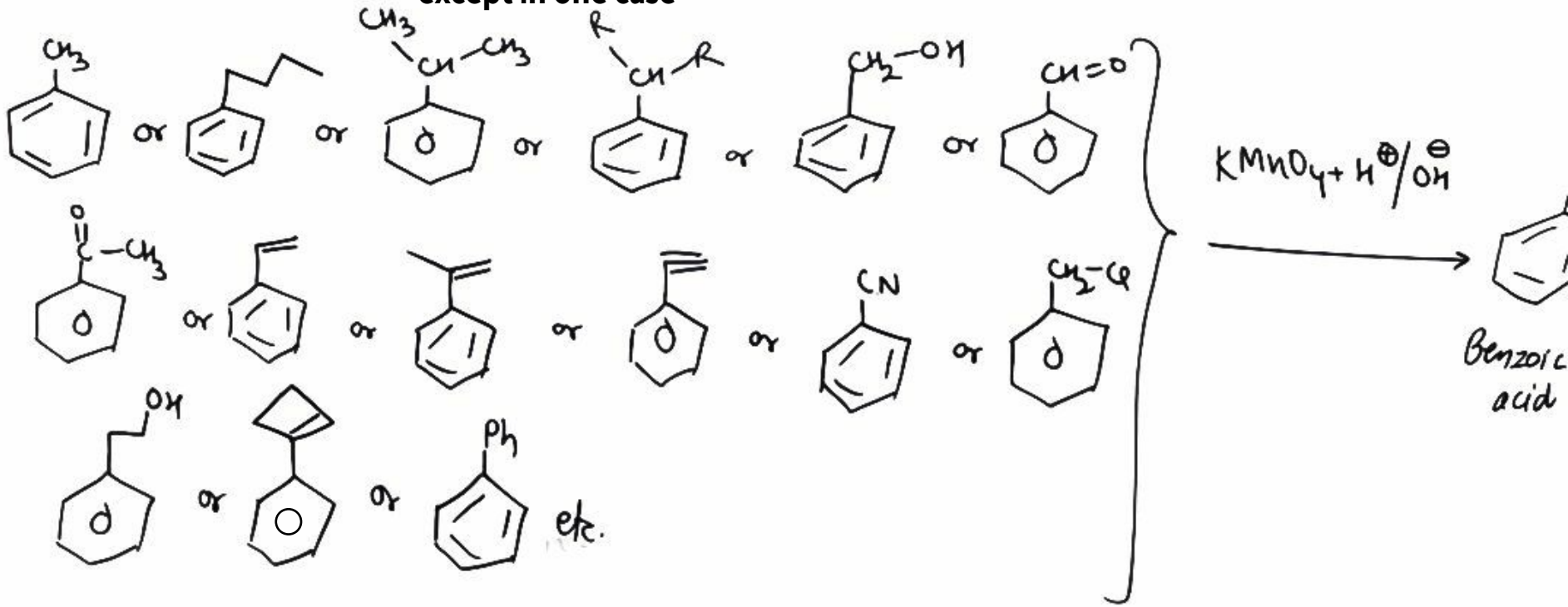
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

(E) Oxidation of aromatic compounds.

(2) oxidation of side chain of aromatic ring :-

benzene should have a directly attached C, rest whatever it is, doesn't matter, except in one case



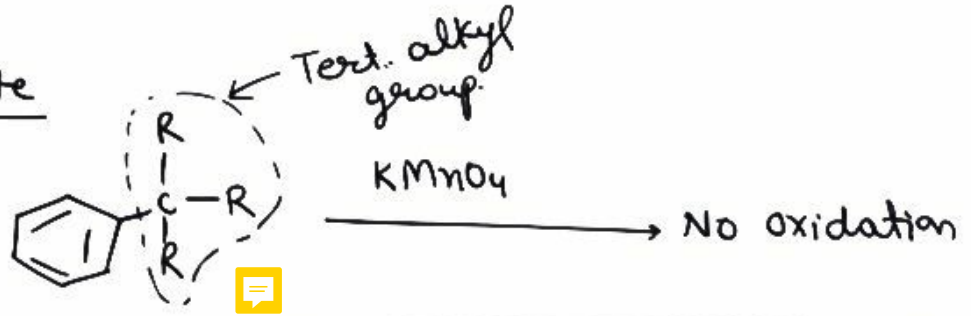
Reduction, Oxidation And Hydrolysis

(II) Oxidation :-

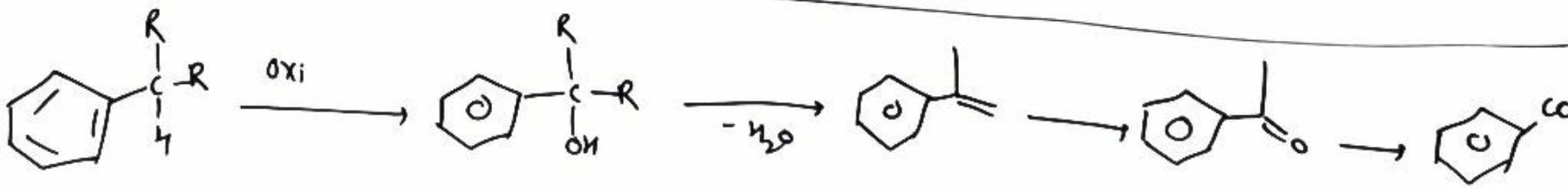
(E) Oxidation of aromatic compounds.

(2) oxidation of side chain of aromatic ring :-

Note



Step.



multi-step process, finally ends up at carboxylic acid

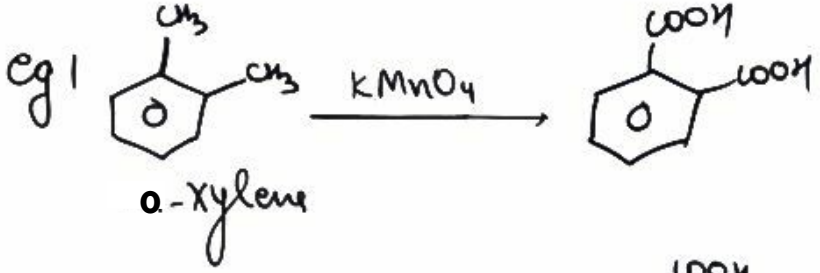
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

(E) Oxidation of aromatic compounds.

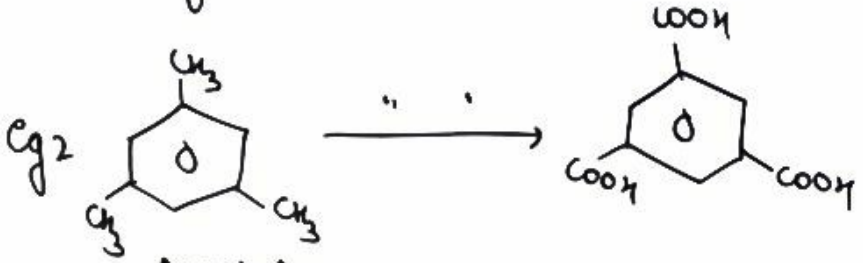
(2) Oxidation of side chain of aromatic ring :-

eg 1



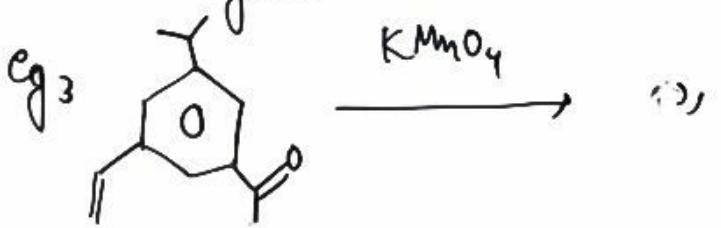
o-Xylene

eg 2

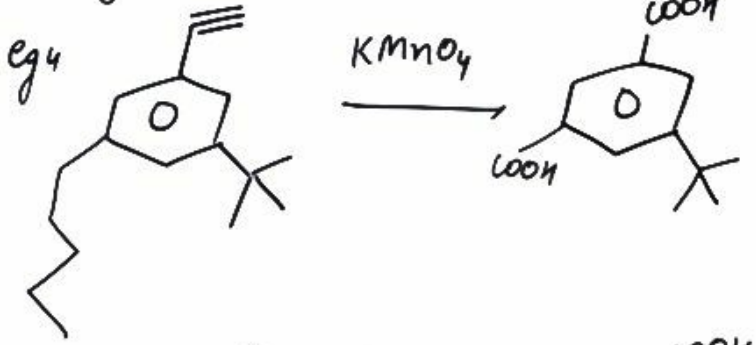


Mesitylene

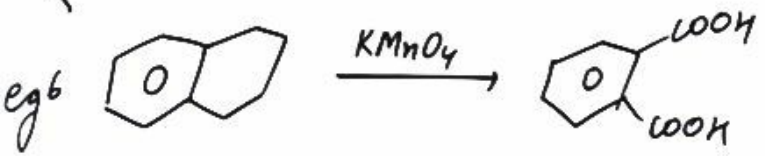
eg 3



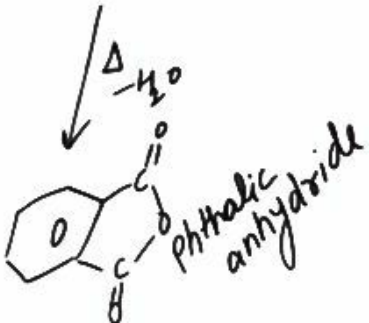
eg 4



eg 6



Phthalic acid



Phthalic anhydride

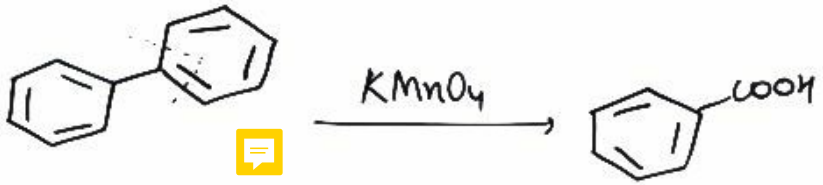
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

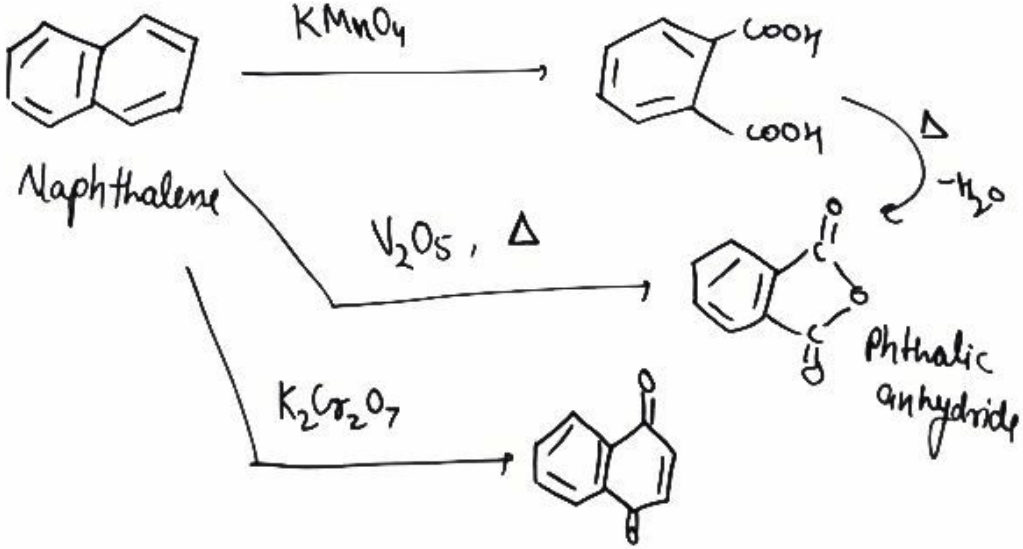
(E) Oxidation of aromatic compounds.

(2) oxidation of side chain of aromatic ring :-

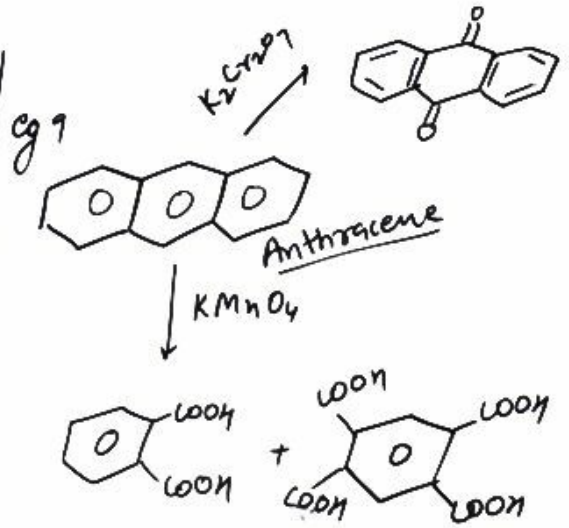
eg 7



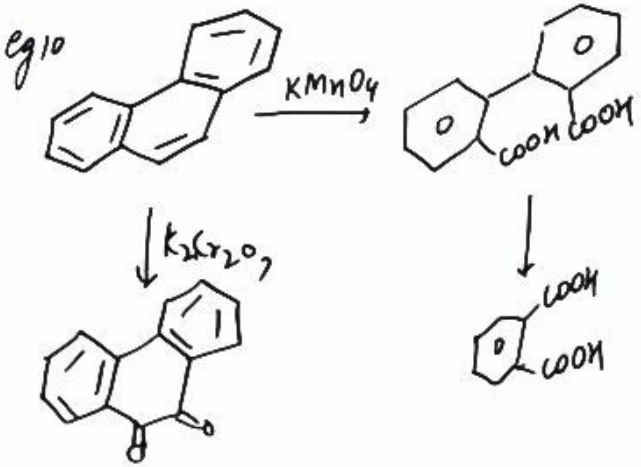
eg 8



eg 9



eg 10



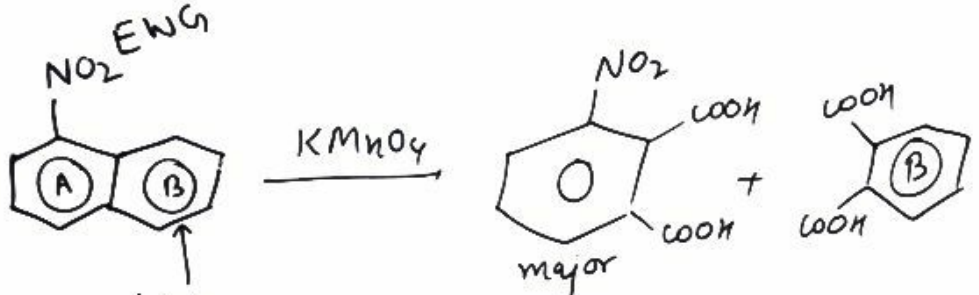
REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

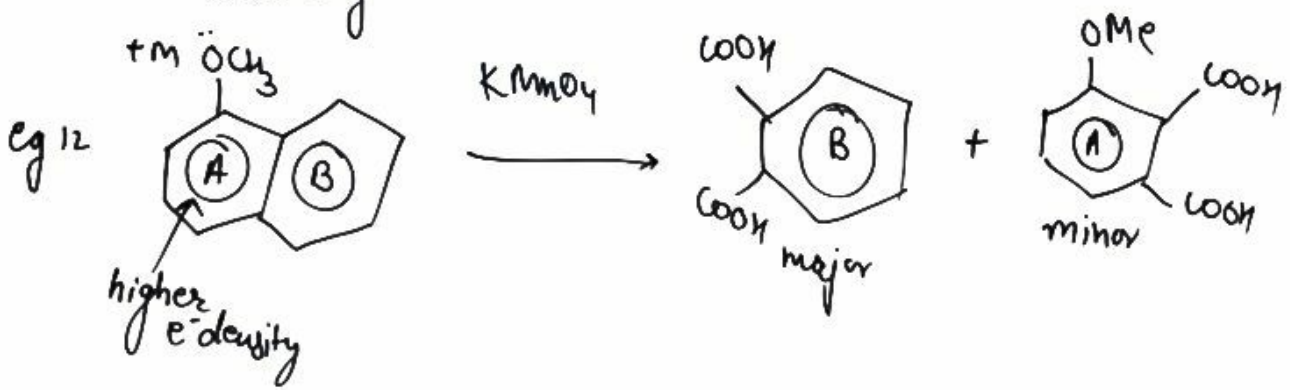
(E) Oxidation of aromatic compounds.

(2) Oxidation of side chain of aromatic ring :-

eg 11



eg 12

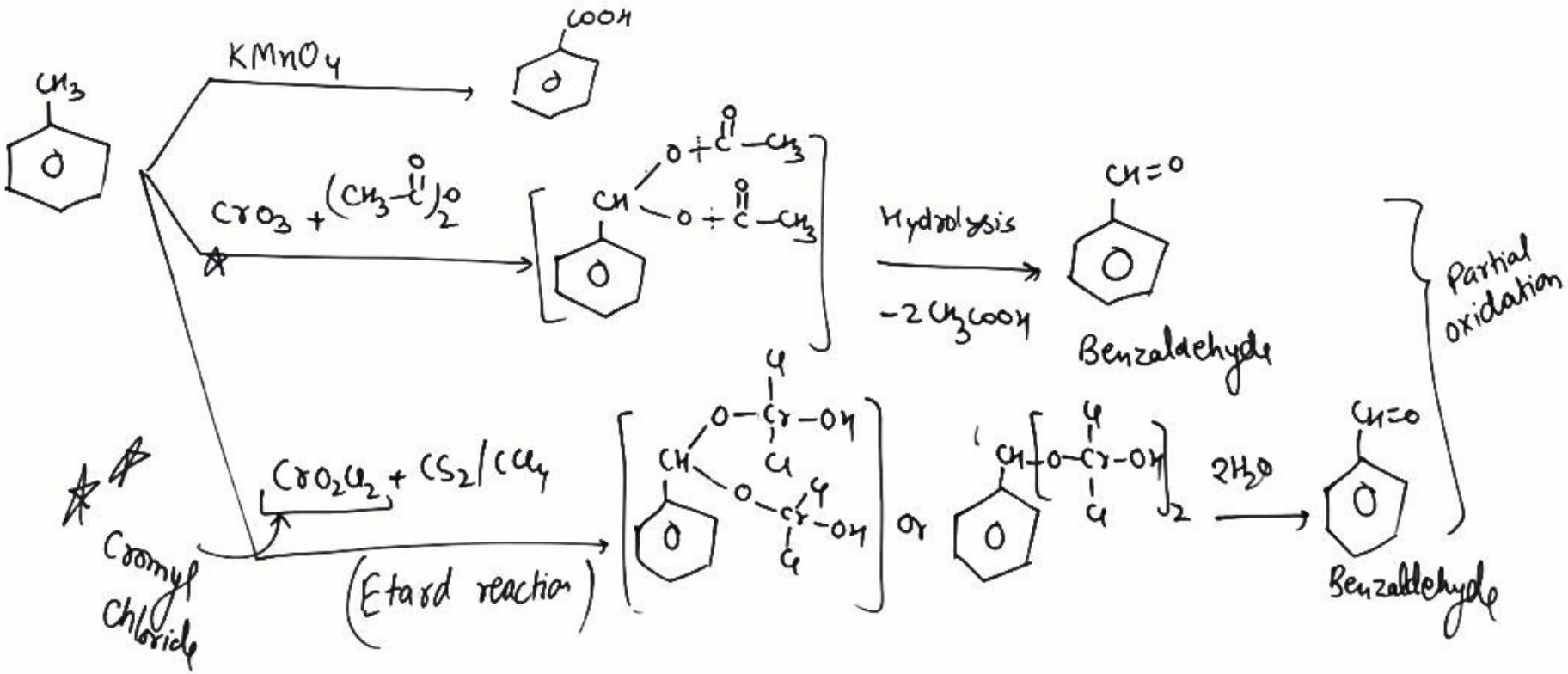


Reduction, Oxidation And Hydrolysis

(II) Oxidation :-

(E) Oxidation of aromatic compounds.

(3) Oxidation of Toluene:-



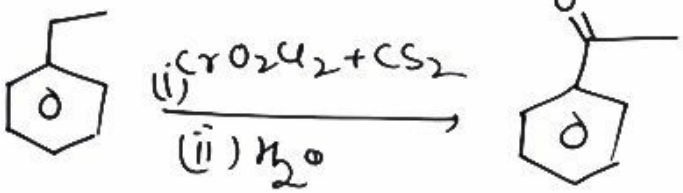
Reduction, Oxidation and Hydrolysis

(II) Oxidation :-

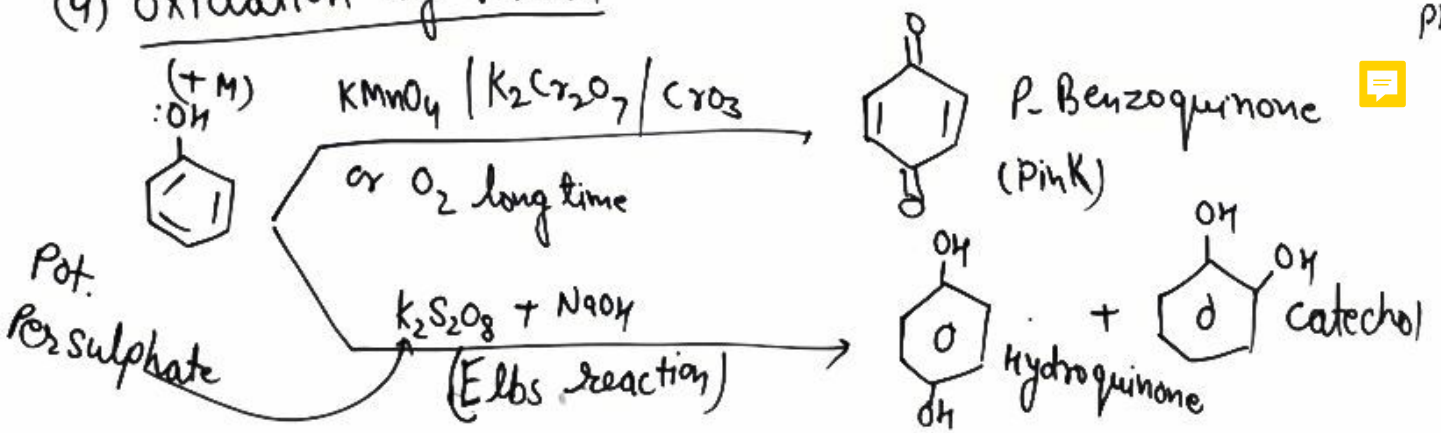
(E) Oxidation of aromatic compounds.

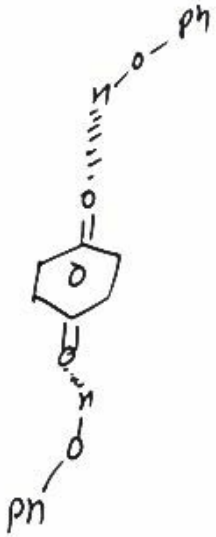
(3) Oxidation of toluene:-

Etard rex. with ethyl benzene



(4) oxidation of phenol





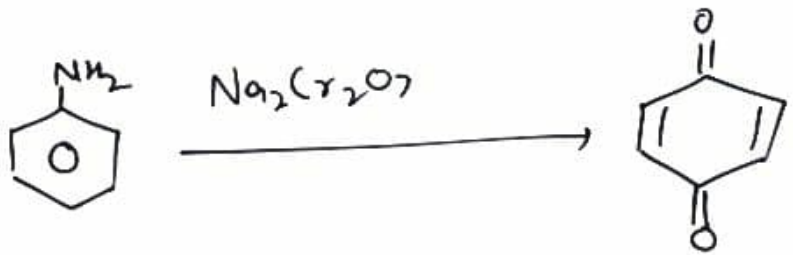
H-bonding b/w phenol and benzoquinone

REDUCTION, OXIDATION AND HYDROLYSIS

(II) OXIDATION :-

(E) Oxidation of aromatic compounds.

(5) Oxidation of aniline



REDUCTION, OXIDATION AND HYDROLYSIS

(III) HYDROLYSIS :–

[A] Hydrolysis of alkyl halides :– (ORM-III)

$$R\text{–}Cl \xrightarrow{H_2O + \overset{\ominus}{O}H} R\text{–}OH$$

Alcohol

[B] Hydrolysis of acid derivative (ORM-I) S_N^2 Th

$$R\text{–}\overset{O}{\overset{\|}{C}}\text{–}Cl \xrightarrow{H_2O + H^{\oplus}/\bar{O}H} R\text{–}COOH + HCl$$

$$R\text{–}\overset{O}{\overset{\|}{C}}\text{–}O\text{–}\overset{O}{\overset{\|}{C}}\text{–}R \xrightarrow{,,\quad :} 2\,R\text{–}COOH$$

$$R\text{–}\overset{O}{\overset{\|}{C}}\text{–}O\text{–}R' \xrightarrow{,,\quad :\ \Delta} R\text{–}COOH + R'\text{–}OH$$

$$R\text{–}\overset{O}{\overset{\|}{C}}\text{–}NH_2 \xrightarrow{,,\quad :\ \Delta} R\text{–}\overset{O}{\overset{\|}{C}}\text{–}OH + NH_3$$

① Hydrolysis of cyanide

$$R\text{–}C\equiv N \xrightarrow[\text{room temp.}]{H_2O + H^{\oplus}/\overset{\ominus}{O}H} \underset{\text{amide}}{R\text{–}\overset{O}{\overset{\|}{C}}\text{–}NH_2}$$

partial hydrolysis

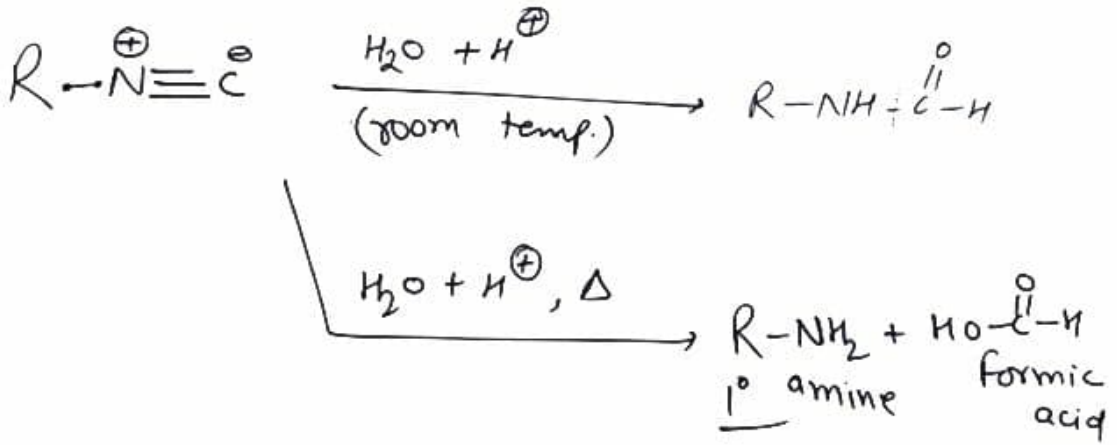
$$\xrightarrow{H_2O + H^{\oplus}/\overset{\ominus}{O}H,\ \Delta} R\text{–}COOH + NH_3$$

complete hydrolysis

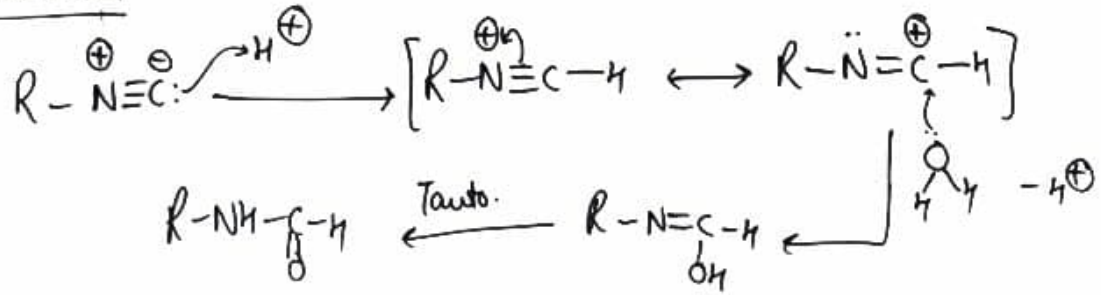
REDUCTION, OXIDATION AND HYDROLYSIS

(III) HYDROLYSIS :-

(D) Hydrolysis of isocyanide (only in acidic medium)



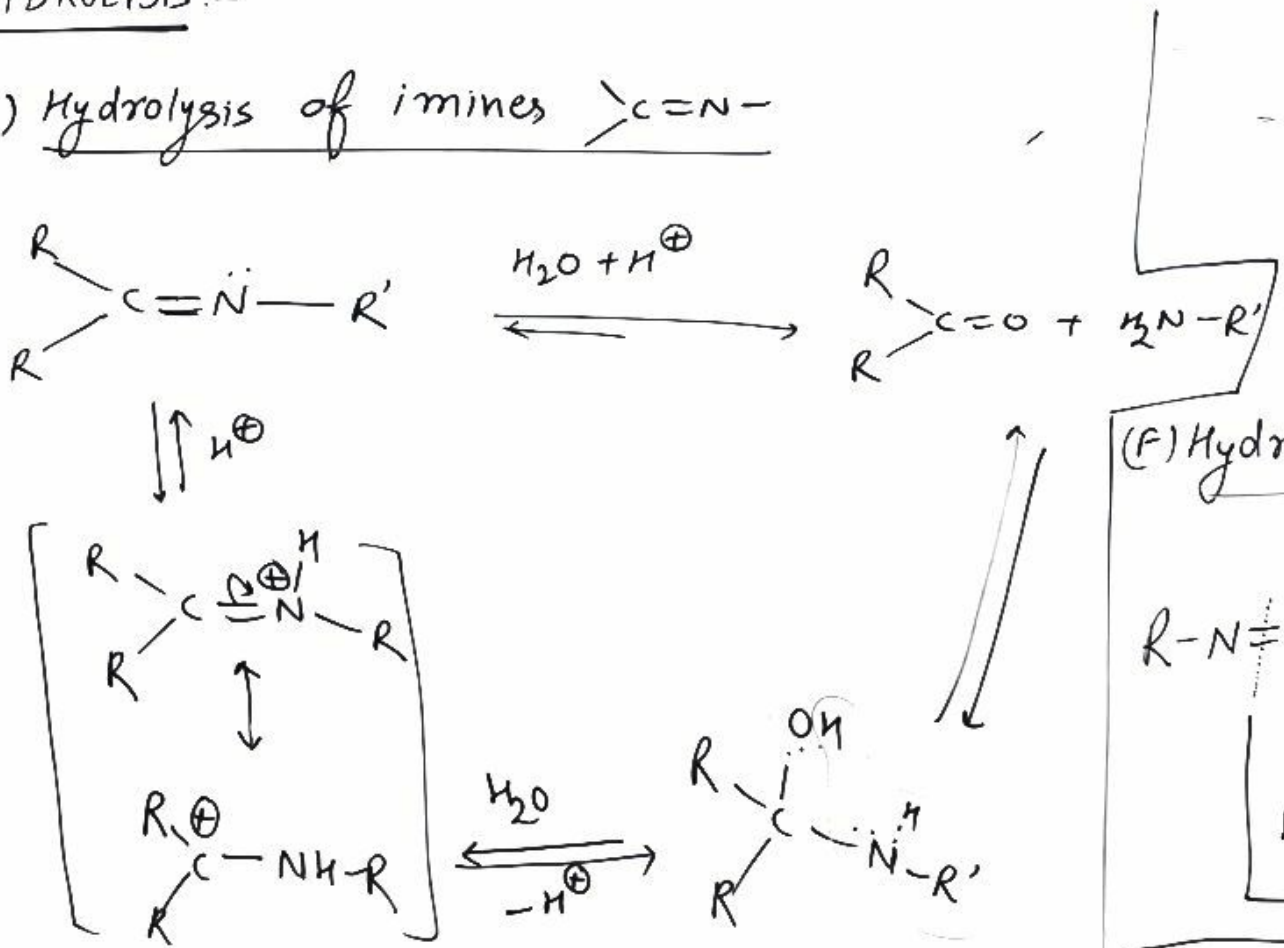
mechanism



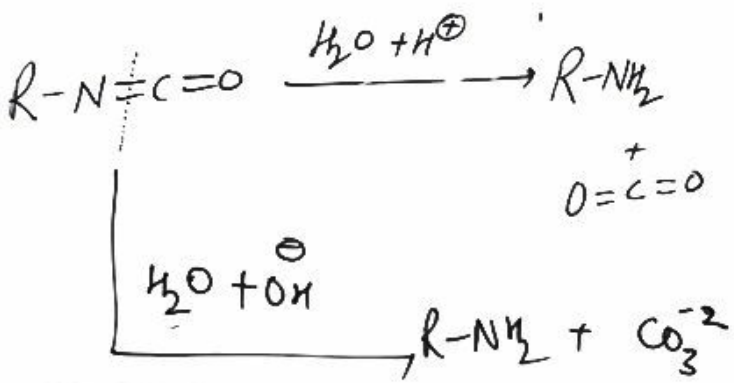
REDUCTION, OXIDATION AND HYDROLYSIS

(III) HYDROLYSIS :-

(E) Hydrolysis of imines $\rangle C=N-$



(F) Hydrolysis of isocyanate



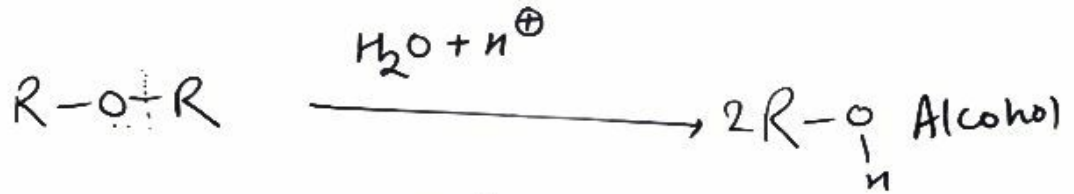
$2NaOH + CO_2 \longrightarrow Na_2CO_3$

REDUCTION, OXIDATION AND HYDROLYSIS

(III) HYDROLYSIS :-

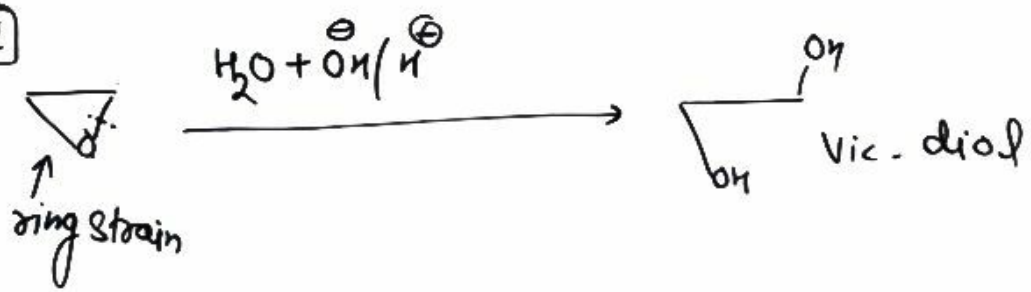
(G) Hydrolysis of ether (R-O-R) (in ORM-III)

- Only in acidic medium



(H) Hydrolysis of epoxide :- (Hydrolysis possible in acidic as well as basic medium)

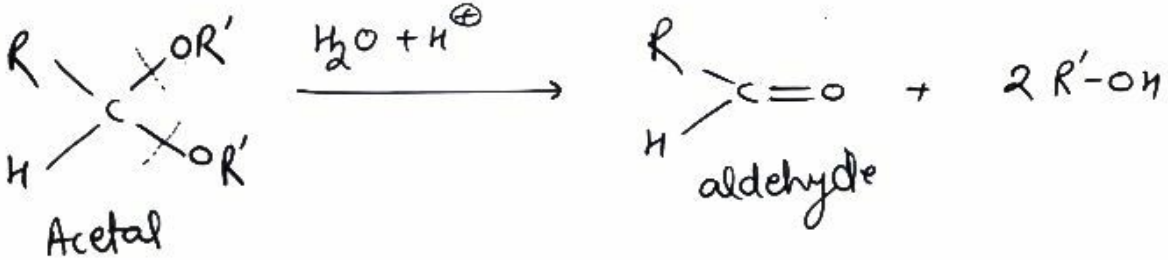
ORM-III

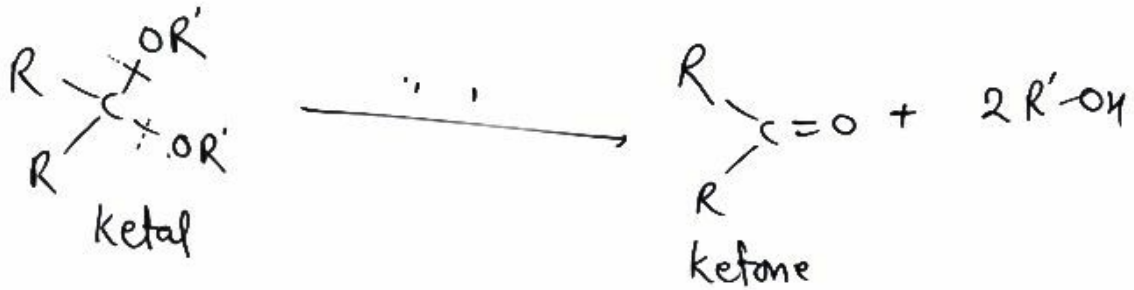


REDUCTION, OXIDATION AND HYDROLYSIS

(III) HYDROLYSIS :–

(I) Hydrolysis of acetal and ketal (Hydrolysis only in acidic medium)

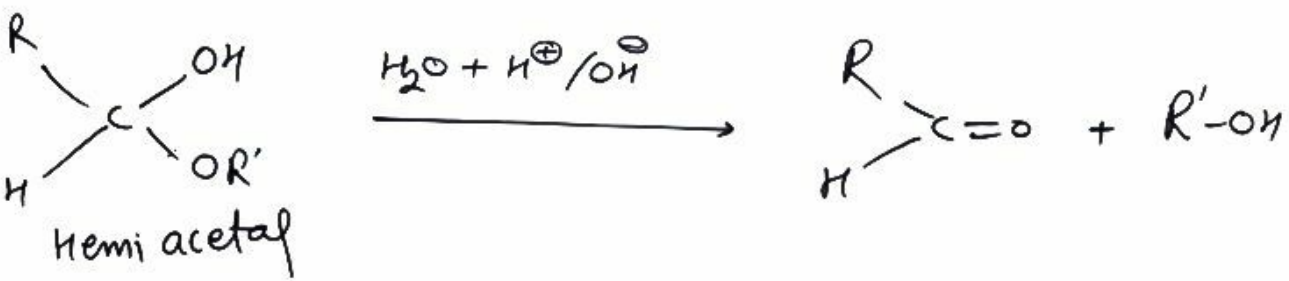


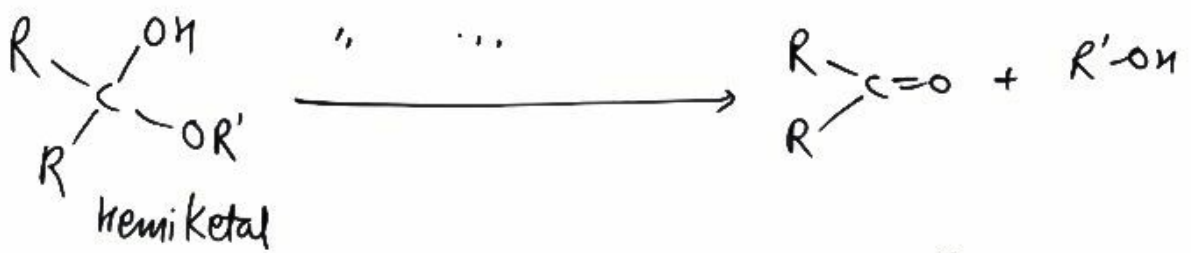


REDUCTION, OXIDATION AND HYDROLYSIS

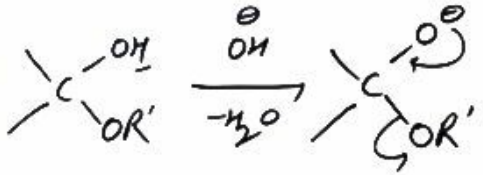
(III) HYDROLYSIS :-

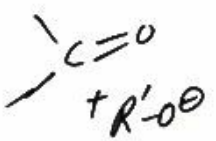
(J) Hydrolysis of Hemiacetal & Hemi Ketal :- (Hydroolysis possible in $H^{\oplus}/OH^{\ominus}$ medium)





Mechanism in basic medium.



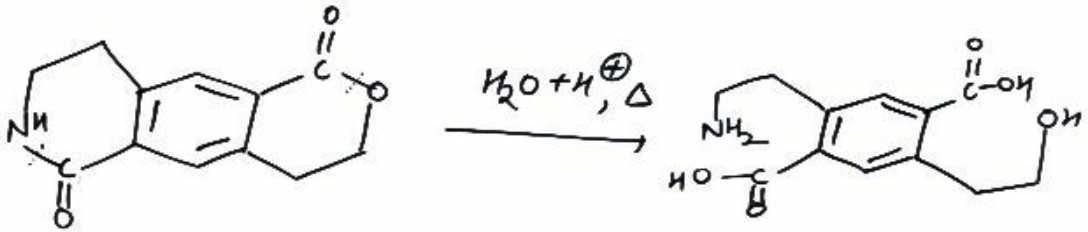


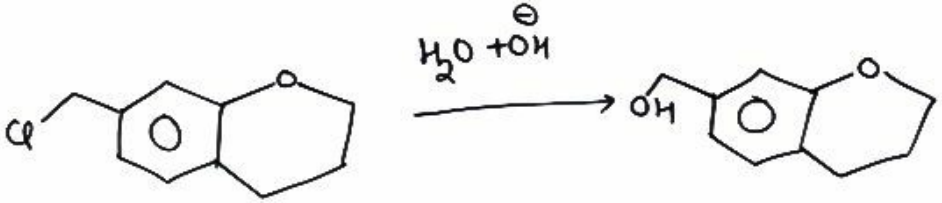


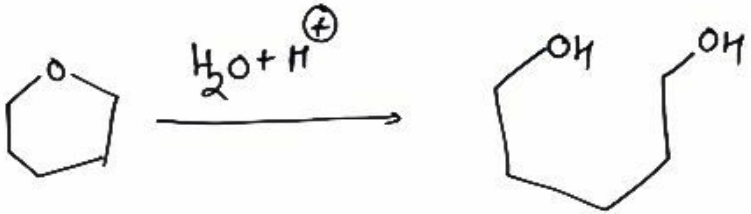
REDUCTION, OXIDATION AND HYDROLYSIS

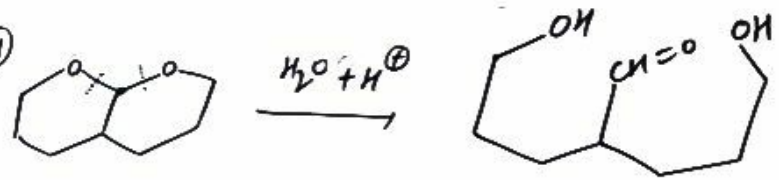
(III) HYDROLYSIS :-

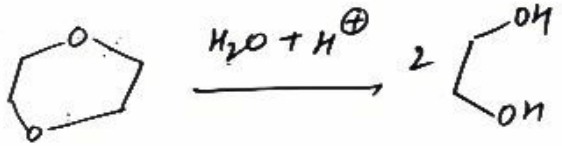
Q.

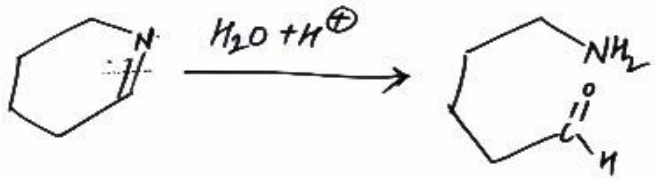
① 

② 

③ 

④ 

⑤ 

⑥ 

⑦ 